Handout

BT504 – Genomics and Proteomics

Mid Term Handout

Prepared by Ahmad Saeed

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Lesson 1 : Study of Omics

Study of Omics

- Genomics: all the genes
- Pharmacogenomics choice of personalized medicine
- Nutrigenomics choice of best diet
- Toxicogenomics prediction of toxicity
- Epigenomics: all epigenetic changes in genome
- Transcriptomics: all the mRNAs
- Proteomics : all the proteins
- Interactomics : all interactions between all proteins
- Metabolomics (or metabonomics): all metabolites

Genomics, Proteomics and Metabolomics

Genome and Genomics

- The complete set of DNA found in each cell is known as the genome and study is called as genomics.

Proteome and Proteomics

- The complete set of proteins found in each cell is known as the proteome.

Proteome and Proteomics

-
- Proteins concentration (and activity) may be different than gene expression due to post-translational modification

Metabolomics

-
  - The complete set of metabolites found in each cell is known as the metabolome.
  - Use of high-throughput mass spectrometry to analyze the metabolic components of cell.

Metabolomics

-
  - Useful for determining the effects of the environment or gene transformation on the metabolism of the plants/animals.

Conclusion

-
  - Genomics, proteomics and metabolomics will give an integrated, wholistic view of the cell.

<u>Lesson 2 : Genomics, Proteomics and metabolomics</u>

Genomics, Proteomics and Metabolomics

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  - The complete set of proteins found in each cell is known as the proteome.

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Metabolomics

-

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Conclusion

-
- Genomics, proteomics and metabolomics will give an integrated, wholistic view of the cell.
- Can be used to monitor or modify organisms in a comprehensive way. Bioinformatics - the key to understand the plethora of information and modeling the cell.

Lesson 3 : Structure of RNA

Structure of RNA

Ribonucleic acid (RNA) is a polymeric molecule essential in various biological roles in coding, decoding, regulation and expression of genes. RNA and DNA are nucleic acids, and, along with lipids, proteins and carbohydrates, constitute the four major macromolecules essential for all known forms of life. Like DNA, RNA is assembled as a chain of nucleotides, but unlike DNA, RNA is found in nature as a single strand folded onto itself, rather than a paired double strand. Cellular organisms use messenger RNA (mRNA) to convey genetic information (using the nitrogenous bases of guanine, uracil, adenine, and cytosine, denoted by the letters G, U, A, and C) that directs synthesis of specific proteins. Many viruses encode their genetic information using an RNA genome.

Some RNA molecules play an active role within cells by catalyzing biological reactions, controlling gene expression, or sensing and communicating responses to cellular signals. One of these active processes is protein synthesis, a universal function in which RNA molecules direct the synthesis of proteins on ribosomes. This process uses transfer RNA (tRNA) molecules to deliver amino acids to the ribosome, where ribosomal RNA (rRNA) then links amino acids together to form coded proteins.

Structure of RNA nucleotides:

Each nucleotide in RNA contains a ribose sugar, with carbons numbered 1' through 5'. A base is attached to the 1' position, in general, adenine (A), cytosine (C), guanine (G), or uracil (U). Adenine and guanine are purines, cytosine and uracil are pyrimidines. A phosphate group is attached to the 3' position of one ribose and the 5' position of the next. The phosphate groups have a negative charge each, making RNA a charged molecule (polyanion). The bases form hydrogen bonds

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between cytosine and guanine, between adenine and uracil and between guanine and uracil. However, other interactions are possible, such as a group of adenine bases binding to each other in a bulge, or the GNRA tetraloop that has a guanine–adenine base-pair.

Structure of RNA

Lesson 4 : DNA transcription

Transcription:

Transcription is a process by which cells are able to express their genes. It is how DNA is re-written into RNA (specifically messenger RNA). mRNA may then direct the synthesis of various proteins.

Controlling the production of mRNA in the nucleus allows the regulation of gene expression. In this article we will look at the process of DNA transcription and how mRNA is processed.

Stages of DNA Transcription:

The process of transcription can be broadly categorised into 3 main stages: initiation, elongation & termination.

Initiation

Transcription is catalysed by the enzyme RNA polymerase. It attaches to and moves along the DNA molecule until it recognises a promoter sequence. There may be multiple promoter sequences in a DNA molecule. They indicate transcription start sites.

It then unwinds a portion of the DNA double helix, exposing the bases on each of the singular DNA strands. One of the strands will act as a template to create the new mRNA strand.

Incoming ribonucleotides are used by RNA polymerase to form the mRNA strand. It does this using complementary base pairing (A to U, T to A, C to G and G to C). The RNA polymerase then binds the ribonucleotides together covalent by phosphodiester bonds.

Eukaryotes vs Prokaryotes

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Understanding transcription in both eukaryotes and prokaryotes allows you to target their metabolic differences. This allows transcription to be disrupted in prokaryotic bacteria without harming the body's eukaryotic cells.

*(present 30 base pairs down stream from coding region)

Elongation

This is the process by which ribonucleotides are added to the template strand, enabling growth of the mRNA transcript. The mRNA transcript is made in a 5' to 3' direction.

Termination

Elongation will continue until the RNA polymerase encounters a stop sequence. At this point, transcription stops and the RNA polymerase releases the DNA template.

Lesson 5 : Protein translation

Protein Translation

Translation is a process by which the genetic code contained within an mRNA molecule is decoded to produce the specific sequence of amino acids in a polypeptide chain.

It occurs in the cytoplasm following transcription and, like transcription, has three stages: initiation, elongation and termination. In this article we will look at the stages of translation and compare the process in prokaryotes and eukaryotes.

Components of Translation

The key components required for translation are mRNA, ribosomes, tRNA and aminoacyl-tRNA synthetases. During translation mRNA nucleotide bases are read as three base codons, each of which codes for a particular amino acid.

The genetic code is described as degenerate because a single amino acid may be coded for by more than one codon. There are also specific codons that signal the start and the end of translation.

Each tRNA molecule possesses an anticodon on the opposite end that is complementary to the mRNA codon. tRNA molecules are therefore responsible for bringing amino acids to the ribosome in the correct order ready for polypeptide assembly.

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Aminoacyl-tRNA synthetases are enzymes that link amino acids to their corresponding tRNA molecules. The resulting complex is charged and is referred to as an aminoacyl-tRNA.

Initiation

At the 5' cap of mRNA the small 40s subunit of the ribosome (with methionyl-tRNA) binds. For translation to start the start codon 5'AUG must be recognised. This is a codon specific to the amino acid methionine (anticodon on tRNA=5'CAU). The large 60s subunit of the ribosome then binds for elongation to occur.

Figure 1. Intitiation of translation

The ribosome has two tRNA binding sites; the P site which holds the peptide chain and the A site which accepts the tRNA.

Figure 2: Elongation

While Met-tRNA occupies the P site, another aminoacyl-tRNA with an anticodon complementary to the next codon comes to occupy the A site. This process requires GTP. The enzyme peptidyl-transferase forms a peptide bond between methionine and the next aminoacyl-tRNA.

The tRNA molecule in the P site becomes uncharged and leaves the ribosome. The ribosome then translocates along the mRNA molecule to the next codon. This opens up the A site for the next aminoacyl-tRNA. The polypeptide chain is built up in the direction from the N terminal to the C terminal.

Termination

One of the three stop codons enters the A site. No tRNA molecules bind to these codons so the peptide and tRNA in the P site become hydrolysed releasing the polypeptide into the cytoplasm.

Figure 3: Termination of translation process

The small and large subunits of the ribosome dissociate ready for the next round of translation.

The polypeptide produced undergoes post-translational modification before becoming the fully active protein.

Lesson 6 : 6_Introduction to Genomics

Genetics and Genomics

What is Genomics

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Genomics is the study of whole genomes of organisms, and incorporates elements from genetics. Genomics uses a combination of recombinant DNA, DNA sequencing methods, and bioinformatics to sequence, assemble, and analyse the structure and function of genomes.

Genome

- Total amount of DNA of a single cell of an organism (haploid cell in the case of a diploid).
- The whole hereditary information of an organism encoded by DNA.

Genomics and Biology of Organisms

- Determination of entire genome sequence is a prerequisite to understand the complete biology of an organism.

Genomics includes

- To generate physical, genetic, and sequence maps of different genomes.
- To sequence the genomes of model organisms.
- To develop new technologies for mapping/sequencing.

Genomics helps in different ways

- Functions of genes
- Organizations of genomes.
- Structural make-up of genomes.
- Functions of coding and non-coding DNA

Study of Genomics helps to understand

- The ethical, social, and legal issues and challenges posted by genomic information.

Genomics Sub-disciplines

- Comparative genomics
- Structural genomics
- Functional genomics
- Population genomics

Genomics Sub-disciplines

- Metagenomics
- Microbial genomics

Genomics - Conclusion

- Total amount of DNA of a single cell of an organism – Genome and study is called as Genomics.

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Lesson 7 : Anatomy and organization of genome

Anatomy and Organization of Genomes

Genome Anatomy

- Anatomy of different genomes differ from each other.
- Eukaryotes and prokaryotes genomes differ very significantly.

Genome Anatomy

- Size of genomes - 1000 fold difference between eukaryotes and prokaryotes.
- ~ 30 fold between genomes of different eukaryotes.

Genome Anatomy

- In humans ~ 23,000
- Bacterial genomes ~ 1,500 – 2,000 genes.

Genome Anatomy - Eukaryotes

- Eukaryotic genomes are full of simple repeats, numerous types of transposable elements and other sequences.

Genome Anatomy - Prokaryotes

- Prokaryotes have a few repeats and transposable elements and their genomes consist of mainly the genes.
- **Genome Anatomy: Segment of human chromosome 7**

1 Gene: TRY4

2 Gene Segments: V28 & V29-1

1 Pseudogene: TRY5

52 genome wide repeat sequences:

LINE, SINE, LTR, & DNA transposon.

Two Microsatellites

Conclusion

- Anatomy of different genomes differ from each other.

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- Eukaryotes and prokaryotes genomes differ very significantly.

Lesson 8 : 8_Gene anatomy

Gene Anatomy

What is Gene

-
  - A piece of DNA (or RNA) that contains the primary sequence to produce a functional biological gene product (RNA or protein).
-
  - Entire nucleic acid sequence necessary for the synthesis of a functional polypeptide (protein chain) or functional RNA

Genetic information is stored in DNA

-
  - Segments of DNA that encode proteins or other functional products are called genes.
  - Gene sequences are transcribed into messenger RNA (mRNA).

mRNA is translated into Proteins

-
  - mRNA intermediates are translated into proteins that perform most of the life functions.

Gene Anatomy

-
  - Three components
  - Open Reading Frame: From start codon (ATG) to stop (TGA, TAA, TAG). The start codon marks the site at which translation into protein sequence begins (i.e ATG), and the stop codon marks the site at which translation ends (i.e TGA, TAA, TAG)
  - Upstream region with binding site. (e.g. TATA box, GC box, CAAT box).
  - Poly-a tail

What is Gene

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  - A piece of DNA (or RNA) that contains the primary sequence to produce a functional biological gene product (RNA or protein).

Lesson 9 : 9_Prokaryotic gene and eukarytic gene

Prokaryotic Gene and Eukaryotic Gene

Bacterial Gene

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-
  - Most do not have introns
  - Many are organized in operons: contiguous genes, transcribed as a single polycistronic mRNA, that encode proteins with related functions

Bacterial Gene

-
  - Polycistronic mRNA encodes several proteins

Bacterial Gene: Polycistronic mRNA encodes several proteins

Eukaryotic Gene: Exons and Introns

- Introns: intervening sequences within a gene that are not translated into a protein sequence.
- Exons: sequences within a gene that encode protein sequence.

Eukaryotic Gene

- Splicing: Removal of introns from the mRNA molecule

Eukaryotic Gene

- Splicing: Removal of introns from the mRNA molecule

Eukaryotic Gene

- Organize expression of genes' (function calls)
- Promoter region (binding site), usually near coding region
- Binding can block (inhibit) expression

Alternative Splicing in Eukaryotic Genes

- Most have introns
- Produce monocistronic mRNA
- Large in size.

Eukaryotic Gene: Alternative Splicing

Eukaryotic Gene

Computational challenges

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Identify binding sites

Correlate sequence to expression

Conclusion

- Most of prokaryotic genes are without introns and in the are polycistronic.
- Eukaryotic genes have introns and alternative splicing.

Lesson 10 : 10_Prokaryotic and eukaryotic gene expression_Difference

Prokaryotic and Eukaryotic Gene Expression - Difference

Prokaryotic and Eukaryotic Gene Expression

- Bacterial genetics are different.
- Prokaryote genes are grouped in operons
- Prokaryotes have one type of RNA polymerase for all types of RNA

Difference between Prokaryotic and Eukaryotic gene Expression

- mRNA is not modified
- The existence of introns in prokaryotes is extremely rare.
- Transcription in bacteria, sigma factors bind to RNA polymerases.
- RNA polymerases/ sigma factors complex bind to promoter about 40 bases prior to coding region of gene.
- In prokaryotes, the newly synthesized mRNA is polycistronic (code for more than one polypeptide chain).

Difference between Prokaryotic and Eukaryotic gene Expression

- In prokaryotes, transcription of a gene and translation of the resulting mRNA occur simultaneously.
- Many polysomes are found associated with an active gene.

Prokaryotic/Eukaryotic Gene Expression

- Prokaryotes-operons simultaneously transcription and translation
- Eukaryotes – no operons, RNA processing, chromatin remodeling
- **Difference - Prokaryotic and Eukaryotic Gene Expression**

Prokaryotes	**Eukaryotes**
• **Operons**	• **No operons, but they still need to**

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○ **27% of E. coli genes (Housekeeping genes not in operons)** ○ **Simultaneous transcription and translation**	**coordinate regulation** ○ **More kinds of control elements** ○ **RNA processing** ○ **Chromatin remodeling** ○ **Histones must be modified to loosen DNA**

Lesson 11 : 11_Gene families and Gene clusters

Genes Families and Genes Clusters

Genes Clusters

- Many genes are arranged in groups along a chromosome - Genes Clusters
- Eukaryotic ribosomal RNA genes: tandem repeats

Genes Families

- Related genes may be organized as several clusters at different locations
- These are known as gene families e.g. globin genes family

Single Copy Genes and Genes Families

- Many eukaryotic genes are present in one copy per haploid set of chromosomes
- The rest of the genes occur in multigene families, collections of identical or very similar genes

Gene Family and Family members -

- Peroxiredoxin family
- *PRDX* is the root symbol
- Family members are *PRDX1*, *PRDX2*, *PRDX3*, *PRDX4*, *PRDX5* and *PRDX6*

Genes Families

- Some multigene families consist of identical DNA sequences, usually clustered tandemly, such as those genes that code for rRNA products

Ribosomal RNA Gene Family

Human Alpha & Beta Globin Gene Families

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Genes Families on Different Chromosomes

Genes Families

- The misalignment of genes during recombination is the most likely cause of gene duplication and clustering.
- Once repeats exist the probability of misalignment increases.

Genes Families: The misalignment - Globins

Super Families

- Super families are much larger than single multigene families.
- Super families contain hundreds of genes, including multiple multigene families and single individual genes

Super Families

- The large number of members allows super families to be widely dispersed with some genes clustered and some spread far apart.

Genes Families and Genes Clusters - Conclusion

- Single individuals Genes
- Gene families
- Gene families in form of clusters
- Gene families as Super families

Lesson 12 : 12_Types of Proteins and Families

Type of Proteins/ Families

Protein Structure

- Amino acids
  - 20 amino acids
  - Hydrophobic / hydrophylic
  - Charged / neutral
- Functions
  - Enzymes
  - Structure protein
  - Channel
  - Other functions

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Amino Acids: Building Blocks of Proteins

Protein Structure: Amino Acid

Protein Structure: Types of Amino Acids

Protein Domains: Domains are units of compact structure, function and evolution and folding

Proteins

- Different method of categorizing proteins
- Three major categories
- Fibrous proteins
- Globular proteins
- Complexes with multiple components including proteins

Fibrous Proteins – Cytoskeletal Proteins

- Actin
- Coronin
- Dystrophin
- Keratin
- Tubulin

Fibrous Proteins – Extracellular Matrix Proteins

- Collagen
- Elastin
- Fibronectin

Globular Proteins - Major Types

- Plasma proteins
- Hemoproteins
- Cell adhesion
- Transmembrane transport proteins
- Hormones and growth factors
- Receptors

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- DNA-binding protein
- Immune system proteins
- Nutrient storage/transport
- Chaperone proteins
- Enzymes

Complexes with multiple components including proteins

- Nucleosome
- Ribonucleoprotein (generic)
- Signal recognition particle
- Spliceosome

Complexes with multiple components including proteins

- Nucleosome
- Ribonucleoprotein (generic)
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Types of Proteins - Conclusion

- Three major categories
- Fibrous proteins
- Globular proteins
- Complexes with multiple components including proteins

Lesson 13 : 13.Genome Informatics

Genome Informatics

Introduction

- Genome sequencing provides the sequences of all the genes of an organism
- A major application of Bioinformatics is analysis of full genomes that have been sequenced

- Challenge is to identify those genes that are predicted to have a particular biological function

Genomics

- Study of all of a person's genes (the **Genome**), including interactions of those genes with each other and with the person's environment **(NHGRI)**

Genome Informatics

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- Genome informatics is the field in which computational and statistical techniques are applied to derive biological information from genome sequences
- Genome informatics includes methods to analyze DNA sequence information and to predict protein sequence and structure

(Iossifov, et al. 2014)

Genome Sequences

Availability of genome sequences facilitates;

- The discovery and utilization of sequence polymorphisms
- Opportunity to explore genetic variability both between and within the organisms

Genome Analysis

Following tasks;

- Sequencing
- Assembly
- Repeat identification and masking out
- Gene prediction
- Looking for EST and cDNA sequences
- Genome annotation

- Expression analysis
- Metabolic pathways and regulation studies
- Functional genomics
- Gene location/gene map identification
- Comparative genomics
- Identify clusters of functionally related genes
- Evolutionary modeling
- Self-comparison of proteome

Model organisms

- E. coli – bacteria
- S. cerevisiae – yeast
- C. elegans – worm
- D.melanogaster – fly
- Danio rerio – zebrafish
- Mus musculus - mouse
- Homo sapiens – you and me
- Arabidopsis - plant

Lesson 14 : 14.Prokaryotic Genome

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Prokaryotic Genome

- Prokaryotes are the organisms whose Genetic material (DNA) is not enclosed in a nuclear membrane
- No membrane bound organelles

Mitochondria evolved from a bacterial endosymbiont

- First prokaryotic Genome sequenced was that of *Hemophilus influenzae*
- Paved the way for sequencing of many other organisms

Selection criteria

Following are the criteria for organisms selected for sequencing;

- They had been subjected to a detailed biological analysis and thus were model organisms
- important human pathogens
- They were of phylogenetic interest
- Sequences were annotated as they were sequenced

Conclusions

- Prokaryotes are simple Genomes
- Easy models to study Biochemistry and Molecular biology of life processes
- Sequencing is done on economically important organisms

Lesson 15 : 15.Eukaryotic Genome

Eukaryotic Genome

Introduction

- Eukaryotes have larger genomes.
- They have tandem repeats. (A tandem repeat is a sequence of two or more DNA bases that is repeated numerous times in a head-to-tail manner on a chromosome. Tandem repeats are generally present in non-coding DNA. In some cases, tandem repeats can serve as genetic markers to track inheritance in families. They can also be useful for DNA fingerprinting in forensic studies.)

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- They have introns in their protein-coding genes. (An intron is a region that resides within a gene but does not remain in the final mature mRNA molecule following transcription of that gene and does not code for amino acids that make up the protein encoded by that gene. Whereas an exon is a region of the genome that ends up within an mRNA molecule. Some exons are coding, in that they contain information for making a protein, whereas others are non-coding. Genes in the genome consist of exons and introns. Most protein-coding genes in the human genome consist of exons and introns.)

- Heterochromatin and euchromatin region (Heterochromatin is defined as the area of the chromosome which is darkly stained with a DNA specific stain and is in comparatively condensed form. Euchromatin is defined as the area of the chromosome which is rich in gene concentration and actively participates in the transcription process. Euchromatin has a less compact structure, and is often described as a 11 nm fiber that has the appearance of 'beads on a string' where the beads represent nucleosomes and the string represents DNA. In contrast, heterochromatin is more compact, and is often reported as being composed of a nucleosome array condensed into a 30 nm fiber.)

Staining with dyes

- Dense heterochromatin
- Light Euchromatin
- Heterochromatic regions are poorly transcribed (expressed)
- Euchromatic regions are highly expressed

Conclusions

- Eukaryotes are distinguished by the presence of prominent nuclei
- have larger genomes, tandem repeats and introns in their protein-coding genes

Lesson 16 : Epichromosomal elements (EEs)

Epichromosomal Elements (EEs):

Introduction

Genome is the total collection of genetic material and is made up of

-
  - Chromosomes
  - Epichromosomal Elements

Prokaryotic EEs

- Plasmids
- Self replicating
- Additional rings
- Bacteriophages

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- Host colonization
- Transposons

Eukaryotic EEs

- Organellar DNA
- Mitochondria (in both plants and animals)
- Chloroplasts (in plants)
- Plasmids
- yeast
- Transposons
- Viral genomes
- Retroviruses membrane-bound organelles
- hundreds to thousands of them
- Site of respiration (mitochondria) or photosynthesis (chloroplast)
- mtDNA (mitochondrial DNA)
- cpDNA (chloroplast DNA)

Endosymbiont hypothesis

- originally proposed in 1883 by Andreas Schimper, but extended by Lynn Margulis in the 1980s
- Mitochondria and Chloroplast are derived from endosymbiotic bacteria

Organelle Genome

- Circular
- Double stranded
- Supercoiled
- No histones
- Multiple copies
- of mtDNA/cell or cpDNA/cell

Genome	**Size & Organization**
Plant plastid	150 kb circle
Plant mitochondria	150 – 2000 kb multipartite

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Human mitochondria	17 kb circle
saccharomyces mitochondria	75 kb circle

Organelle Genome

- Encode only a subset of genes required to elaborate a functional organelle
- rRNAs, tRNAs, ribosomal proteins, membrane-associated respiratory or photosynthetic components
- Other components encoded by nuclear genome, translated in the cytosol and imported into the organelle
- 10% of nuclear genes devoted to mitochondrial function; 15% to plastid function

Conclusions

- Organelle genome is similar to prokaryotes
- High copy number
- mtDNA and cpDNA depends on Nuclear DNA

Lesson 17 : Genome repeats

Genome Repeats:

Sequence Repeats

- Repeats skew the base composition
- Contribute to differences in buoyant densities
- Repeat containing DNA can be separated as **satellite DNA** on the bases of these densities

Satellite DNA

- One to several thousand base pairs long
- Tandem array of 100 million bases long
- Near centromere and telomere
- Minisatellite and Nicrosatellite
- **Minisatellite**
- 15 bases long in an array of several hundred to thousands of kb
- Typically in euchromatin
- VNTR e.g, used to identify human individuals in forensics

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- **Microsatellite**
- 2-6 bases long in arrays of 10-100 bases
- Inherited to offspring
- Useful markers for genetic analysis and evolutionary studies
- Found in telomeres TTAGGG
- SSRs and STRs
- **Transposable Elements (TEs)**
- Large portion of eukaryotic genome
- Thought to play important role in the evolution of these genomes
- Move (Jump) from one location to another faster than chromosome replicate
- **Transposable Elements (TEs)**
- Have a potential to increase in number
- Make up a large proportion of eukaryotic genome
- Detectable until blend into genome due to mutations

Conclusions

- Large proportion of eukaryotic genome is composed of repeats
- Different repeats act as markers to detect genetic variation and are also used to study evolution

Lesson 18 : Transposable Elements (TEs)

Transposable Elements (TEs)

Introduction

Any segment of DNA, able to transpose

- **Insertion Sequences (IS elements)**
  - The simplest transposable elements
  - Code only for the ability to transpose
  - prokaryotes

IS elements

- usually very small (< 1 Kb to 2 Kb)
- Flanked by inverted repeat sequences (IRs)
- Encode at least one gene that provides their own transposition functions

IS elements

- Do not code for noticeable (phenotypic) traits
- Can cause mutations by transposition into genes

Transposase Gene

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Transposons

-
  - More complex transposable elements
  - Code for the ability to transpose
  - Code for other detectable traits as well as ability to transpose.
  - A transposable element (TE, transposon, or jumping gene) is a DNA sequence that can change its position within a genome, sometimes creating or reversing mutations and altering the cell's genetic identity and genome size. Transposition often results in duplication of the same genetic material.

Eukaryotic TEs

- 2 Classes; Class I and Class II
- Class I
  - Encode reverse transcriptase
  - Use RNA-mediated mechanisms of transcription

Eukaryotic TEs

- **Class I**

Three classes

-
  - LTR (Long Terminal Repeats) retrotransposons
  - Retrotransposons
  - Retrovirus like Elements

Eukaryotic TEs

- **SINES** (Short Interspersed Nuclear Elements
  - 80-300 bp
  - ***Alu* repeats**
- **LINES** (Long Interspersed Nuclear Elements)
  - 6-8 KB long

Eukaryotic TEs

- **Class II TEs**
  - Ac-Ds in maize
  - P elements in Drosophila
- **MITES** (Miniature Inverted repeat TEs)

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- Features of both Class I and II
- 400 bp

Conclusions

- Transposable elements make up a significant part of eukaryotic genome
- Move within and across genomes
- Cause genome expansion

Lesson 19 : Eukaryotic Gene Structure

Eukaryotic Gene Structure

Eukaryotic genes

- Eukaryotic genes are complicated
- They possess exons which are protein coding regions and are interrupted with introns

- Whole gene is transcribed as a large transcript
- Introns are spliced leaving an ORF that is translated

Human Beta globin Gene

- Introns start on the 5' ends as GT and end as AG towards 3' end **GT----------AG**
- This trend is highly conserved

Introns proportion

- A small proportion in yeast, only 239 introns in genome
- Human genes have hundreds, often 95% of genes
- Might have embedded genes

Intron origin

- **Intron-early**
  - used to assemble first genes from ancient conserved introns
- **Intron-late**
  - Broke up previously continuous genes by inserting into them

Number of Genes

- Degree of compactness contributes to variation in gene density
- Compact genomes have higher genome density

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Organism	Genome size (haploid MB)	Predicted genes
A. thaliana (plant)	130	~25,000
C. elagans (worm)	100	18,424
Drosphila melangaster	180	13,601
Escherichia coli	4.7	4,288
Homo sapiens (human)	3000	45,000 – 120,000
S. cerevisiae (yeast)	13.5	6,241

Pseudogenes

- Non functional genes
- Derived from functional genes through mutations following genome duplication
- Processed pseudogenes lack introns and promoters

Gene families

- Set of genes having similar sequences and functions
- Gene families arise from gene duplication and subsequent divergence

Conclusions

- Eukaryotic genes have exons and introns
- Introns make up a significant portion of Human genome
- Pseudogenes are non functional genes
- Similar genes make gene families

Lesson 20 : Comparative Genomics

Comparative Genomics

Introduction

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- Comparison of gene number, gene content and gene location in both prokaryotic and eukaryotic groups of organisms
- Availability of genome makes possible a comparison of all the proteins (proteome) encoded by one organism with those of another

Orthologs

- Genes in two organisms that are so similar that they must have the same function and evolutionary history are called orthologs (Fitch, 1970)

Paralogs

- Gene families originating from rare gene duplication events over the evolutionary time are called paralogs
- Unlike pseudogenes, 2nd copy of gene remains functional

***Drosophila* and yeast**

- *Drosophila* has core proteome only twice the size of that of yeast
- *Drosophila* proteome is more similar to mammalian proteomes than worm or yeast

Drosophila* and *C. elegans

- Despite the large differences between fly and worm, they use a core proteome of similar size
- Nearly 30% of fly genes have putative orthologs in the worm

***Drosophila* and Human**

- Some human disease genes absent in *Drosophila*
- A number of previously unknown counterparts to human cancer and neurological disorders genes

Conclusions

- Comparative genomics reveals the relationship among different organisms
- Fruit fly has more similarities with mammals

Lesson 21 : Functional Genomics

Functional Genomics

- Functions of genes, their regulation and end products.
- Functional genomics analyzes all genes in genomes to determine their functions and their gene control and expression.
- Classically, genetics analysis begins with a phenotype and moves for identification of genes.

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- New approaches are needed to work in the opposite direction, from genes to phenotype.

Functional Genomics relies on Molecular Biology, Biochemistry, Genetics and Bioinformatics tools

- Functional genomics relies on molecular biology lab research and sophisticated computer analysis by bioinformatics tools.

Functional Genomics

- Fusion of biology with maths and computer science is used for many things. Examples:
- Finding genes within a genomic sequence.
- Aligning DNA/proteins sequences.

Functional Genomics – what is included

- Subtracted cDNA libraries
- Differential display
- Representational difference analysis
- Suppression subtractive hybridization

Functional Genomics – what is included

- cDNA Microarrays
- Serial analysis of gene expression
- 2-D Gel electrophoresis

Functional Genomics - Conclusion

-
  - Function of gene products.
  - Describing interactions between genes and gene products in the cell, between cells and between organisms.
  - Considering phylogenetic relationships.

Lesson 22 : Structural genomics

Structural Genomics

-
  - The ultimate goal of genomic studies is to determine the nucleotides sequences of entire genomes of organisms.
  - The genetic and physical mapping and sequencing of chromosomes.

Genetic Mapping

-

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- Genetic mapping - approximate locations of genes, relative to the locations of other genes, based on the rates of recombinations.

Physical Mapping

-
- Physical mapping is based on the direct analysis of DNA.
- Physical mapping places genes on the genomes in relation to distances measured in bp, kbp, and mbp.

Structural Genomics incudes

-
- Distinct components of genomes
- Abundance and complexity of mRNA
- Genome sequences
- Gene numbers
- Coding and non-coding DNA

Structural Genomics – Complex Genomes

-
- Complex genomes have roughly 10x to 30x more DNA than is required to encode all the RNAs or proteins in the organism
- Introns in genes
- Regulatory elements of genes
- Multiple copies of genes, including pseudogenes
- Intergenic sequences
- Interspersed repeats

Structural Genomics – Transposable Elements

Vast majority of TEs can be classified into four families:

LINEs (Long Interspersed Nuclear Elements, autonomous)

SINEs (Short Interspersed Nuclear Elements, use LINE proteins for life cycle)

LTR elements (Long Terminal Repeats; derived from retroviruses)

DNA transposons (replicate without RNA intermediary)

Structural Genomics - Conclusion

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- To determine the nucleotides sequences of genomes.
- To study genetic maps
- To study physical maps.

Lesson 23 : 23_Concepts of comparative genomics

Comparative Genomics

We can do comparison of gene numbers, gene locations and biological functions of genes, in the genomes of different organisms. We can also identify and compare groups of genes that play a unique biological role in different organisms.

Homology

Homology is the relationship of any two characters (genes or proteins) that have descended, usually through divergence, from a common ancestor/ ancestral character.

Homologues

Homologues are thus components or characters (such as genes/proteins with similar sequences) that can be attributed to a common ancestor of the two organisms during evolution.

Homologoues can be

-
  - Orthologues
  - Paralogues
  - Xenologues
  - Analogues
- **Orthologues**
  - Orthologues are homologues that have evolved from a common ancestral gene by speciation.
  - They usually have similar functions.
- **Paralogues**
  - Paralogues are homologues that are related or produced by duplication within a genome followed by subsequent divergence.
  - They often have different functions.
- **Xenologues**
  - Xenologues are homologous that are related by an interspecies (horizontal transfer) of the genetic material for one of the homologues.
  - The functions of the xenologues are quite often similar.
- **Analogues**
  - Analogues are non-homologues genes/proteins that have descended convergently from an unrelated ancestor.
  - Similar function, different sequence or structure.

Comparative Genomics – Evolutionary Trends

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Comparison of genomes belonging to relatively similar group, like mammals, reveals some evolutionary trends.

Comparative Genomics – Functional elements

Provides a powerful and general approach to identify functional elements without previous knowledge of functions.

Comparative Genomics – Ancestral Genome

Reconstruction of an ancestral genome for a group of organism.

Conclusion

Comparative genomics is the comparison of gene numbers, gene locations and biological functions of genes, in the genomes of different organisms.

Lesson 24 : Population genomics

Population Genomics

- Study of genomes of a specific population, strains, varieties or organisms.
- Study about the genetic diversity.
- Understanding new insights into disease and drug response.
- Deeper insights about genomic size.
- Further complexities in the genomes of the organisms.
- Variations between individuals or strains.
- Human genome is 200 times larger than yeast but 200 times smaller than Amoeba.
- Less than 2% of human genome is coding sequence

1000 Genomes - Population Genomics

- International research consortium sequenced the genomes of at least 1000 people from around the world.
- Detailed and medically useful pictures of human genome variations.
- Any two humans are more than 99% identical at genetic level.
- Genetic variations may explain individual differences in susceptibility to diseases, responses to drugs.

HapMap Project - Population Genomics

- The HapMap project has already discovered many regions of the genome containing genetic variations associated with common human diseases.
- **Population Genomics - HapMap Project and 1000 Genomes Project**

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Goals of Population Genomics

- Produce a catalog of variants present at 1% or greater frequency in the human population.
- Down to 0.5 percent or lower within genes.
- Increase sensitivity of disease discovery.
- Provide better understanding of very rare genetic diseases.
- Understand contribution of common variants to common diseases like diabetes and heart diseases.
- Identify SNP but also large differences like rearrangements, deletions or duplications.

Lesson 25 : Metagenomics

Metagenomics

- Metagenome-environmental genome.
- Collection of genes sequenced from the environment could be analyzed in a way analogous to the study of a single genome.

Why do Metagenomics

Objectives of Metagenomics

- Examining phylogenetic diversity using 16s rRNA
- Diversity patterns of microorganisms for monitoring and predicting environmental conditions/change.
- Examining genes/operons for desirable enzymes (cellulases, lipases, antibiotics, other natural products).
- Exploited for industrial or medical applications.
- Examining secretory, regulatory, and signal transduction mechanisms associated with samples or genes of interest.
- Examining bacteriophage or plasmid sequences. These potentially influence diversity and structure of microbial communities.
- Examining potential lateral gene transfer events. Knowledge of genome plasticity may give us an idea of selective pressures for gene capture and evolution within a habitat.
- Examining metabolic pathways.
- Directed approach towards designing culture media.
- Examining genes that predominate in a given environment compared to others.

Metagenomics Data

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- Metagenomic data can be used towards designing low and high throughput experiments focused on defining the roles of genes and microorganisms in the establishment of dynamic microbial community.

Conclusion

- Metagenome-environmental genome

Lesson 26 : 26_Why Sequence Genomes

Why Sequence Genomes?

-
  - To identify gene numbers, their locations on genomes, and to study their functions.
  - Genes regulation
  - DNA sequence
  - Genome organization
  - Chromosomal structure and organization
  - Noncoding DNA types, amount, distribution and functions.
  - Coordination of gene expression, protein synthesis, and post-translational events.
  - Interaction of proteins in complex molecular machines
  - Predicted vs experimentally determined gene function
  - Evolutionary conservation
  - Proteins structure and function.
  - Proteomes (total protein content and function) in organisms.
  - Correlation of SNPs with health and disease
  - Disease-susceptibility prediction based on gene sequence variation
  - Genes involved in complex traits and multigene diseases

Novel Diagnostics

-
  - Complex systems biology, developmental genetics.
  - To provide platform for microchips and DNA microarrays.
  - Gene expression – RNA
  - Complex systems biology, developmental genetics and genomics
  - Complex systems biology, developmental genetics and genomics
  - Drug target discovery
  - Rational drug design
  - Molecular docking
  - Gene therapy
  - Stem cell therapy

Understanding Metabolism

-
  - To understand the metabolism of cells and tissues within different organisms.

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Understanding mechanism of diseases

-
  - Inherited diseases
  - Infectious diseases
  - Pathogenic bacteria
  - Viruses

Conclusions

-
  - Better understanding of the genomes would be possible by sequencing of the genomes.

Lesson 27 : 27_Genome characteriazation - Techniques used

Major Techniques used for Genomes Characterization

-
  - Cloning
  - Hybridization
  - PCR amplification
  - Sequencing
  - Computational tool

Genomes Characterization Techniques – Cloning:

-
  - Genomes digested with restriction enzymes and inserted in vectors to produce genomic libraries.
  - BACs
  - YACs

Genomes Characterization Techniques - Hybridization

-
  - To arrange large contigs of genomes to produce genetic maps and physical maps of genomes.

Genomes Characterization Techniques – Hybridization:

- To arrange large contigs of genomes to produce genetic and physical maps
-

Genomes Characterization Techniques – PCR

-

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- Technique to amplify the DNA. Different variants of the technique used

Genomes Characterization Techniques – DNA Sequencing

-
- One of the important technique used to characterize the genomes
- To study structure and function of genomes.

- **Genomes Characterization Techniques – DNA Sequencing: DNAs are amplified and sequenced**

Genomes Characterization Techniques – Computational Tools

-
- Used to align the sequenced DNA to produce physical maps of the genomes.

Genomes Characterization Techniques – Conclusion

-
- Different techniques used for genomes characterizations.

Lesson 28 : 28_Genome Analysis Steps

Steps of Genomes Analysis

-
- Genome sequence assembled
- Identify repetitive sequences – mask out
- Gene prediction – train a model for each genome.
- Look for EST and cDNA sequences
- Genome annotation
- Microarray analysis
- Metabolic pathways and regulation
- Protein 2D gel electrophoresis.
- Functional genomics
- Gene location/gene map
- Self-comparison of proteome
- Comparative genomics
- Identify clusters of functionally related genes
- Evolutionary modeling

Lesson 29 : 29_Benefits of genome analysis

Benefits of Genomes Research

Genomes Research- Molecular medicine

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- Improve diagnosis of disease
- Detect genetic predispositions to disease (cancer, diabetes etc)
- Create drugs based on molecular information
- Use gene therapy and control systems as drugs

Genomes Research- Risk Assessment

- Evaluate the health risks faced by individuals who may be exposed to radiations and to cancer causing chemicals and toxins

Genomes Research- Bioarchaeology, Anthropology, Evolution and Human Migration

- Study evolution through genetic variants in lineages
- Study of migration of different populations
- Study mutations on the Y chromosome to trace lineage and migration of males
- Evolution of mutations with ages of populations

Genomes Research- DNA Forensics

- Identify potential suspects whose DNA may match evidence left at crime scenes
- Exonerate persons wrongly accused of crimes
- Identify catastrophe victims
- Establish paternity and other family relationships
- Identify endangered and protected species as an aid to wildlife officials

Genomes Researchs

- Detect bacteria and other organisms that may pollute air, water, soil and food
- Match organ donors with recipients in transplant programs
- Determine pedigree for seed or livestocks

Genomes Research- Disease resistant crops and disease resistant animals

- Grow disease/insect resistant and drought resistant crops
- Breed healthier, more productive, disease resistant farm animals

Genomes Research- Agriculture, Livestock, Breeding and Bioprocessing

- Develop biopesticides
- Incorporate edible vaccines incorporated into food products

Genomes Research- Microbial Genomics

- Rapidly detect and treat pathogens (disease-causing microbes)
- Develop new energy sources (biofuels)
- Monitor the environment to detect pollutants
- Protect populations from biological and chemical warfare

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- Clean up toxic waste safely and efficiently

Lesson 30 : 30_Size of Genomes – Genes

Genes and Size of Genomes

Size of Genomes

-
  - Genomes of most bacteria and archaea range from 1 to 6 million base pairs (Mb).
  - Genomes of eukaryotes are usually larger.
  - Most plants and animals have genomes greater than 100 Mb.
  - Humans have genome size of 3,000 Mb
  - Within each domain there is no systematic relationship between genome size and phenotype

Conclusion

-
  - Although most eukaryotes have large size of genomes.
  - Within each domain there is no systematic relationship between genome size and phenotype

Lesson 31 : 31_Viral Genomes

Viral Genomes

Genomes of Viruses

-
  - Viral genomes can be
  - ssRNA
  - dsRNA
  - ssDNA
  - dsDNA
  - Linear
  - Ciruclar

Viruses Genomes

-
  - A viral genome is the genetic material of the virus.

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- Also termed the viral chromosome.
- Viral genomes vary in size -few thousand to more than a hundred thousand nucleotides.

Viruses with RNA Genomes

-
- Almost all plants viruses and some bacterial and animal viruses
- Genomes are rather small (a few thousands nucleotides)

Viruses with DNA Genomes

-
- Often a circular genome
- lambda = 48,502 bp

Replicative form of Viral Genomes

-
- All ssRNA viruses produce dsRNA molecules
- Many linear DNA molecules become circular

Viruses and Kingdoms

- Many plants viruses contain ssRNA genomes.
- Many fungal viruses contain dsRNA genomes.
- Many bacterial viruses contain dsDNA genomes.

Genomes in Virions: The genomes of viruses can be composed of either DNA or RNA, and some use both as their genomic material at different stages in their life cycle. However, only one type of nucleic acid is found in the virion of any particular type of virus.

Viruses and Number of Genes

Genomes in Virions

Genome of Tobacco Mosaic Virus

- Single, 6400 nucleotides RNA, 3 Essential Genes

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Genome of Poxvirus –A typical large dsDNA Virus

- 180 kb DNA, >100 Essential Genes

Genome of Polio Virus: Single-stranded positive-sense RNA genome that is about 7500 nucleotides long

Genome of Pox Virus

- Linear dsDNA 130-375 kbp; covalently closed termini.
- Large hairpin structure at each terminus - up to 10 kb total at each end is repeat sequence.
- Encode 150-300 proteins.
- Coding regions are closely spaced, no introns.
- Coding regions are on both strands of genome, and are not tightly clustered with respect to time of expression or function.

Viral Genomes

Viral genomes can be

ssRNA, dsRNA, ssDNA, dsDNA, Linear, Ciruclar

Lesson 32 : 32_Bacterial Genomes

Bacterial Genomes

- Small organisms carry high coding density (85-90%)
- 1 gene per 1000 bases in prokaryotes
- Large variation in genome size between bacteria

Genomes of Bacteria – Large Variation

- *Tremblaya princeps* 140kb, 121 coding sequences

- *Sorangium cellulosum*
- 14000kb
- 11599 coding sequences

Bacterial Genomes - Conclusion

- Small organisms carry high coding density.

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- Large variation in genome size between bacteria.

Lesson 33 : 33_Yeast Genome

Yeast Genome

- The nuclear genome consists of 16 chromosomes.
- In addition, there is a mitochondrial genome and a plasmid, 2 micron circle.
- The haploid yeast genome consists of ~ 12.1 Mb
- Yeast genome was completely sequenced by 1996.
- Small and compact
- Small intergenic sequences
- Few transposable elements
- Few introns
- Limited RNA interference
- The yeast genome is predicted to contain about 6,200 genes
- 274 tRNA
- 287 introns
- Small percentage of yeast genes have introns
- The intergenic space between genes is only between 200bp - 1,000bp

Yeast Genome: Genome of Yeast Cell

Yeast Genome

- The largest known regulatory sequences are spread over about 2,800bp
- MUC1/FLO11
- Yeast genes have names consisting of three letters and up to three numbers
- GPD1, HSP12, PDC6
- Usually they are meaningful

Yeast Genome – Genes Nomenclature

- Wild type genes are written with capital letters in italics: *TPS1*, *RHO1*, *CDC28*
- Recessive mutant genes are written with small letters in italics: *tps1, rho1, cdc28*
- Three letters provides information about a function, mutant phenotype, or process related to that gene.
- CDC - Cell Division Cycle ; ADE-ADEnine biosynthesis

- **Yeast – Mitochondrial DNA**

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Yeast – Plasmid DNA

- The 2u circle is a 6.3 kb
- 50 to 100 copies per haploid genome of the yeast cells
- ARS, the FLP gene, the three genes which encode proteins required for regulation of FLP expression (REP2, REP1, and D)
- Set of small direct repeats (called "STB") required for partitioning into daughter cells during mitosis and meiosis.
-

Yeast Genome

- Yeast nuclear genome has 16 chromosomes.
- A mitochondrial genome.
- A plasmid.

Lesson 34 : 34_Mitochondrial genome

Mitochondrial Genome

- Multiple identical circular chromosomes
- ~15-16 Kb in animals
- ~ 200 kb to 2,500 kb in plants
- Over 95% of mitochondrial proteins are encoded in the nuclear genome.
- Often A+T rich genomes.

Human Mitochondrial Genome

- Circular, double stranded, 16.6 kb
- The two strands are notably different in base composition, leading to one strand being heavy (H strand) and the other light (L strand).
- Both strands encode genes, although more are on the H strand.
- A short region (1121 bp), the D loop is a DNA triple helix: two overlapping copies of the H strand.
- The D loop is also the site where most of replication and transcription is controlled.
- Genes are tightly packed, with almost no non-coding DNA outside of D loop.
- Human mitochondrial genes contain no introns, although introns are found in the mitochondria of other groups (plants)
- 37 Genes
- 22 tRNAs
- 2 rRNAs
- 13 polypeptides
- tRNA: only 60 of the 64 codons code for amino acids.
- **Mitochondrial Genome - Humans**

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Mitochondrial Genome – Yeast

Mitochondrial Genome – Human and Yeast

Universal Code and Mitochondrial Code

Mitochondrial Genome – Tumors due to mutations

Mitochondrial DNA polymorphisms track human migrations

All humans descend from a small group of Africans

This group originated in central Africa ~200,000 years ago

The founding group was small (10^2-10^4 people)

Conclusion

- Multiple identical circular chromosomes
- 15-16 Kb in animals
- 200 kb to 2,500 kb in plants

Lesson 35 : 35_Genomes comparisons

Genomes Comparisons

Genomes vary in size

- Genomes of most bacteria and archaea range from 1 to 6 million base pairs (Mb)
- Most plants and animals have genomes greater than 100 Mb; humans have 3,000 Mb

Genomes vary in genes numbers

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- Free-living bacteria and archaea have 1,500 to 7,500 genes
- Fungi have about 5,000 genes and multicellular eukaryotes upto 40,000 genes

Genomes vary in genes numbers

- Number of genes is not correlated to genome size
- Nematode
  C. elegans has 100 Mb and 20,000 genes, while *Drosophila* has 165 Mb and 13,700 genes

Genomes vary in genes numbers

- Vertebrate genomes can produce more than one polypeptide per gene because of alternative splicing of RNA transcripts

Humans and Mammals have low gene density

- Humans and other mammals have the lowest gene density, or number of genes, in a given length of DNA. Multicellular eukaryotes have many introns within genes and noncoding DNA between genes

Multicellular eukaryotes have much noncoding DNA and multigene families

- Most of eukaryotic genomes neither encodes proteins nor functional RNAs
- Evidence indicates that noncoding DNA plays important roles in the cell

Human Genome: Distribution of coding and non-coding DNA

Comparing Genomes

- Significant similarity between genomes of "distant" species (Man – Yeast 23%)
- Similarity increases for taxonomically close species.
- Closely related species help us understand recent evolutionary events
- Distantly related species help us understand ancient evolutionary events

Comparing Genomes

- Human and chimpanzee genomes differ by 1.2%, at single base-pairs, and by 2.7% because of insertions and deletions

Lesson 36 : 36_Comparing distantly and closely related species

Comparing distantly/closely related species

Comparing distantly related species

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- Highly conserved genes have changed very little over time
- These help to clarify relationships among species that diverged from each other long ago
- Bacteria, archaea, and eukaryotes diverged from each other 2 and 4 billion years ago
- Highly conserved genes can be studied in one model organism.

Comparing closely related species

- Genetic differences between closely related species can be correlated with phenotypic differences
- Genetic comparison of several mammals with non-mammals helps to identify what make mammals
- Human and chimpanzee genomes differ by 1.2%, at single base-pairs, and by 2.7% because of insertions and deletions
- Several genes are evolving faster in humans than chimpanzees
- Genes involved in defense against malaria and tuberculosis and in regulation of brain size, genes code for transcription factors
- Humans and chimpanzees differ in the expression of the *FOXP2* gene, whose product turns on genes involved in vocalization
- Differences in the *FOXP2* gene may explain why humans but not chimpanzees communicate by speech

Conclusion

- Highly conserved genes have changed very little over the time
- These help to clarify relationships among species that diverged from each other long ago

Lesson 37 : 37_Genome mapping

Genome Mapping

- Different types
- Genetic mapping
- Physical mapping

Genetic Mapping of Genomes

- Genetic mapping is based on the use of genetic techniques to construct maps showing the positions of genes and other sequence features on a genome.

Genetic Mapping

- Genetic techniques include cross-breeding experiments
- Case of humans, the examination of family histories (pedigrees).

Physical Mapping of Genomes

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- Physical mapping uses molecular biology techniques to examine DNA molecules directly in order to construct maps showing the positions of sequence features, including genes.

Mapping of Genomes

Mapping Strategy	Requires	Resolution	How to increase resolution
Genetic	**Polymorphic Markers, and Pedigrees**	**Medium to High**	**Increase number of marke s or people**
Restriction	**Restriction Enzymes**	**High**	**Increase number of enzym**
Somatic Cell Hybrid	**Somatic Hybrid Panel, and STSs**	**Low to Medium**	**Increase number of de tion hybrids**
Radiation Hybrid	**Radiation Hybrid Panel, and STSs**	**High**	**Use additional hybrids, or ake a new panel**

Genome Mapping

- Different types
- Genetic mapping
- Physical mapping

Lesson 38 : 38_Genetic mapping of Genomes

Genetic Mapping of Genomes

- Linkage analysis is the basis of genetic mapping
- Recombination fraction is a measure of the distance between two loci.
- Two loci that show 1% recombination are defined as being 1 centimorgan (cM) apart on a genetic map.
- 1 map unit = 1 cM centimorgan
- Genetic mapping involves determining, the location of genes on a chromosome relative to other genes, using genetic crosses and pedigree analysis.

Genetic Mapping

- Genetic map of the human genome has 24 different maps; one for each autosome, X and Y.
- Marker alleles in genetic crosses help to determine crossover rate between linked genes

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Genetic Mapping of Genomes - Procedure

- Individuals with different alleles at two or more loci are crossed, and their offspring examined.
- Most of the offspring will have phenotypes corresponding to the linked alleles. A few progeny will be recombinant.
- The frequency of the recombinant phenotype is calculated as a percentage of the total offspring, giving the recombination frequency or genetic distance.

Genetic Mapping in Humans

- Experimental crosses are not done in humans and so genetic mapping relies on pedigree analysis, and is limited by rarity of large, multigenerational pedigrees showing segregation of defined linked traits.
- Usually, the lod (logarithm of odds) score method is used for statistical analysis of pedigree data.

Genetic Mapping of Genomes

- A lod score compares the expected distributions of traits if they are linked or not linked.
- The lod score is the log10 of the ratio of the two probabilities. The higher the lod score, the closer the two genes.
- The map distance for linked markers is computed from the recombination frequency
- Mapped features that are not genes are called as DNA markers.
- A DNA marker must have at least two alleles to be useful.
- Mostly, three types RFLP, SSLP, SNP
- To test linkage between the genes for two traits, certain types of matings are examined whether or not the pattern of the combinations of traits exhibited by the offspring follows the law of independent assortment.
- If not, the gene pairs for those traits must be linked, that is they must be on the same chromosome pair.
- **Recombinations Frequencies (RF)**
- Two genes that undergo independent assortment have RF of 50 % and are located on non-homologous chromosomes
- They are unlinked
- Genes with recombination frequencies less than 50 percent are on the same chromosome
- They are linked
- **Genetic Mapping of Genomes – Recombinations Frequency Calculated as:**

Genetic Mapping of Genomes

- Linkage analysis is the basis of genetic mapping
- Recombination fraction is a measure of the distance between two loci.

Lesson 39 : 39_Physical mapping of genomes

Physical Mapping of Genomes

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- A physical map of a chromosome or genome that shows the physical locations of genes and other DNA sequences

Genetic Mapping provides in-sufficient information about exact locations of genes

- Genetic Information by genetic map rarely sufficient for directing the sequencing phase of a genome project.

Genetic Mapping – limited accuracy

- Two reasons
- The resolution of a genetic map depends on the number of crossovers that have been scored.
- Genetic maps have limited accuracy

Physical Mapping

- In humans, even the detailed genetic map lacks the required resolution.
- Therefore, a physical map derived directly from genomic DNA rather than analysis of recombinants has been generated.

Cytogenetic Map, Genetic Map, Physical Map

Genetic Map and Physical Map

Physical Map

Physical Mapping of Genomes

- As in human, there are 24 physical maps, 22 autosomes plus X and Y.
- Types of physical maps are presented in order of increasing resolution:

Physical Mapping Techniques

- Cyotogenetic map
- A restriction map
- Fluorescent in situ hybridization
- Sequence tagged site (STS) map
- Nucleotide sequence map

Physical Mapping Techniques

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- Plasmid
- Phagemid
- Cosmid
- YACs
- BACs

Physical Map of Genomes

- A physical map of a chromosome or genome that shows the physical location of genes/DNA
- Low resolution -banding patterns; highest-resolution nucleotide sequence

Lesson 40 : 40_Cytogenetic mapping

Cytogenetic Mapping of Genomes

- Microscopic examination of stained chromosome reveals a banding patterns
- Regions - designated based on their chromosomal position relative to the centromere

- Chromosomal banding pattern.
- These methods allow a rough determination of locations, but not to yield a direct measure of distance.

- Regions designated "q" are on the chromosome's long arm. Regions designated "p" are on the short arm
- Regions are numbered from the centromere outward, with q1 and p1

- Cytogenetic
  - Visual study of chromosomes at microscopic level
- Karyotype
  - Chromosome complement

– also applied to picture of chromosomes

- Idiogram
  - Stylised form of karyotype
- Classified according to position of centromere
- Central centromere - metacentric
- Sub-terminal centromere - acrocentric
  - have satellites which contain multiple copies of genes for ribosomal RNA on short arm
- Intermediate centromere – submetacentric
- Terminal centromere - Telocentric

Cytogenetic Mapping of Genomes: Gene is located at Chromosome 1p36.1

- Microscopic examination of stained chromosome reveals a pattern of bands.

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- Regions are designated based on their chromosomal position relative to the centromere

Lesson 41 : 41_FISH mapping

FISH Mapping of Genomes

- Fluorescence in situ hybridization (FISH) is a type of mapping that uses fluorescent probes that bind to only those parts of the chromosome with a high degree of sequence complementarity.

- It is used to detect and localize the presence or absence of specific DNA sequences on chromosomes.
- Fluorescence microscopy used to find out where fluorescent probe is bound to the chromosomes.
- Used in genetic counseling, medicine and species identification.
- Individual metaphase chromosomes are probed in situ with specific fluorescently labeled DNA sequences, identifying homologous sequences in the chromosome
- Different probes labeled with different fluorescent dyes may be used in same experiment.
- Fluorescence microscopy provides data for computer imaging analysis to determine binding site for each probe.
- With a resolution of 2-5 Mb in metaphase chromosomes, FISH can localize markers to sub-regions of chromosomal bands.
- Less condensed chromosomes may be resolved in the 5-700 kb range.

FISH Mapping of Genomes

- Advantages

Highly specific, Microdeletions /Microduplications

- Disadvantages

500-600 probes needed to match the power of karyotyping

Lesson 42 : 42_Restriction mapping of genomes

Restriction Mapping of Genomes

- A restriction map is a map of known restriction sites within a sequence of DNA.
- Restriction mapping requires the use of restriction enzymes.
- Restriction enzymes are used that cut DNA due to the recognition sequence in the DNA/genome under study
- To construct a restriction map is to compare the fragment sizes produced when a DNA molecule is digested with different restriction enzymes that recognize different target sequences.
- **Restriction Mapping of Genomes**

Restriction Mapping of Genomes

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- A map showing positions of restriction sites in a DNA sequence
- If DNA sequence is known then construction of restriction map is a trivial exercise
- In early days of molecular biology DNA sequences were often unknown

Limitations of Restriction Mapping

- Restriction mapping is more applicable to small rather than large molecules.
- In practice, if a DNA molecule is less than 50 kb in length it is usually possible to construct an restriction map.
- To construct map for large genomic size is difficult. The limitations of restriction mapping can be eased slightly by choosing enzymes expected to have infrequent cut sites(rare cutter) in the target DNA molecule.

Lesson 43 : 43_Radiation hybrid mapping of genomes

Radiation Hybrid Mapping of Genomes

- A radiation hybrid is a rodent cell line carrying a small genomic DNA molecule from another organism e.g. a human.

- Exposure to X rays breaks the DNA in human cells.
- The fragments become smaller with more X ray exposure
- Fragments length determines the map resolution.
- Irradiation kills the human cells, which are then fused with rodent cells, rescuing chromosomal fragments that are typically a few Mb in length.

Radiation Hybrid

-
  - Human DNA in the RH is analyzed for gene and/or DNA markers.
  - Closer the two markers to each other – chromosome; the more likely they are to be found together in an RH.

Radiation Hybrid Mapping of Genomes

Radiation Hybrid Mapping: Methodology

-
  - Standard somatic cell fusions contain entire human chromosomes. To locate a gene more closely, you need to use chromosome fragments.
  - Start by irradiating human cells with a controlled dose of X-rays: chromosomes break up. Then, fuse the cells to rodent cells.
  - The human chromosome fragments get integrated into mouse/rodent chromosomes
  - Hybrid cell lines contain random human chromosome fragments.

Radiation Hybrid Mapping

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- Radiation hybrid mapping is a method for high-resolution mapping.
- Exploits the ability of rodent cells (hamster cells) to stably incorporate genetic material from fused cells.
- Resolution is tunable
- Relatively cheap
- Difficult to compare results from different groups

Lesson 44 : 44_Clone contigs mapping of genomes

Clone Contigs Mapping of Genomes

- A partial restriction digest produces a set of large, overlapping DNAs, which are cloned into YAC/BAC vector
- Shearing may also be used to make high-molecular-weight DNA that is blunt-end cloned into a YAC/BAC.
- An entire genome or single chromosome may be represented in a YAC clone library
- YAC clones are then assembled into a map.
- Maps can be generated either by matching with a FISH-generated chromosome map or by DNA fingerprinting
- Assembly of clone contigs is based on clone overlaps.
- Non-polymorphic short tandem sequence are especially useful for YAC contig mapping
- A complete library should yield a complete contig map that indicates the order in which the cloned fragments occur in the chromosome
- Problems arise when some of the YAC inserts contain DNA from more than one chromosomal location.
- Complicated the efforts at generating a YAC contig map of human chromosome
- Many labs have switched to BAC vectors with a capacity of 300 kb and the ability to replicate in *E. coli* as a resource for their sequencing projects.
- **YAC Contigs Map of Chromosome 7**

Clone Contigs Mapping of Genomes

- A contig is a set of partially overlapping clones, a contiguous set of clones. No gaps between them.
- Contigs allow you to build up the sequence of the chromosome over much larger regions than any single clone.
- The first reasonably complete physical map of the human genome involved contigs generated by YACs.

Clone Contigs Mapping of Genomes: Yac48-R

- An entire genome or single chromosome may be represented in a YAC/BAC clone librar

Lesson 45 : 45_Generating sequence of genomes

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Generating Sequence of Genomes

- Dideoxy sequencing used to sequence the genomes
- One sequencing reaction is limited to on average 500 nucleotides, and for accurate sequences both strands were sequenced.

Two competing sequencing strategies used

- Clone-by-clone
- Whole-genome shotgun

Generating Sequence of Genomes: Clone-by-Clone Shotgun Sequencing

- Map construction
- Clone selection
- Sub-clone library construction
- Random shotgun phase
- Directed finishing phase and sequence authentication

Generating Sequence

- Human genome sequencing by the mapping approach used BACs
- BAC insert too large to sequence in one reaction.
- BAC inserts were sequenced using a shotgun approach
- Each insert is cut from the vector, sheared into fragments that will be partially overlapping and cloned into a plasmid vector.
- Each subclone is sequenced, and overlaps are used by a computer to assemble the data into one contiguous sequence representing the BAC insert.
- Using the chromosomal map for BAC clones, the BAC insert sequences are put in order to yield the complete chromosome sequence

Lesson 46 : 46_Human genome project

Human Genome Project

- 1990 – Human Genome Project was started (NHGRI)
- Later many institutes of UK, France, Japan, Germany, China involved.
- In 1998, Celera announced a 3-year plan to sequence human genome

Challenges to Sequence Human Genome

- Size
- Polymorphism
- Repeats - smaller repeats are technically difficult to sequence
- Some DNA sequence are repeated all over the genome.
- Relies on cloning (Some regions are difficult to clone - Heterochromatin).
- Some sequences rearrange or are deleted when cloned.

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Human Genome Project - Objectives

-
  - Create a genetic and physical map of the 24 human chromosomes (22 autosomes, X & Y)
  - Identify the entire set of genes & map them all to their chromosomes.
  - Determine the nucleotide sequence of the estimated 3 billion base pairs
  - Analyze genetic variations in humans
  - Map and sequence the genomes of model organisms

Time line to sequence Human Genome

-
  - Two strategies used to sequence human genome
  - Hierarchical Shotgun Sequencing
  - Whole Genome Shotgun Sequencing

Human Genome Project - conclusion

-
  - Determine the nucleotide sequence of the estimated 3 billion base pairs

Lesson 47 : 47_Human genome project - Hierarchical shotgun sequencing

HGP – Hierarchical Shotgun Sequencing

HGP – Hierarchical Shotgun Sequencing

- Map Construction
- Clone selection
- Sub-clone library construction
- Random shotgun phase
- Directed finishing phase and sequence authentication

-

Hierarchical Shotgun Sequencing: Map Construction

- Clone genomic DNA in YACs (~1MB) or BACs (~200-300Kb)
- Map the relative location of clones
  - Sequenced-tagged sites (STS, e.g. EST) mapping

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    - PCR or probe hybridization to screen STS
    - Restriction site fingerprint
- Most time consuming
    - 1990-98 to generate physical maps for human

Hierarchical Shotgun Sequencing: Resolve Clone Relative Location

Hierarchical Shotgun Sequencing: Clone Selection:

Based on clone map, select authentic clones to generate a minimum tiling path.

Subclone Library Construction: DNA fragmented by sonication or RE cut; Fragment size ~ 2-5 kb

Random Shotgun Phase: Dideoxy termination reaction; Informatics programs; Coverage and contigs

HGP – Hierarchical Shotgun Sequencing

- Map Construction
- Clone selection
- Sub-clone library construction
- Random shotgun phase
- Directed finishing phase and sequence authentication

Lesson 48 : 48_Human genome project - whole genome shotgun sequencing

HGP – Whole Genome Shotgun Sequencing

- Whole genome randomly digested three times
    - Plasmid library constructed with ~ 2kb inserts and ~10 kb inserts
    - BAC library with ~ 200 kb inserts
- Computer program assembles sequences into chromosomes
- No physical map construction
- Reduces problems of repeat sequences

HGP – Whole Genome Shotgun Sequencing

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- **HGP – Whole Genome Shotgun Sequencing**
-
- **HGP – Whole Genome Shotgun Sequencing**

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HGP – Whole Genome Shotgun Sequencing

- After the sequence is shotgunned, 10 million fragments of the genomic sequences need to be recompiled into the readable base pairs in the proper order.
- The Celera Assembler was one of the core competencies and makes the task possible.
- Shotgunned fragments are compared against each other and equivalent sequences greater than 40 base pairs long are identified.
- These 40 base pairs matches are statistically impossible to occur by chance. These matches are then determined to be true or repeat induced.
- True matches are overlapping sections and are the desired fragments

HGP – Whole Genome Shotgun Sequencing

HGP – Whole Genome Shotgun Sequencing

- The assembler then searches for overlapping fragments that have a common sequence and are not contested elsewhere in the dataset.
- The uncontested data is assembled into unitigs containing approximately 30 fragments
- These assembled unitigs are 99 % accurate
- Unitigs passing this filter are ready for ordering.
- By looking at these contigs and orientation the scaffold become complete
- Mapped scaffolds were arranged to prepare chromosome maps

Lesson 49 : 49_Characteristics of our genome (A)

Human Genome - Characteristics (A)

- Human Genome contain ~3 billion bp.
- The average gene consists of 3000 bases, but sizes vary.
- Largest known human gene dystrophin at 2.4 million bases.
- The total number of genes is estimated at around 23,000-much lower than previous estimates of 30,000.
- Almost all (99.9%) nucleotide bases are exactly the same in all people.
- Human genome's gene-dense areas are predominantly composed of the DNA building blocks G and C.
- Gene poor areas are rich in the DNA building blocks A and T.
- GC- and AT-rich regions can be seen through microscope as light and dark bands-chromosomes.
- Genes appear to be concentrated in random areas along the genome
- Large areas of noncoding DNA between the genes

- Stretches of up to 30,000 C-G bases repeating over and over often occur adjacent to gene-rich areas.
- CpG islands are believed to regulate gene activity.
- Chromosome 1 has the most genes ~ (2100), and the Y chromosome has the fewest ~ (458).
- 3,000 Mb
- ~ 23,000 genes
- Exons ~ 1.5%

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- Introns ~ 3.5%
- Repeats ~ 45%

Lesson 50 : 50_Characteristics_human genome (B)

Human Genome -Characteristics

- ~ 1.5% of the genome codes for proteins.
- Repeated sequences make up at least 45 % of the human genome
- Average size of an exon is 120 -145bp.
- Average number of exons is 7-9.
- Average coding sequence encodes proteins of 370 - 440 amino acids

Human Genome - Human genes categorized by function of the transcribed proteins, given both as number of encoding genes and percentage of all identified genes

- The human genome has many different regulatory sequences which are crucial to controlling gene expression.
- Conservative estimates indicate that these sequences make up 8% of the genome.
- About 8% of the human genome consists of tandem DNA arrays or tandem repeats
- Mobile elements:
- LTR retrotransposons (8.3% of total genome)
- SINEs (13.1% of total genome) including Alu elements
- LINEs (20.4% of total genome)
- Pericentromeric and sub-telomeric regions of chromosomes filled with large transposable elements
- Chimpanzee genome differs from that of the human genome by 1.23% in direct sequence comparisons
- Male mutation rate about twice female
- Most mutations occurs in males
- Recombination rates much higher in distal regions of chromosomes

Lesson 51 : 51_Genome browsers_UCSC

Genome Browser - UCSC

Genome Browsers

- The genomes are so large that useful information is hard to find.
- Researchers at UCSC decided to make a computational microscope to help scientists search the genomes.

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- Researchers can use the UCSC Genome Browser to find information in the human genome and other genomes that have been sequenced
- UCSC Genome Browser
- *http://genome.ucsc.edu*
- **The UCSC Genome Browser - Homepage**
- **The UCSC Genome Browser**

Single Base/Nucleotide View

- Researchers can use the UCSC genome browser to find information in the human genome and other genomes that have been sequenced

Lesson 52 : 52_ Genome browser_Ensembl

Genome Browser - Ensembl

Ensembl Genome Browser

- Ensembl is a joint project between 3 organizations
  - EMBL
  - EBI
  - WTSI
- Ensembl does not gather any genome project directly
- Works in relation with the sequencing centers that generate the genome assembly
- 4 Main Databases
  - Ensembl Core Database
  - Ensembl EST Database
  - Ensembl Compara Database
  - Ensembl Variation Database
- Ensembl core databases store genome sequence and annotation information
- Gene, transcript, and protein models
- Databases stores information about cDNA and protein alignments
- Within one genome: regulatory elements, gene order etc
- Comparative studies: Evolution, conserved regions rearrangements
- Gene quality and prediction
- Ensembl Genome browser
- *http://www.ensembl.org*
- **Ensembl Genome Browser – Home page**

Ensembl Genome Browser

- Homo Sapiens - Human
- Pan troglodytes- Chimpanzee
- Macaca mulatta - Rhesus monkey
- Mus musculus - Mouse

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- Rattas norvegicus - Rat
- Canis familiaris - Dog
- All data generated is free for download
- Includes genes sequence, transcript, protein predictions
- Ensembl provides a dedicated Export View page, can be exported into HTML, text or zipped format

Lesson 53 : 53_HGP_ethical and legal issues

HGP- Ethical, Legal and Social Issues

- The ability to identify human genes raises complex ethical issues involving
- The right to information about one's own genome
- Right to genetic information by employers, insurance companies and government agencies, and concerns about the ability to diagnose but not treat genetic disorders
- Privacy and confidentiality of genetic information
- Fairness in the use of genetic information by insurers, employers, courts, schools, adoption agencies
- Psychological impact and discrimination due to an individual's genetic differences.
- Reproductive issues including adequate and informed consent and use of genetic information in reproductive decision making.
- Clinical issues including the education of doctors and other health-service providers
- People identified with genetic conditions, and the general public about capabilities, limitations, and social risks.
- Uncertainties associated with gene tests for susceptibilities and complex conditions (e.g., heart disease, diabetes, and alzheimer's disease).
- Fairness in access to advanced genomic technologies.
- Health and environmental issues concerning genetically modified (GM) foods and microbes.
- Commercialization of products including property rights (patents, copyrights, and trade secrets) and accessibility of data and materials.

Lesson 54 : 54_Genome annotations - protein coding genes

Genome Annotations – Proteins Coding Genes

Genome Annotations

- To make use of the genome sequences, all components of genomes need to understand.
- Assigning identities and functions to sequences within the genome is called genome annotation.
- Genes transcribed, which means that we can identify them
- Traditionally cDNA /EST sequencing
- More recently by microarray
- Structural Annotation – Finding genes and other biologically relevant sites
- Annotation/function can be mapped to different levels:
- Organism, cellular, molecular level

Genome Annotations

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-
- Genes can be identified *in vitro* using computational methods
- Protein-coding genes have recognizable features
- Open reading frames
- Codon bias
- Known transcription and translational start and stop motifs
- Splice consensus sequences at intron-exon boundaries
- Protein-coding genes recognizable feature
- Software to scan the genome and identify these features
- Some of these programs work quite well, in bacteria and simple eukaryotes
- Harder for the higher eukaryotes where there are a lot of long introns, genes can be found within introns of other genes

Validating Predictions & Refining Gene Models

- Validation of predictions include;
- Match to previously annotated cDNA
- Match to EST from same organism
- Similarity of nucleotide or translated protein sequence to GenBank
- Protein structure prediction match - PFAM domain
- Associated with recognized promoter sequences, i.e TATA box, CpG island

Finding Non-Protein-Coding Genes

- Harder to find
- No poly-A tailed—
- No ORF
- Rely on sequence divergence at nucleotide not protein level, so homology is harder to detect

Gene Discovery Programs

- Most gene-discovery programs makes use of machine learning algorithm.
- Two approaches;
- Artificial Neural Networks and Hidden Markov Models.

Gene Discovery Programs- Diagram for an HMM for Gene Discovery

-
- **Examples of known gene models; provide the probability that a stretch of sequence is a gene.**

Conclusion:

- Assigning identities and functions to sequences within the genome is called genome annotation.

Lesson 55 : 55_Genome annotations - transcription factors

Genome Annotations – Transcription Factors

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Genome Annotations

- Other than protein coding genes
- Structural sequences
- Regulatory sequences
- Non-functional junk
- Annotate regulatory sequences such as transcription factor binding sites (Cis-)
- Transcription factors (TFs) are proteins that bind to the DNA and help to control gene expression.
- Sequences to which TF bind called transcription factor binding sites (Cis-acting elements)
- Transcription factors bind to specific DNA sequences
- Most transcription factors can bind to a range of similar sequences.
- Once we know the binding site, we can search the genome to find all of the (predicted) binding sites.

Genome Annotations

- Frequency matrix and its graphical depiction, a sequence logo

Genome Annotations - Conclusion

- Genome annotations can also be performed by searching transcription factor binding sites

Lesson 56 : 56_Arabidopsis genome

Arabidopsis Thaliana Genome – A Model Organism in Plant Biology and Genetics

- Arabidopsis Thaliana – A model Plant
- Arabidopsis has 5 chromosomes (2n=10)
- Arabidopsis Thaliana Chromosomes

Arabidopsis Thaliana Genome

- Contains about 135 megabases of sequence
- Encodes approximately 27,000 genes and 35,000 proteins.
- Has 35% unique genes
- Has 37% genes that exist as members of large gene families (families of 5 or more members)
- An estimated 58-60% of the Arabidopsis genome exists as large segmental duplications
- Arabidopsis genome contains genes encoding RNA polymerase subunits not seen in other eukaryotic organisms.

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- Arabidopsis has genes unique to plants – approximately 150 unique protein families were found, including 16 unique families of transcription factors
- **Segmental duplications in *Arabidopsis* thaliana.**
- **Colored bands serve as representations of duplicated segments.**
- Arabidopsis has 5 chromosomes (2n=10)
- Contains about 135 Mb of sequence
- Encodes ~ 27,000 genes and 35,000 proteins.

Lesson 57 : 57_Mouse genome

Mouse Genome – A Model Organism in Animal Biology and Genetics

Mouse Genome

- Number of chromosome 2n=40
- Human and mouse genomes have conserved blocks of genetic material
- Humans and mice suffer from similar diseases
- Estimated genome Size 2.6 billion bp
- Number of estimated coding genes 22,500

At the nucleotide level, approximately 40% of the human genome aligned to the mouse genome.

Mouse Genome: Chromosomes and their Sizes

Mouse Genome: Chromosomes and their Sizes

- Mouse genome contains fewer CpG islands (15,500)
- The (G+C) content for each of the mouse chromosome is relatively similar
- Approximately 99% of mouse genes have a homologue in the human genome
- **Human and Mouse Chromosomes**

Human and Mouse: Chromosomes Orthologs

- Most human genes have mouse orthologs
- Coding exons usually correspond 1-1
- Coding sequence similarity ~ 85%

Mouse Genome Contigs

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- Most mouse BAC contigs contained multiple mouse markers.
- Coverage of the mouse genome (2.6 billion bp) in mapped BACs is virtually complete: 296 contigs of average size 9.3 Mb.
- All mouse chromosomes are acrocentric, with the centromeric end at the top of each chromosome.

Cytogenetic View of Chromosomes

Human–Mouse Homology Clone Map

Alignment between part of human chromosome 6 and mouse chromosome 4 . 1.6-Mb interval is enlarged, showing part of 16.1 aligned to a 1.3-Mb mouse BAC contig. 11 of the 15 segments of human sequence match to 29 of segments of mouse BAC contig

Mouse Gene and Human Gene

- Number of chromosome 2n=40
- Humans and mice suffer from similar diseases
- Estimated genome Size 2.6 billion bp
- Number of estimated coding genes 22,500

Lesson 58 : 58_ Comparison of human and Mouse genome

Genome Comparison -Human and Mouse

Comparison

- The mouse genome is about 14% smaller than the human genome
- Over 90% of mouse and human genomes can be partitioned into corresponding regions of conserved synteny
- At the nucleotide level, 40% of the human genome can be aligned to the mouse genome
- Mouse and human genomes each seem to contain ~ 23,000 protein-coding genes.
- The proportion of mouse genes without any homologue currently detectable in the human genome (and vice versa) is < 1%.
- Despite marked differences in activity of transposable elements, similar types of repeat sequences have accumulated in the corresponding genomic regions
- **Comparison - Human and Mouse Genome**
- A typical 510-kb segment of mouse chromosome 12 that shares common ancestry with a 600-kb section of human chromosome 14.
- Blue lines connect the reciprocal unique matches in the two genomes.
- Mouse genome contains fewer CpG islands than human.
- (G+C) content and density of CpG islands shows more variability in human than mouse

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- Human chromosomes show more variation (G+C) on chromosomes 16, 17, 19 and 22 have higher (G+C) content, and chromosome 13 lower (G+C) content.
- The density of CpG islands is relatively homogenous for all mouse chromosomes and more variable in human.
- Approximately 99% of mouse genes have a homologue in the human genome
- **Comparison – Human and Mouse Genome**

Comparison - Conclusion

- The mouse genome is about 14% smaller than the human genome
- Mouse and human genomes each seem to contain ~ 23,000 protein-coding genes.

Lesson 59 : 59_Mapping of Disease Genes

Mapping of Disease Genes

-
  - Disease gene mapping is one of the main objectives of genotyping
  - Different approaches for identification of disease genes.

Position Independent Method

-
  - No information about the location of the gene.
  - Starting at the phenotype, determining which protein was involved and getting to gene through the protein.
  - Starting from the approximate location of gene, to finding the gene itself, then translating it to learn about the protein and its function.
  - Positional cloning or reverse genetics
  - Mostly identifying and isolating the protein product of the gene.
  - Such genes usually produce large amounts of well-known and well studied proteins.
  - Two common methods used
  - Gene-specific oligonucleotides
  - Antibody Methods
  - Recombination mapping and/or somatic cell mapping to defined the region of interest as tightly as possible.
  - Chromosomes are inherited as intact units, so it is reasoned that the alleles of some pairs of genes will inherit together because they are on the same chromosome.
  - Different approaches are used for identification of disease genes

Lesson 60 : 60_Microarrays

DNA Microarray

- Microarray technology evolved from Southern blotting, where fragmented DNA is attached to a substrate and then probed with a known DNA sequence.

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- There are several synonyms of DNA microarrays such as
- DNA chips
- Gene chips
- DNA arrays
- Gene arrays
- Biochips
- The principle of DNA microarrays lies on the hybridization between the nucleotides.
- Presence of one genomic or cDNA sequence in 100,000 or more sequences can be screened in a single hybridization.
- DNA microarray allows us to analyze thousands of genes in one experiment
- DNA microarrays are solid supports; glass or silicon, upon which DNA is attached

Simultaneous Studies of Many Genes

- Patterns or clusters of genes are more informative regarding total cellular function than looking at one or two genes – can figure out new pathways
- Just sequencing genomes is not sufficient
- Thousands of genes remain without an assigned function
- Each spot of DNA, (probe), represents a single gene

Microarray Use

- Determine which genes are active in a cell and at what levels
- Compare the gene expression profiles of a control vs treated
- Assessing gene expression levels
- Genome-wide studies and genotyping
    - Evaluating microRNA levels
- Determine which genes have biological significance in a system
- Discovery of new genes, pathways, and cellular trafficking

Lesson 61 : 61_Types of microarray

Types of Microarrays

- DNA microarrays
- Exon arrays
- Comparative Genomic Hybridization array
- DNA microarrays- such as cDNA microarrays and oligonucleotide microarrays, SNPs
- Exon arrays-Alternative splice variant detection
- CGH arrays- Comparative genomic hybridization

Oligonucleotide Arrays

cDNA Arrays

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Types of Microarray

- Protein microarrays (protein-protein interactions)
- Tissue microarrays
- Cellular microarrays (also called transfection microarrays
- Expression Arrays
- Chemical compound microarrays
- Antibody microarrays (proteomics)
- Carbohydrate arrays (glycoarrays)
- Re-sequencing arrays

Microarray Require

- Microarrays combine genomics, silicon chip manufacturing, DNA and Protein chemistry, signal and image processing, statistics, software skills and traditional molecular biology experiments.

Microarray and Genes Expression Profiling

- Gene expression profiling can be monitored for thousands of genes simultaneously
- CGH arrays assesses genome content in different cells or related organisms

Microarray and SNPs

- SNP array: Identifying single nucleotide polymorphism among alleles within or between populations

Methylation Microarrays

- Methylation arrays determine methylated DNA
- Determining which regions of DNA are methylated ultimately to study epigenetics

Lesson 62 : 62_Microarrays Formats_some errors

Microarray Formats

Main Formats

- Cartridge-based

  Spotted

  Electronic

- Spotted Glass Slide

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- Tissue Section Slide

Cartridge-Based Chips

- High density arrays of DNA oligos within a plastic housing
- One sample = One chip (Affymetrix, Agilent, Applied Biosystems etc)
- Generally used with expression and DNA arrays

Cartridge-Based Expression Microarrays

- Involves Fluorescently tagged biotinylated cRNA
- One chip per sample
- Uses single fluorescent dye

Spotted Glass Arrays

- Uses cDNA, Oligonucleotide, protein, antibody
- Robotically spotted cDNAs or Oligonucleotides
- Printed on Nylon, Plastic, or Glass microscope slide

Spotted cDNA and Oligo Glass Arrays

- Involves two dyes on the same slide

  Red dye-Cy5

  Green dye-Cy3

- Control and experimental cDNA

Main Formats

- Cartridge-based

  Spotted

  Electronic

- Spotted Glass Slide
- Tissue Section Slide

Lesson 63 : 63_Microarray – procedure

Microarray Procedure

- Collect Samples
- Isolate mRNA

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- Create Labelled DNA
- Hybridization
- Microarray Scanner
- Analyze Data

Samples:

This can be from a variety of organisms. Two samples – cancerous human skin tissue & healthy human skin tissue

- Extract the RNA from the samples.
- After isolating the RNA, isolate the mRNA from the rRNA and tRNA.
- mRNA has a poly-A tail, use a column containing beads with poly-T tails to bind the mRNA
- Rinse with buffer to release the mRNA from the beads. The buffer disrupts the pH, disrupting the hybrid bonds.

Microarray Procedure - Create Labeled cDNA

- Add a labelling mix to the RNA. The labelling mix contains poly-T (oligo dT) primers, reverse transcriptase (to make cDNA), and fluorescently dyed nucleotides.
- Add cyanine 3 (fluoresces green) to the healthy cells and cyanine 5 (fluoresces red) to the cancerous cells

Microarray Procedure - Hybridization

- Apply the cDNA to a microarray plate.
- When comparing two samples, apply both samples to the same plate.
- The ssDNA will bind to the cDNA already present on the plate

Microarray Procedure – Lasers

- Laser scans array and produces images
- One laser for each color, e.g. one for green, one for red

Microarray Procedure – Microarray Scanner

- The scanner has a laser, a computer, and a camera.
- The laser causes the hybrid bonds to fluoresce.
- The camera records the images produced when the laser scans the plate.
- The computer allows to immediately view the results and it also stores the data.

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Microarray Procedure – Microarray Scanner

- GREEN – the healthy sample hybridized more than the diseased sample.
- RED – the diseased/cancerous sample hybridized more than the nondiseased sample.

Microarray Procedure – Microarray Scanner

Microarray Procedure – Microarray Scanner

- YELLOW - both samples hybridized equally to the target DNA.
- BLACK - areas where neither sample hybridized to the target DNA.

Microarray Procedure – Microarray Scanner

- By comparing the differences in gene expression between the two samples, we can understand more about the genomics of a disease**.**

Microarray Procedure

- Collect Samples
- Isolate mRNA
- Create Labelled DNA
- Hybridization
- Microarray Scanner
- Analyze Data

Lesson 64 : 64_Microarray chips

Microarray Chips

- There are two types of DNA Chips/Microarrays:
- cDNA based microarray
- Oligonucleotide based microarray

cDNA Based Microarray Chips

- Chips are prepared by using cDNA- cDNA chips or cDNA microarray.
- The cDNAs are amplified. Then these immobilized on a solid support made up of nylon filter of glass slide

- The probe DNA are loaded by capillary action.
- Small volume of this DNA is spotted on solid surface
- DNA is delivered mechanically or in a robotic manner.
- When one DNA spotting is done, the pin is washed and loaded with fresh DNA to start the second cycle

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DNA Based Microarray Chips

- Almost similar type procedure
- Gene expression profiling
- Discovery of drugs
- Diagnostics and genetic engineering
- Alternative splicing detection
- Functional genomics
- DNA sequencing
- Toxicological research (Toxicogenomics)

Microarray Chips

- There are 2 types of DNA Chips/Microarrays:
- cDNA based microarray
- Oligonucleotide based microarray

Lesson 65 : 65_Microarray applications

Microarray Applications

Microarray -Advantages

- Provides data for thousands of genes
- One experiment instead of many
- Fast and easy to obtain results
- Closer to discovering cures for diseases and cancer
- Different parts of DNA can be used to study gene expression

Disadvantages

- The biggest disadvantage of DNA chips is that they are expensive to create.

The production of too many results at a time requires long time for analysis, which is quite complex in nature.

- The DNA chips do not have very long shelf life, which proves to be another major disadvantage of the technology.

Microarray -Limitations

- Cross-hybridization of sequences with high identity
- Chip to chip variation
- The real limitation is Bioinformatics
- Expensive – repeat experiments

Gene Discovery and Disease Diagnostics

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- Gene discovery and disease diagnosis
- Classify the types of cancer on the basis of the patterns of gene activity in the tumor cells

Pharmacogenomics

- Study of correlations between therapeutic responses to drugs and the genetic profiles of the patients

Toxicogenomics

- Microarray technology allows us to research the impact of toxins on cells.
- Some toxins can change the genetic profiles of cells.

Biological Applications

- Many biological discovery
- New and better molecular diagnostics
- New molecular targets for therapy
- Finding and refining biological pathways

Lesson 66 : 66_Types of RNA and RNomics

Types of RNA and RNomics

Types of RNA

- Wide range of RNAs
- Some of them are coding while others are noncoding

Types of RNA - Classification

Types of RNA – Housekeeping RNAs

Types of RNA – Small Non-coding RNAs

Types of RNA – Long Non-coding RNAs

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RNomics is the study

- Identification
- Expression
- Biogenesis
- Structure
- Regulation of expression
- Targets
- Biological functions of RNAs

Computational RNomics

- Searching conserved intronic sequences by comparative analysis of introns
- Searching conserved intergenic sequences

Experimental RNomics

- PAGE separation of non-coding RNAs and sequencing
- Non-coding RNA enriched cDNA libraries and sequencing

Types of RNA

- Wide range of RNAs
- Some of them are coding RNA while others are noncoding.

Lesson 67 : 67_Noncoding DNA

Non-coding DNA

Types of Non-coding DNA

- Non- coding functional RNA
- Cis- and Trans- regulatory elements
- Introns
- Pseudogenes
- Repeat sequences, transposons and viral elements
- Telomeres

Non-coding Functional RNA

- The RNA molecules which are not translated into proteins.
- Examples: Ribosomal RNA, transfer RNA & micro RNA

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Cis- and Trans- Regulatory Elements

- Those sequences that control the transcription of a nearby gene.
- Located within 5' or 3' untranslated regions or within introns.

Introns

- They are non-coding sections of a gene.
- Transcribed in the precursor mRNA sequence but is ultimately removed by RNA splicing.
- Most of the introns appear to be mobile genetic elements.

Pseudogenes

- They are related to known genes, that have lost their protein-coding ability or are otherwise no longer expressed in the cell.
- Arise from retrotransposition or genomic duplication of functional genes.

Repeat Sequences, Transposons & Viral Elements

- Transposons and Retrotransposons are mobile genetic element
- Retrotransposons: LINEs, SINEs

Telomeres

- Telomeres are regions of repetitive DNA
- Located at the end of a chromosome
- They provide protection from chromosomal deterioration during DNA replication

Functions

- Essential for chromosome structure
- Genome Protection
- Enhancers, silencers, promotors, insulator
- Genetic switches
- Regulation of gene expression

Lesson 68 : 68_Noncoding RNA

Non-coding RNA

- Non-coding RNA is a RNA molecule that functions without being translated into a protein
- Large number of genes of noncoding RNA
- Noncoding RNA -different functions

Types of Non-coding RNA

- Ribosomal RNA

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- Transfer RNA
- Small nuclear RNA
- Small nucleolar RNA
- Short interfering
- Micro RNA
- Long non-coding RNA

Functional Diversity of Non-coding RNA

Non-coding RNAs in Human Genome

Non-coding RNA

- Non-coding RNA (ncRNA) is a RNA molecule that functions without being translated into a protein

Lesson 69 : 69_miRNA

MicroRNA (miRNA)

MicroRNA (miRNA)

- Small non-coding double stranded RNAs

- Approximately 19-22 nucleotides long
- Repress activity of complementary mRNAs
- Regulate 30% of mammalian genes
- Described in invertebrates and vertebrates: worms, fungi, plants, and mammals
- Many are conserved between vertebrates and invertebrates
- miRNA originates from ssRNA that forms a hairpin secondary structure
- miRNA regulates post-transcriptional gene expression and is often not 100% complementary to the target
- Originate from capped & polyadenylated full length precursors (pri-miRNA)
- Hairpin precursor ~70 nt (pre-miRNA)
- Mature miRNA ~22 nt (miRNA)
- Drosha and Pasha are part of the "Microprocessor" protein complex (~600 - 650kDa)
- Drosha and Dicer are RNase III enzymes
- Pasha is a dsRNA binding protein
- Exportin 5 is a member of the karyopherin nucleocytoplasmic transport factors
- Argonautes are RNase H enzymes
- **MicroRNA (miRNA)**

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- 1900 discovered in Homo sapiens
- Mouse and human have highly conserved miRNA genes

MicroRNA (miRNA) Conclusion

- Small non-coding double stranded RNAs

- Approximately 19-22 nucleotides long
- 30% of mammalian mRNAs are regulated by miRNAs

Lesson 70 : 70_Biogenesis of miRNA

Biogenesis of MicroRNA (miRNA)

MicroRNA (miRNA)

- Small non-coding double stranded RNAs

- Approximately 19-22 nucleotides long
- miRNA originates with ssRNA that forms a hairpin secondary structure
- Originate from capped & polyadenylated full length precursors (pri-miRNA)

Biogenesis of miRNA

- Primary-miRNA is transcribed in the nucleus, and is usually several kilobases long; having 5' cap and a poly-A tail.
- Cleaved in the nucleus by Drocha enzyme to 70nt hairpin transcript (pre-miRNA).
- Transported to the cytoplasm by Exportin 5 through nuclear pores.
- Cleaved by Dicer enzyme (RNase III enzyme) into 19-22nt ds-transcripts.

Illustration of miRNA Processing - Another View

- Small non-coding double stranded RNAs

Approximately 19-22 nucleotides long

Lesson 71 : 71_Fucntions of miRNA

Functions of miRNA

Functions of miRNA

- Involved in the post-transcriptional regulation of gene expression

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- Important in development
- Metabolic regulation (miR-375 & insulin secretion)

Functions of miRNA: Processing bodies are sites of storage and/or degradation of mRNA

Functions of miRNA

Tumor Suppressors and Oncogenes

- miRNAs can function as tumour suppressors and oncogenes.
- Gene therapies that use miRNAs might be an effective approach to blocking tumour progression.
- miRNAs such as let-7, which has been shown to negatively regulate the Ras oncogenes, and miR-15 and miR-16, which negatively regulate BCL2, are promising candidates for cancer treatment.

Functions of miRNA

- Involved in the post-transcriptional/translational regulation of gene expression
- Metabolic regulation

Lesson 72 : 72_Mode of action miRNA

miRNA: Mode of Action

Mode of Action of miRNA

- Three different mode of action of miRNA
- mRNA degradation
- Transcriptional regulation
- Translational regulation

Structure of miRNA at Nucleotide level

miRNA Binding to Target (mRNA)

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Mode of Action of miRNA - mRNA Degradation

Common Plants

Mode of Action - Translational Regulation

- **Common in Animals**

Mode of Action – Transcriptional Regulation

- **Common in Yeast and Plants**
- **Sometimes in Animals**

Mode of action of miRNA

- Three different mode of action of miRNA
- mRNA degradation
- Transcriptional regulation
- Translational regulation

Lesson 73 : 73_Non coding RNA and X chromosome Silencing

Non-coding RNA and Silencing of X Chromosome

Silencing of one X chromosome

- X chromosome silencing is mediated by Xist – a 16,000 nt long ncRNA

- Xist ncRNA recruited complex has one entry site in X chromosome, corresponding to Xist gene itself
- Xist appears to recruit a specific histone isoform which maintains the chromosome in inactive state
- Additionally, Xist containing complexes recruit histone deacetylases and methylases
- Xist activity is regulated by another 40,000 nt long ncRNA – Tsix, which contains anti-sense sequence of Xist and therefore is able to regulate Xist activity by base-pairing to it.

shRNA and Genetic Imprinting

- Activity of small heterochromatic RNAs (shRNAs) appear to be essential for establishing and maintaining the imprinted status of genes

Lesson 74 : 74_RNA Induced Silencing Complex'

RNA Induced Silencing Complex

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- RISC is a large (~500-kDa) RNA multi-protein complex, which triggers mRNA degradation in response to siRNA

RISC

- Unwinding of double-stranded siRNA by ATP independent helicase.
- Active components of RISC are endonucleases called argonaute proteins which cleave the target mRNA strand.

RNA Induced Silencing Complex - Structure

RNA Induced Silencing Complex

RISC is a large (~500-kDa) RNA-multiprotein complex, which triggers mRNA degradation

Some components have been defined by genetics, but function is unknown, e.g. - unwinding of double-stranded siRNA (Helicase) ribonuclease component cleaves mRNA (Nuclease).

- RISC bound to partially complementary mRNA induces translational repression: results in destabilization of target mRNA
- The more RISC bound; greater the inhibitory effect
- RISC bound to perfectly complementary mRNA (to miRNA) are cleaved and degraded by RISC in a process similar to RNAi
- RISC is a large (~500-kDa) RNA-multiprotein complex, which triggers mRNA degradation in response to siRNA

Lesson 75 : 75_Dicer and Drosha

Dicer and Drosha

Dicer

- Dicer or helicase with RNase motif, is an enzyme of RNase III family
- In humans, it is encoded by the DICER1 gene.
- Dicer cleaves double-stranded RNA (dsRNA) and pre-microRNA into short double-stranded RNA fragments called small interfering RNA and microRNA
- It is able to digest dsRNA into uniformly sized small RNAs (siRNA)
- Dicer family proteins are ATP-dependent nucleases.
- RNase III enzyme acts as a dimer
- Dicer homologs exist in many organisms including C. elegans, Drosphila, yeast and humans

Dicer is a Ribonuclease (RNase III family) with 4 Distinct Domains

Drosha

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- Drosha is a class 2 ribonuclease III enzyme that in humans is encoded by the gene DROSHA (formerly RNASEN) gene
- The RNase III drosha is the core nuclease that executes the initiation step of microRNA (miRNA) processing in the nucleus

Dicer and Drosha - Humans

- Dicer is an enzyme that is part of the RNase III family
- Drosha belongs to class 2 ribonuclease III enzyme family

Lesson 76 : 76_siRNA

Small Interfering RNA (siRNA)

- Small interfering RNA (siRNA), sometimes known as short interfering RNA, are 20-25 nucleotide-long RNA molecules that interfere with the expression of genes.
- They are naturally produced as part of the RNA interference (RNAi) pathway by the enzyme Dicer.
- Exogenously introduced by investigators to bring about knockdown of a particular gene.
- siRNA's have a well defined structure.
- A short (usually 21-nt) double-strand of RNA (dsRNA) with 2-nt overhangs on either end, including a 5' phosphate group and a 3' hydroxy (-OH) group.
- **siRNA:** short (usually 21-nt) double-strand of RNA (dsRNA) with 2-nt overhangs on either end.
-

siRNA: Short double-strand of RNA (dsRNA) with 2-nt overhangs on either end.

- Small interfering RNAs that have an integral role in the phenomenon of RNA interference(RNAi),a form of post-transcriptional gene silencing
- A single base pair difference between the siRNA template and the target mRNA is enough to block the process.
    - Several different methods of expression
    - Several different methods of delivery
    - Dicer (type III RNAse III) cleaves long dsRNA into siRNA 21-25nt dsRNA from exogenous sources

Small interfering RNA: siRNA design

- Small interfering RNA (siRNA), sometimes known as short interfering RNA, are a class 20-25 nucleotide-long RNA molecules that interfere with the expression of genes.

Lesson 77 : 77_Design of siRNA

Design of siRNA

- 21-23 nt dsRNA,

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- GC% slightly < 50%
- Perfect complimentary to target mRNA
- Targeting 3'UTR works better than 5' UTR

Design of siRNA

-
- **siRNA/miRNA: Difference in mode of action**

Different Design of siRNA

Design of siRNA

- 21-23 nt dsRNA,
- Perfect complimentary to target mRNA
- Targeting 3'UTR works better than 5' UTR

Lesson 78 : 78_Applications of siRNA

Applications of siRNA

- Determining protein function
    - Easier than preparing knockout
    - Used for partial knockdowns
    - Applications for cancer prevention, infections and developmental defects

siRNA

-
    - Exogenously delivered
    - 21-23mer dsRNA
    - Acts through RISC
    - Induces homologous target cleavage
    - Perfect sequence match
    - Results in target degradation

miRNA

- Endogenously produced
- 21-23mer dsRNA
- Acts through RISC
- Induces homologous target cleavage
- Imperfect sequence match
- Results in translation arrest

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Applications of siRNA

- Determining protein function
    - Easier than a knockout
    - Used for partial knockdowns

Lesson 79 : 79_Transcriptome

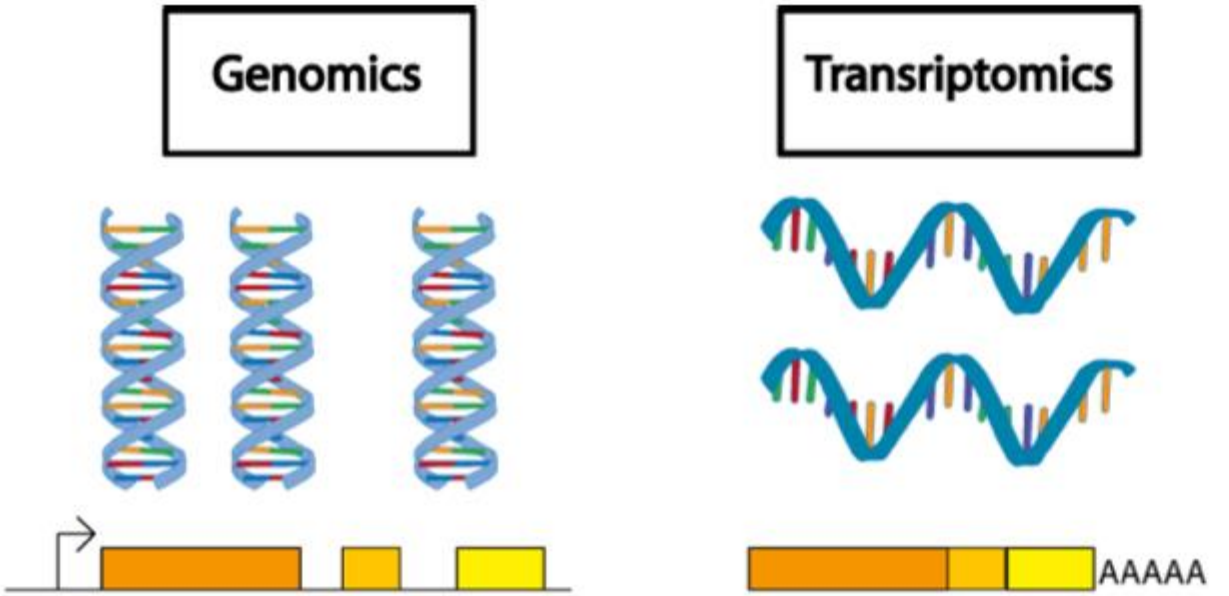
Transcriptomics

- The population of mRNAs expressed by a genome at any given time
- The complete collection of transcribed elements of the genome

Transcriptomics

- The study of characteristics and regulation of the functional RNA transcript population of a cell/s or organism at a specific time

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- genomic DNA Sequencing
- annotations

-microarray hybridization

-RNA-seq

Genes sequences

mRNA sequences
mRNA expression level

- Percentage of the genetic code that is transcribed into RNA molecules (depends on development, environment, time of the day, tissue)
- Collection of all gene transcripts present in a given cell/tissue at a given time
- Genes and pathways involved in biological processes
- Genes with similar expression may be functionally related and under the same genetic control mechanism
- Help elucidating the function of unknown genes based on their spatial and temporal expression
- Identifies marker genes for diagnosis of diseases

Lesson 80 : 80_Sequencing Techniques

Key Points

i. Genome sequencing will greatly advance our understanding of genetic biology and has vast potential for medical diagnosis and treatment.

ii. DNA sequencing technologies have gone through at least three "generations": Sanger sequencing and Gilbert sequencing were first-generation, pyrosequencing was second-generation, and Illumina sequencing is next-generation.

iii. Sanger sequencing is based on the use of chain terminators, ddNTPs, that are added to growing DNA strands and terminate synthesis at different points.

iv. Illumina sequencing involves running up to 500,000,000 different sequencing reactions simultaneously on a single small slide. It makes use of a modified replication reaction and uses fluorescently-tagged nucleotides.

v. Shotgun sequencing is a technique for determining the sequence of entire chromosomes and entire genomes based on producing random fragments of DNA that are then assembled by computers which order fragments by finding overlapping ends.

DNA Sequencing Techniques

While techniques to sequence proteins have been around since the 1950s, techniques to sequence DNA were not developed until the mid-1970s, when two distinct sequencing methods were developed almost simultaneously, one by Walter Gilbert's group at Harvard University, the other by Frederick Sanger's group at Cambridge University. However, until the 1990s, the sequencing of DNA was a relatively expensive and long process. Using radiolabeled nucleotides also compounded the problem through safety concerns. With currently-available technology and automated machines, the process is cheaper, safer, and can be completed in a matter of hours. The Sanger sequencing method was used for the human genome sequencing project, which was finished its sequencing phase in 2003, but today both it and the Gilbert method have been largely replaced by better methods.

Sanger Sequencing

The Sanger method is also known as the dideoxy chain termination method. This sequencing method is based on the use of chain terminators, the dideoxynucleotides (ddNTPs). The dideoxynucleotides, or ddNTPSs, differ from deoxynucleotides by the lack of a free 3' OH group on the five-carbon sugar. If a ddNTP is added to a growing DNA strand, the chain is not extended any further because the free 3' OH group needed to add another nucleotide is not available. By using a predetermined ratio of

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deoxyribonucleotides to dideoxynucleotides, it is possible to generate DNA fragments of different sizes when replicating DNA in vitro.

A Sanger sequencing reaction is just a modified in vitro DNA replication reaction. As such the following components are needed: template DNA (which will the be DNA whose sequence will be determined), DNA Polymerase to catalyze the replication reactions, a primer that basepairs prior to the portion of the DNA you want to sequence, dNTPs, and ddNTPs. The ddNTPs are what distinguish a Sanger sequencing reaction from just a replication reaction. Most of the time in a Sanger sequencing reaction, DNA Polymerase will add a proper dNTP to the growing strand it is synthesizing in vitro. But at random locations, it will instead add a ddNTP. When it does, that strand will be terminated at the ddNTP just added. If enough template DNAs are included in the reaction mix, each one will have the ddNTP inserted at a different random location, and there will be at least one DNA terminated at each different nucleotide along its length for as long as the in vitro reaction can take place (about 900 nucleotides under optimal conditions.)

The ddNTPs which terminate the strands have fluorescent labels covalently attached to them. Each of the four ddNTPs carries a different label, so each different ddNTP will fluoresce a different color.

After the reaction is over, the reaction is subject to capillary electrophoresis. All the newly synthesized fragments, each terminated at a different nucleotide and so each a different length, are separated by size. As each differently-sized fragment exits the capillary column, a laser excites the flourescent tag on its terminal nucleotide. From the color of the resulting flouresence, a computer can keep track of which nucleotide was present as the terminating nucleotide. The computer also keeps track of the order in which the terminating nucleotides appeared, which is the sequence of the DNA used in the original reaction.

Lesson 81 : 81_Automated Sequencing

Automated Sequencing

- Florescent dyes are used to label DNA
- Primers for each reaction (ddA, ddT, ddG, ddC) are tagged with separate dyes
- All reactions in same gel lane
- Band colors help in detection

A flourimeter and computer are hooked up to the gel and they detect and record the dye attached to the fragments as they come off the gel

Capillary Sequencing

Samples run through capillary instruments

Sequence analysis software analyzes sequence in text and electropherogram form

Peak patterns reflect mutations or sequence changes

Conclusions

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Use of flourescent dyes and an automated system improved the effeciency of sequencing

Lesson 82 : 82_Shot Gun Sequencing

Shotgun sequencing

The DNA is broken up randomly into numerous small segments, which are sequenced

Named by analogy with the rapidly expanding, quasi-random firing pattern of a shotgun

- **Clone by clone shotgun sequencing**

It is also known as hierarchical shotgun sequencing or map based shotgun sequencing. The target DNA is first analyzed by clone based physical mapping methods. Individual clones that together span the region of interest are selected and subjected to shotgun sequencing.

- **Whole Genome Shotgun Sequencing (WGS)**
- Whole genome is broken down, cloned into vectors and then sequenced.

Shotgun sequencing

The DNA is broken up randomly into numerous small segments, which are sequenced

Lesson 83 : 83_Next Generation Sequencing

Next-generation sequencing (NGS)

Historical Perspective

The Human Genome sequencing was accomplished by sanger's method

Sanger sequencing had been a gold standard for nucleic acid sequencing for about 25 years

The Lynx Therapeutics (now Solexa) published and marketed "Massively parallel signature sequencing", or MPSS, in 2000

It led to the development of Next Generation Sequencing Techniques

Evolution of sequencing technologies

- 1953 : Discovery of DNA structure by Watson and Crick
- 1973 : First sequence of 24 bp published
- 1977 : Sanger sequencing method published
- 1980 : Nobel Prize Wally Gilbert and Fred Sanger

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- 1982 : Genbank started 1983 : Development of PCR
- 1987 : 1st automated sequencer : Applied
- Biosystems Prism 373
- 1996 : Capillary sequencer : ABI 310
- 1998 : Genome of Caenorhabditis elegans sequenced
- 2000 : Human genome sequenced
- 2005 : 1st 454 Life Sciences Next Generation Sequencing system : GS 20 System 2006 : 1st Solexa Next Generation Sequencer : Genome Analyzer
- 2007 : 1st Applied Biosystems Next Generation Sequencer : SOLiD
- Evolution of sequencing technologies
- 2009 : 1st Helicos single molecule sequencer : Helicos Genetic Analyser System
- 2011 : 1st Ion Torrent Next Generation Sequencer : PGM
- 1st Pacific Biosciences single molecule sequencer : PacBio RS Systems
- 2012 : Oxford Nanopore Technologies demonstrates ultra long single molecule reads

What is Next Generation Sequencing (NGS)?

The high-throughput or Massively parallel signature sequencing is the catch-all term used to describe a number of different modern sequencing platforms.

NGS Technologies

These techniques are used by different companies and they have developed their different systems i.e.

- Illumina (Solexa) sequencing
- Roche 454 sequencing
- Ion torrent: Proton / PGM sequencing
- SOLiD sequencing

What these technologies do?

NGS platforms perform massively parallel sequencing, during which millions of fragments of DNA from a single sample are sequenced in unison. It is high-throughput sequencing and it produces thousands or millions of sequences concurrently. We can have more than 500,000 sequencing-by-synthesis operations run in parallel in these machines.

Generations of sequencers

- **First generation**

  (Sanger sequencing)

- **Second/Next generation**

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(454, Illumina, SOLiD)

- **Third generation/ Next Next Generation**

  (Helicose, PacBio, Ion Torrent, Oxford Nanopore)

Different sequencers used are given in figure below. Illumina is oftenly used.

Advantage

These recent technologies allow us to sequence DNA and RNA much more quickly and cheaply than the previously used Sanger sequencing

Disadvantage

Read sizes are small.

General NGS workflow

Conclusions

NGS technologies sequence millions of segments in parallel

Sequence DNA much more quickly and cheaply than the previously used Sanger sequencing

Resources

1. http://www.ebi.ac.uk/training/online/course/ebi-next-generation-sequencing-practical-course
2. http://technology.illumina.com/technology/next-generation-sequencing.ilmn
3. http://www.ebi.ac.uk/training/online/course/ebi-next-generation-sequencing-practical-course/what-next-generation-dna-sequencing/ion-torre
4. http://www.nature.com/subjects/next-generation-sequencing#research-and-reviews
5. https://www.lifetechnologies.com/pk/en/home/life-science/sequencing/next-generation-sequencing.html

Lesson 84 : 84_NGS Platforms

NGS Platforms

NGS Technologies

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- ROCHE, 454
- ILLUMINA
- Applied Biosystems
- Ion torrent

454:

Founded june 2000 by jonatha M. Rothberg

Subsidiary of Roche Since 2007

No more support from Roche by 2016

Brandford Connecticut mu454.com

Lesson 85 : 85_454 Pyrosequencing

454 Pyrosequencing

Sequencing in 454 machines is performed by pyrosequencing

Principle

• Based on the generation of light signal through release of pyrophosphate (PPi) on nucleotide addition **DNAn+ dNTP ◊DNAn+1 + PPi**

• PPi is used to generate ATP from adenosine phosphosulfate (APS)

• APS + PPI ◊ATP

• ATP and luciferase generate light by conversion of luciferin to oxyluciferin

Only one of the four nucleotides will generate a light signal that will be recorded on a pyrogram

Loading

• After amplification, the emulsion shell is broken and the clonally amplified beads are ready for loading onto the fibre-optic PicoTiterDevice for sequencing

Sequencing

• The PicoTiterPlate is loaded with one fragment carrying bead per well and smaller beads with the necessary enzymes

• Each well is just big enough to hold a single bead

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Sequencing

• The plate is repeatedly washed with each of the four dNTPs, plus other necessary reagents, in a repeating cycle

• The plate is coupled to a fiber optic chip

CCD camera records light flashes from each well

Conclusions

• Sequencing in 454 machines is performed by pyrosequencing

Lesson 86 : 86_Illumina sequencing

Illumina Sequencing

• After the amplification step, a flow cell with more than 40 million clusters is produced

• Each cluster with ~1000 clonal copies of a single template molecule

Sequencing

Illumina uses reversible terminators with flourophores

• All four labelled terminator nucleotides, primers and polymerase flow through each lane

• Base is incorporated • Fluorescent signal

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• Imaging

• Remove flourophore and block• Repeat

Conclusions

• Illumina performs sequencing by synthesis (SBS) using reversible terminators with flourophores

Lesson 87 : 87_Ion Semiconductor

ION SEMICONDUCTOR SEQUENCING

NGS issues

In NGS machines there is phasing, dephasing, addition and removal of nucloeotides. So, in all this a lot of chemicals are being used. In addition to that PCR, where single fragments are taken and amplified also have some issues which are related with itself. So there are some technologies which are not purely NGS or 2nd generation sequencers, they are considered in between 2nd and 3rd generation so 3rd gen is where the amplification or pcr have been removed so there is just single molecule and there is no amplification and that those molecules are used for sequencing.

These 2 technologies are classified here are:

Ion torrent (we still place them in NGS)

Helicose that is will take that in the 3rd gen seq but tbh these in between 2 and 3rd gen seq.

Ion torrent is non optical sequencing. There is no picture taking, no camera movement, so the time is saved. So that but still its using emulsion pcr that is a property of NGS.

Helicose genetic analysis system uses single molecule seqeuncing. But at the same time its kind of using approach which is similar to illumina.

In ion smeiconductor sequencing or in ion torrent, we see there are some special shapes which design these chips look like this where samples are loaded and these are specially designed silica chips.

sample is added here.

There are 3 types of semiconductor chips i.e;

314- 20Mb data

316- 200Mb data

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318- 1Gb data

If we see a cross section as above we can see micro machine wells where the samples are loaded and we do have an ion sensor layer down there and then we have another layer of proprietary ion sensor.

So the beads are loaded into the wells of these sensor and the bead contain those amplified single molecules which are of same type these molecules are then exposed to nucloeotide and whenever a nucleotide is flown and incorporated, it release a hydrogen ion, then that ion goes through different senstivity layers and then detected.

So here we see a polymerase coming and nucleotide is being incorporated, so then there is a release of hydrogen ion and signal is shown below.

If there is no hydrogen ion generated and nucleotide went there, but it did not get incorporated, so there is no hydrogen ion so it means this nucleotide was not there.

In a situation where two hydrogen ions come that means two same nucleotides were incorporated, so the signal strength is stronger and double peaks will appear as shown in abve figure.

So, ion semiconductor sequencing is known to be fastest and we save time while using it. The base read lengths are between 200-250bp. It works on a change in pH by H ion released after incorporation of a nucleotide in a growing chain of DNA.

Lesson 88 : 88_3rdGenSequencing

3RD GENRETION SEQUENCING

Introduction

-
  - Single molecule sequencing
  - No PCR biases
  - Less input samples and reagents (no washing)
  - Longer reads
  - Better quality
  - Higher throughput

3GS Plateforms

-
  - Helicos
  - Pacific Biosciences
  - Oxford nanopore

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Helicos

- A publically traded coy till June, 2012
- Cambridge Massachusetts
- Co founded by stanley Lapidus, Stephen Quake and Noubar Afeyan
- Delisted from NASDAQ in 2010

Pacific Biosciences

- Pacbio RS
- Founded 2004
- Menlo Park, CA
- **SMRT S**ingle **M**olecule **R**eal **T**ime **S**equencing

Oxford Nanopore

- Nanopore sequencing
- Spin out from Oxford University UK
- Since 2005
- had raised over £145 million in investment
- Products are under testing and evaluation phase

Advantages of single molecule sequencing

- Less sample preparation (no PCR)
- No amplification
- no PCR errors
- fewer contamination issues
- no GC-bias
- analyze every sample (unPCRable / unclonable)
- analyze low quality DNA (museum, archeological, forensics samples)
- Absolute quantification
- Sequence RNA directly

Lesson 89 : 89_Helicos

Helicose sequencing

- Images the extension of individual DNA molecules using a defined primer and individual fluorescently labeled nucleotides, which contain a "Virtual Terminator" preventing incorporation of multiple nucleotides per cycle

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- The "Virtual Terminator" technology was developed by Dr. Suhaib Siddiqi

Conclusions

-
- Helicos was an effort towards single molecule sequencing using illumina like approach
- It was an important step towards third generation sequencing

Lesson 90 : 90_PacBioSequencing

Pacbio sequencing

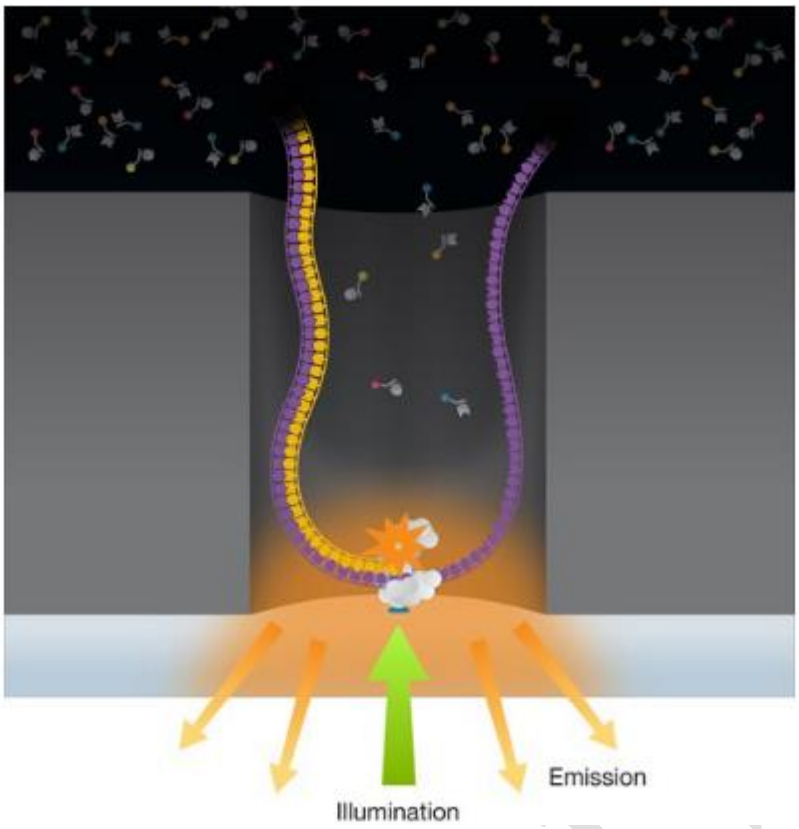
SMRT sequencing

-
- Single Molecule Real Time Sequencing
- utilizes the zero-mode waveguide (ZMW), developed in the laboratories of Harold G. Craighead and Watt W. Webb at Cornell University

SMRT sequencing

-
- ZMW guides light energy into a volume that is small in all dimensions compared to the wavelength of the light
- Nanophotonic structure, a circular hole in an aluminum film on a silica substrate
- With an active polymerase immobilized at the bottom of each ZMW, nucleotides diffuse into the ZMW chamber.
- A, C, G and T are labeled with a different fluorescent dye having a distinct emission spectrum
- Nucleotides held by the polymerase prior to incorporation emit an extended signal that identifies the base being incorporated

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- Single Molecule Real Time Sequencing
- utilizes the zero-mode waveguide (ZMW), developed in the laboratories of Harold G. Craighead and Watt W. Webb at Cornell University

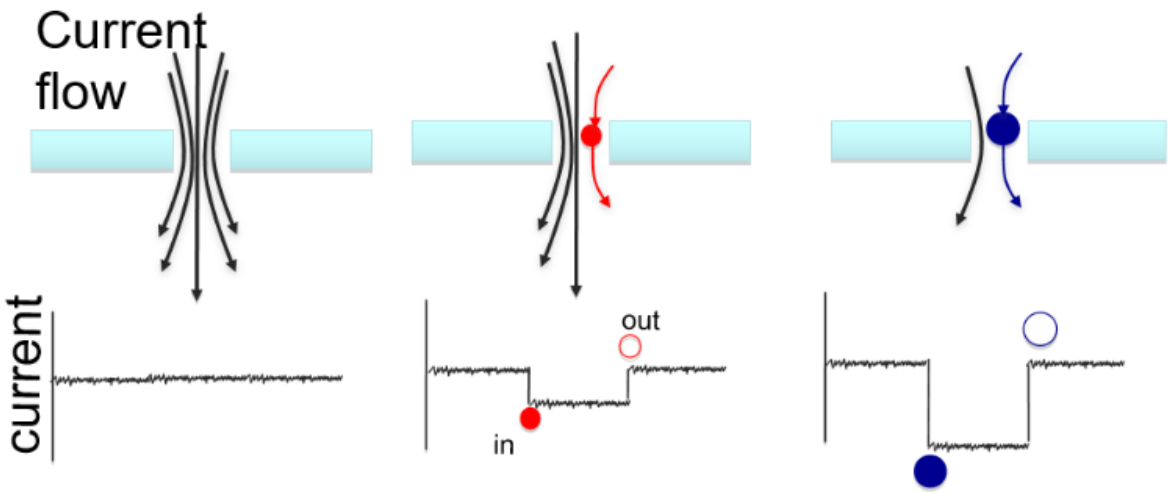
Lesson 91 : 91_OxfordNanopore

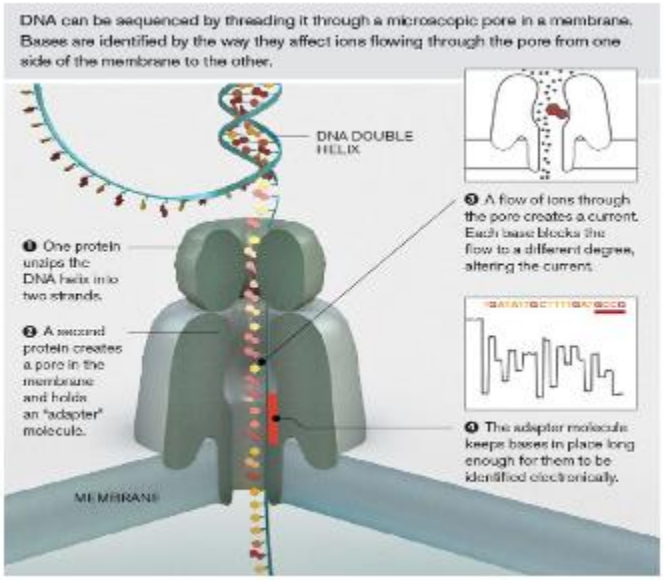
Oxford Nanopore sequencing

Nanopore sequencing

-
  - Latest sequencing technology in development
  - Size of USB drive
  - May drive the next revolution in genomics
  - Whole genome sequencing in 15 minutes for less than $1,000
  - Expected to be available by the end of this year

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Conclusions

-
  - Based on sequencing via change in electric pulse caused by interaction of ions with nucleotides
  - Expected to be available by the end of this year

Lesson 92 : 92_Comparison of NGS Methods

Comparison of NGS Methods

Which Plate-form to use?

It depends on:

-
  - Biological question
  - Budget
  - Analysis team's expertise

Method	Sanger	454	Illumina	SOLiD
Read length	400-900 bp	400-700 bp	100-250 bp	75 bp
Accuracy	99.9%	99.9%	98%	99.9%
Reads per run	N/A	0.1-1 million	up to 3 billion	1.2 to 1.4 billion
Time per run	20 min - 3 hours	10-24 hours	1- 10 days	1 - 2 weeks
Cost/Mb	$2400	$10	$0.05 to $0.15	$0.13
Advantages	Long reads.	Long read size. Fast.	Potential for high sequence yield,	Low cost per base.
Drawbacks	impractical for larger sequences	expensive. Homopolymer errors.	Equipment expensive.	Slower than other methods.
Instrument Cost	3730XL $95K	$500 K	128K-MiSeq 650K-HiSeq	$495K

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Method	Sanger	Illumina	Helicose	PacBio
Read length	400-900 bp	100-250 bp	35 bp	3000 bp
Accuracy	99.9%	98%	99%	99%
Reads per run	N/A	up to 3 billion	1 billion	1 billion
Time per run	20 min - 3 hours	1- 10 days	8 days	30 min
Cost/Mb	$2400	$0.05 to $0.15	2$?	?
Advantages	Long reads.	Potential for high yeild	Long reads	Longest reads
Drawbacks	impractical for larger sequences	Equipment expensive.	Equipment expensive	Yield vs Quality
Instrument Cost	3730XL $95K	128K-MiSeq 650K-HiSeq		$695K

Conclusions

Different NGS Plate-forms have their own strengths and weaknesses, which one to choose from depends on

-
  - Biological question
  - Budget
  - Analysis team's expertise

Lesson 93 : 93_Defining Genes

Defining Genes

Finding genes

After assembly, we have a draft genome in the shape of completely or partially assembled [pseudo]chromosomes or contigs

Next comes the big question, how many genes are there?

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Issues

Defining a gene is problematic also because small genes can be difficult to detect, one gene can code for several protein products

some genes code only for RNA, two genes can overlap, and there are many other complications

GENE is a piece of DNA, a discrete unit of genetic information, which encodes RNA or protein (polypeptide) molecule performing some function in the cell (alone or together with other RNAs and/or proteins)

Historical viewpoint

A tangled knot;

One gene – one function

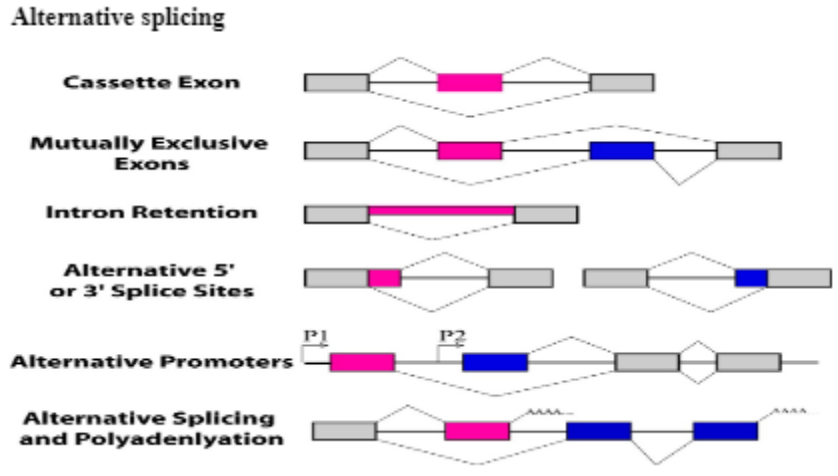
One gene – one enzyme

One gene – one polypeptide

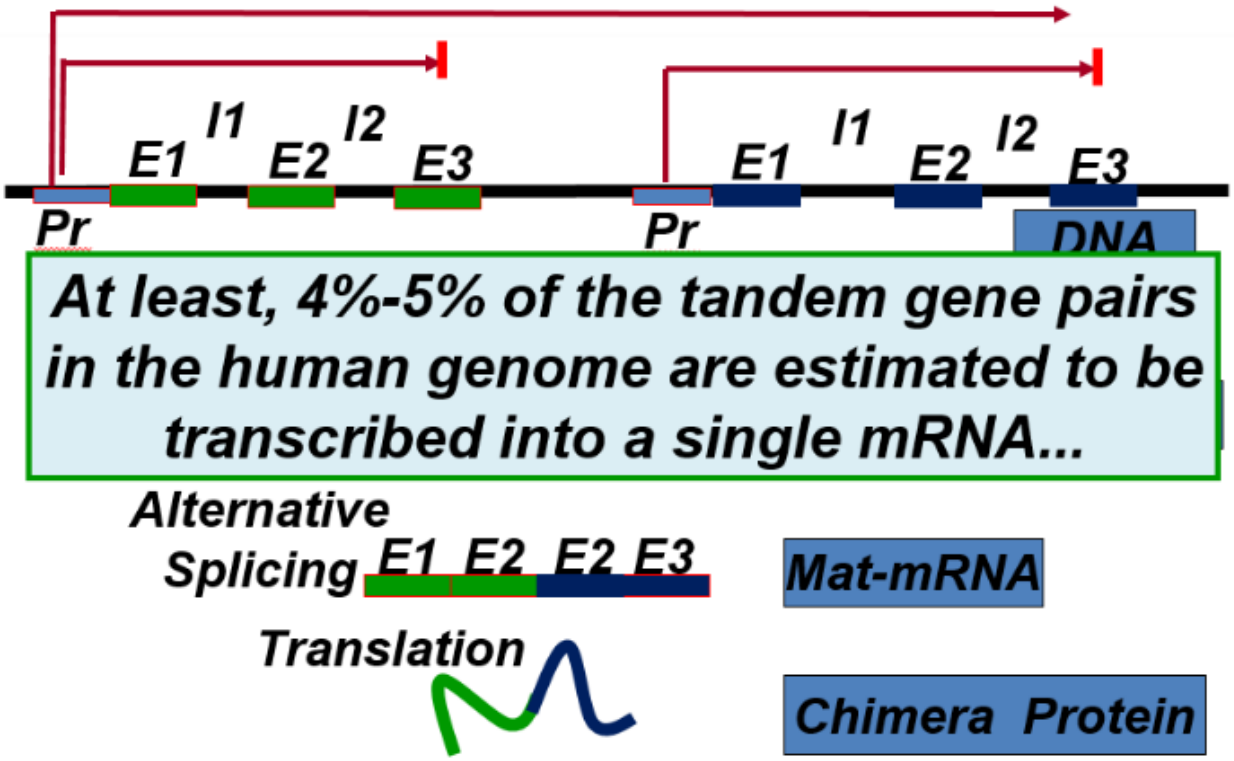
One gene – A number of polypeptides

Number of genes – one polypeptide

Alternative splicing



Muhammad Saeed



Conclusions

Gene is a DNA sequence that's transcribed to produce one or more

functional product(s)

Defining gene is a difficult task due to issues with complicated gene structure and function

Lesson 94 : 94_Finding Genes

Finding Genes

Approaches

Predictions are derived from different computational methods

Two famous approaches;

***"Ab initio"* gene finding**

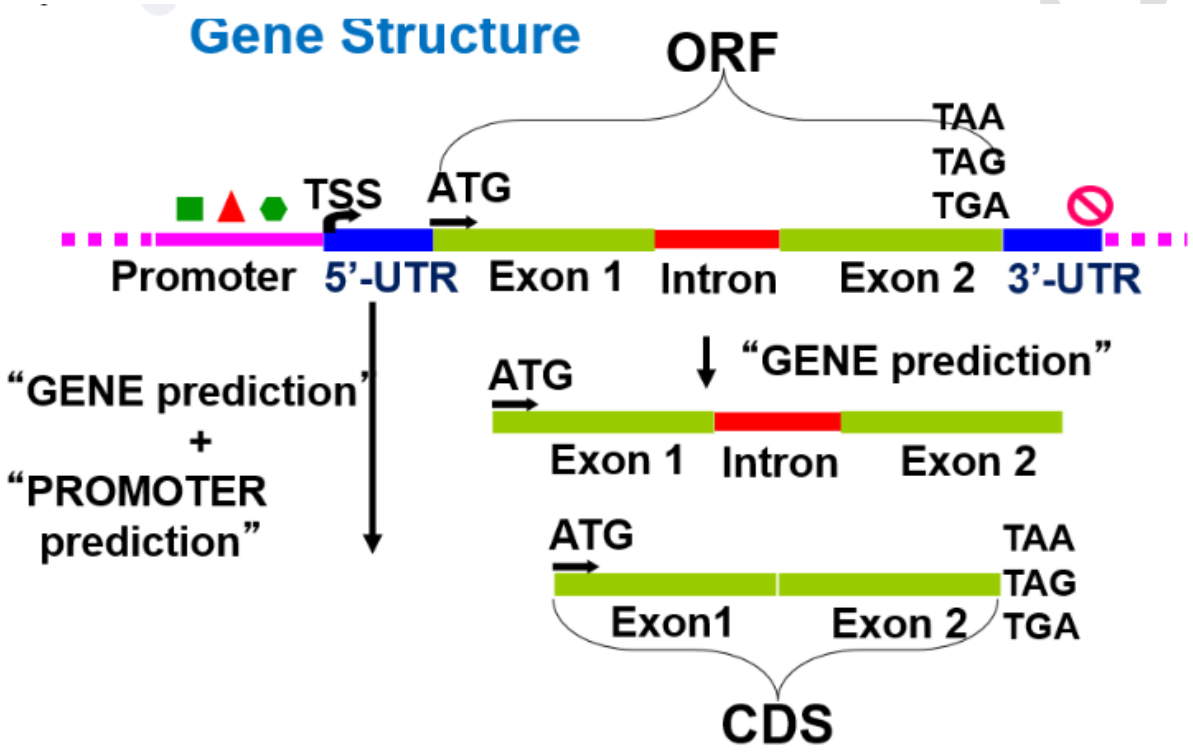
Comparative Approach

Ahmad

***"ab initio"* gene finding**

Detect genes by looking for distinct patterns that define where a gene begins and ends

- *ab initio* gene finding tends to overestimate gene numbers by counting any segment that looks like a gene
- ***Comparative* gene finding**
- Look for genes by comparing segments of sequence with those of known genes
- Tends to underestimate since limited to recognize only genes similar to what have been seen before



Issues

Only about 2-3% of human DNA encodes functional genes

Genes are interspersed among long stretches of non-coding DNA

Repeats of unknown function occupy 40% and more of a genome

Great variation of genes' lengths

We have to distinguish pseudo-genes and true gene duplication(s)

We have to take into attention exon-intron structure of genes in the most of known eukaryotic genomes

99% of yeast genes are intronless

Non-consensus splice sites

other than **GT-----AG**

Where is the true first 5' exon?

cDNA data is incomplete and confusing

Alternative splicing

TIC gene

Alternative Promoters

Conclusions

There are two famous approaches for gene predictions

***"Ab initio"* gene finding**

Comparative Approach

Finding eukaryotic genes is complicated

Lesson 95 : 95_Gene Prediction

Gene Prediction

Gene prediction Involves:

-
  - Prediction of coding regions
  - Prediction of translation starts of gene
  - Prediction of splice junctions

(eukaryotic genomes)

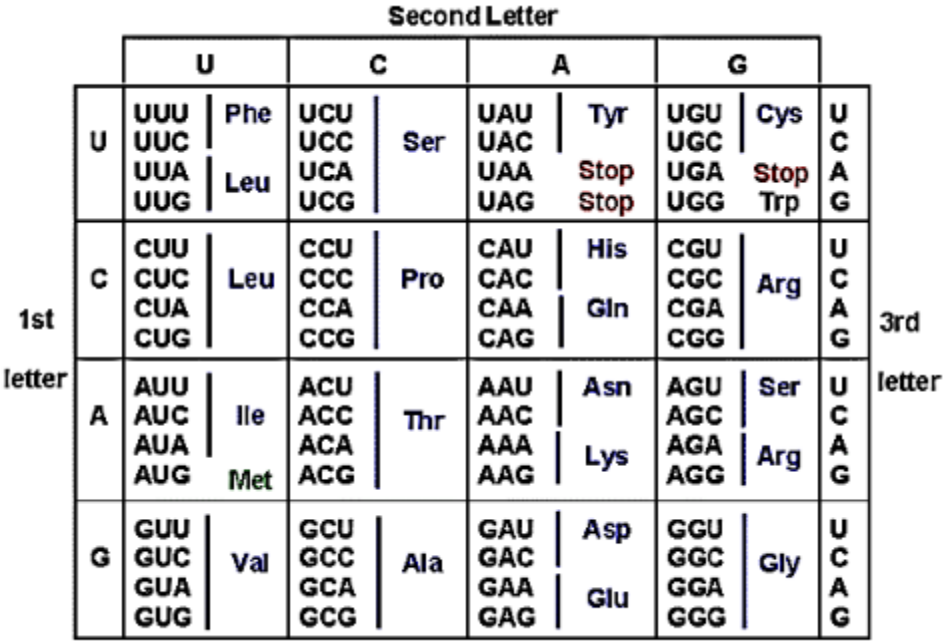
Coding Sequence or CDS

Ahmad Saeed

(**coding DNA**

sequence)

-
  - The portion of a gene's DNA or RNA, composed of exons, that codes for protein
  - The region is bounded nearer 5' end by a start codon and nearer 3' end by stop codon
- Any full mRNA sequence (obtained from cDNA sequencing) will have a full coding sequence
- **ORF (Open Reading Frame)**
- It is the part of gene that has a potential to code for proteins comes in triples called **codons**, beginning and ending with a unique translation **start** (ATG) and one of three **stop** (TAA, TAG, TGA) codons



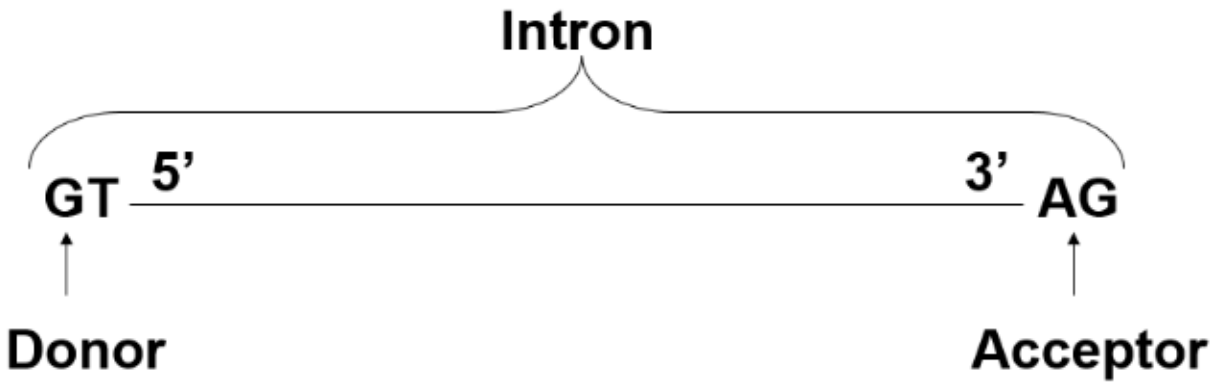
ORF vs CDS

CDS is transcribed and is coding for something

ORF is usually predicted based on DNA sequence and not proven to be transcribed

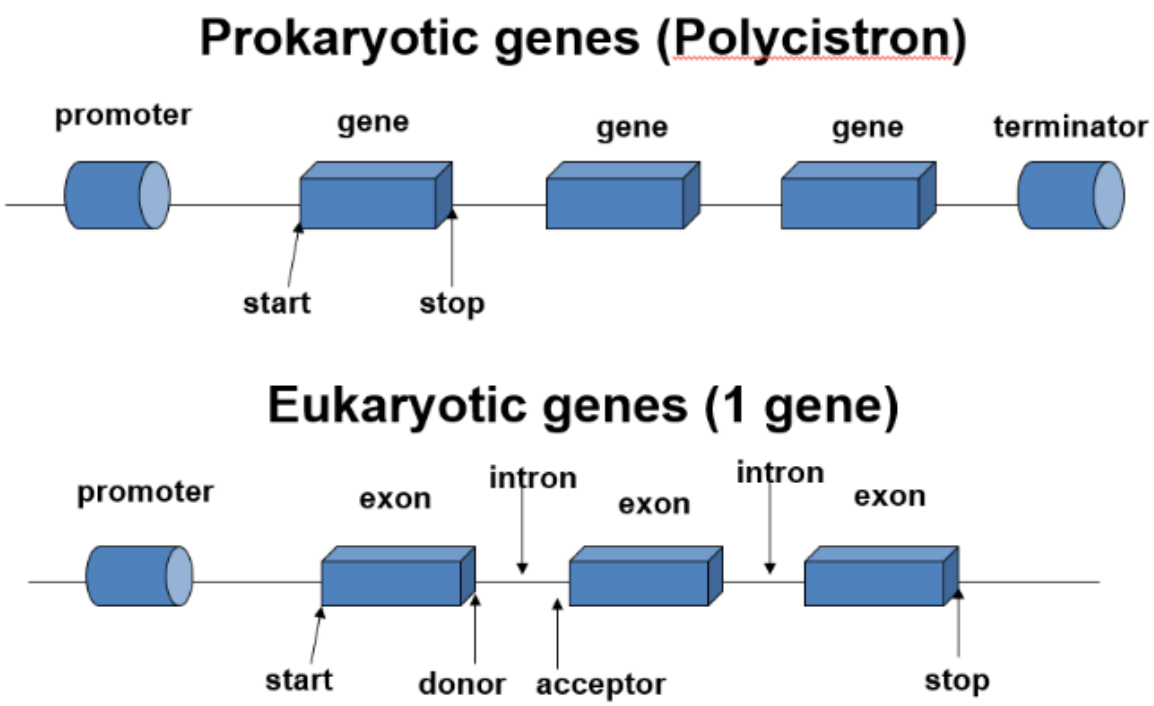
Splice Junction

- Donor site prediction
- Acceptor site prediction



Splice Junction **Intron-Splicing Site-Exon**

GTCTTTTTGAGATAGAAAACAAAAAAGAACTTAT
AGCTCCAAATATCTTTCTTTGGATAACAACTCAA
ATAATATTTGCTTTCTTGTGATTGTAGATGGTTTG
GGAACCGAAGTCAATTTGGACGGGATTGCTTAC
TACAATAATGTTATTAATGCTCTTCTAGAGAAAG
GTATTGAAGCATTCCTAATAAATTTGCTCCTAGTA
TAATTGTGGTTTATGCATAAGAACTTCTCCTGCC
AGGTATTCAGCCTTATGTAACTTTGTACCACTGG
GACCTTCCTTTGCATCTCCAGGAATCAATGAATG
GATGGTTAAATAGGAAGATTGTGTAAGTATTAAA
ATGCCAGGTGTTTAAATATGAAGTTACTGCATAT



Information fusion

Involves;

combining multiple pieces of information for the whole gene prediction:

TSS(s), UTRs Alternative splicing variants, Gene product destination and function(s)

Signal vs Content

A small pattern within the genomic DNA is referred to as a **signal**, whereas a region of genomic DNA is a **content**

Signal

Splice sites, **starts** and **ends** of transcription or translation, **branch points**, **(TBS)** transcription factor binding sites, etc

Content

exons, introns, UTRs, promoter regions

Conclusions

Ahmad

-
    - Gene prediction involves prediction of coding regions, translation start site and splice junction
    - CDS is the actual region of DNA, translated to form proteins while the ORF may contain introns

Lesson 96 : 96_Genes Predictions in prokaryotes

Genes Predictions in Prokaryotes

Genes Predictions

- Locating genes within genomic sequence.
- Defining initiation termination sites of genes.
- Extracting the coding region of each gene.
- Identifying function for coding region.

Gene

- A region of the genome that codes for a functional component such as an RNA or protein

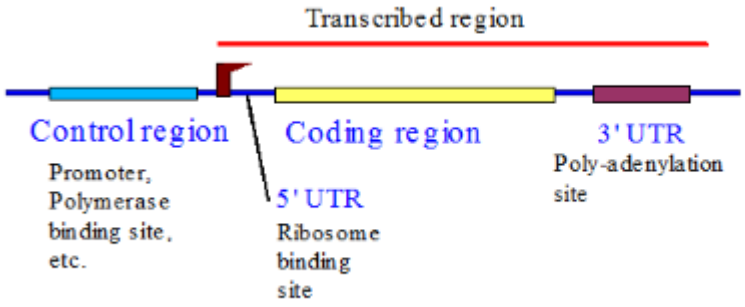
Genes Predictions - Gene

- Identify coding regions computationally
- from raw genomic sequence data

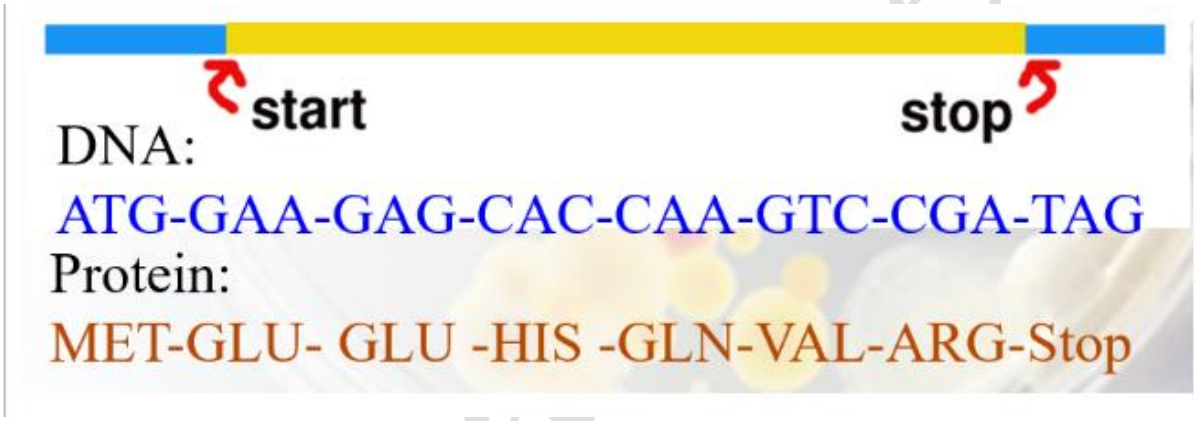
Gene

- Translation utilizes a trinucleotide coding system: codons
- Translation begins at a start codon
- Translation ends at a stop codon
- Most organisms use ATG as a start codon
    - A few bacteria also GTG and TTG
    - Regardless of codon used, the first amino acid in every translated peptide chain is Methionine
- Almost all organisms use TAG, TGA and TAA as stop codons
- The major exception are the mycoplasmas

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In Bacteria and Archaea, the coding region is in one continuous sequence known as an open reading frame (ORF).



- Translation region is contiguous in prokaryotes, gene finding focuses on identifying ORFs

ORF-finder takes a syntactic approach to identifying putative coding regions

Genes Predictions – ORF Finder and Gene Glimmer

- ORF-finder is available from NCBI
  - GLIMMER 2.0 is a more sophisticated program that attempts to model codon usage, average gene length and other features before identifying putative coding regions

Genes Predictions – ORF Finder Approach

- Identifies every stop codon in the genomic sequence.
  - Scan upstream to the farthest, in-frame start codon.
    - Locates ORFs that begin with ATG as well as GTG and TTG

ORF Finder

Advantages	Disadvantages
• Can identify every possible ORF. • Minimum length constraint ensures that many false positives are discarded prior to human review.	• Does not eliminate overlapping ORFs. • Even with a length constraint, there are often many false positives. • Cannot take into account organism-specific idiosyncrasies

Glimmer Approach

Glimmer Approach

-
  - The model includes information about:
  - Average length of coding region
  - Codon usage bias (which codons are preferentially used)
  - Evaluates the frequency of occurrence of higher order combinations of nucleotides

– Glimmer Output

-
  - For each ORF, GLIMMER assigns a likelihood score or probability that the ORF resembles a known gene.
  - High scoring ORFs that overlap significantly with other high scoring ORFs are reported but highlighted.

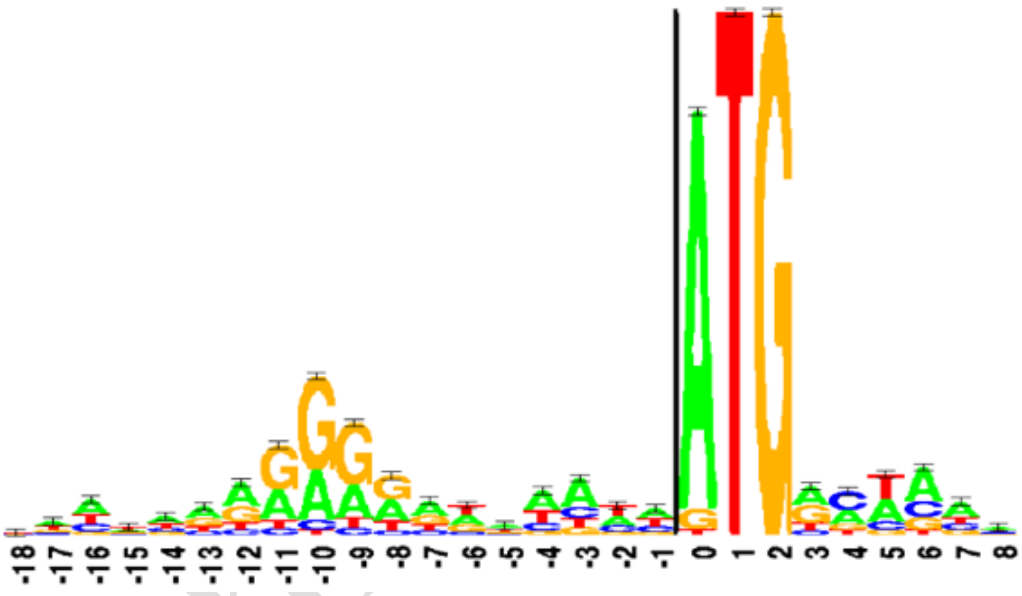
GLIMMER 2.0 is reported to be 98% accurate on prokaryotic genomes

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Glimmer

Advantages	Disadvantages
• Fewer false positives because ORFs are evaluated for likelihood of coding. • Organism-specific because model is built on known genes. • User can modify many parameters during search phase.	• Requires approximately 500+ known genes for proper training. • Genuine coding regions with unusual codon composition will be eliminated. • Reported accuracy difficult to reproduce.

Genes Predictions – RBS



Genes Predictions - Gene

- ORF-finder
- Glimmer

Lesson 97 : 97_Genes Predictions in Eukaryotes

Genes Predictions in Eukaryotes

Genes Predictions - Eukaryotes

- Tools for finding genes in eukaryotes
- Genie
- Fgenes
- Genscan

Ahma

Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Genie

- The model includes information about:
- Average length of exons and introns.
- Compositional information about exons and introns.
- A neural-net derived model of splice junctions and consensus sequences around splice junctions.
- Splice junction information can be further improved by including results of homology searches.

Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Genie

- Genie is approximately 60-75% accurate on eukaryotic genomes.

Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Genie

Actual Gene Structure



Initial Prediction by Genie



Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Genie

Sequence Homology Alignments



Corrected Prediction



Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Genie

Advantages

- Extra predicted exons can be eliminated based on evidence from homology searches.
- Likelihood scores provided for each predicted exon.

Disadvantages

- No organism-specific training is possible.
- Works best on mammalian genomes, not other eukaryotes.
- Reliance on homology evidence can result in oversight of novel genes unique to the organism of interest.

Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Fgenes

- Identifies putative exons and introns.
- Scores each exon and intron based on composition.
- Uses dynamic programming to find the highest scoring path through these exons and introns.
- The best-scoring path is constrained by several factors including that exons must be in frame with each other and ordered sequentially.

Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Fgenes

- Identifies putative exons and introns.
- Scores each exon and intron based on composition.
- Uses dynamic programming to find the highest scoring path through these exons and introns.
- The best-scoring path is constrained by several factors including that exons must be in frame with each other and ordered sequentially.

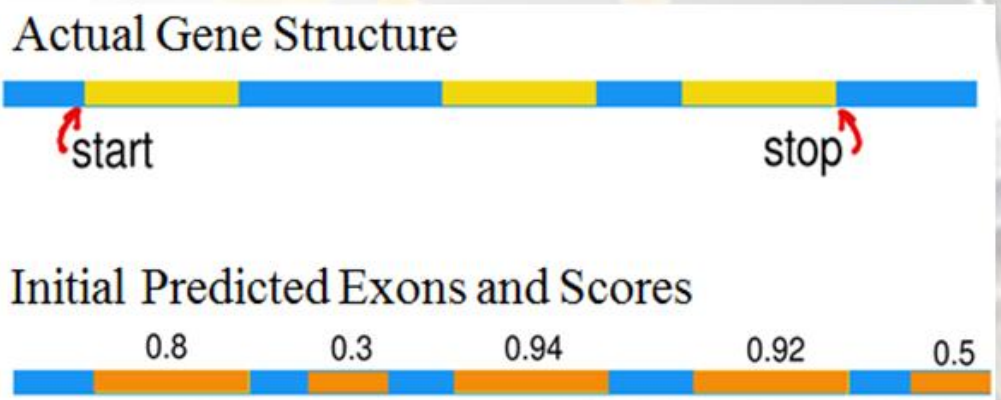
Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Fgenes

- Fgenes is about 70% accurate in most mammalian genomes.

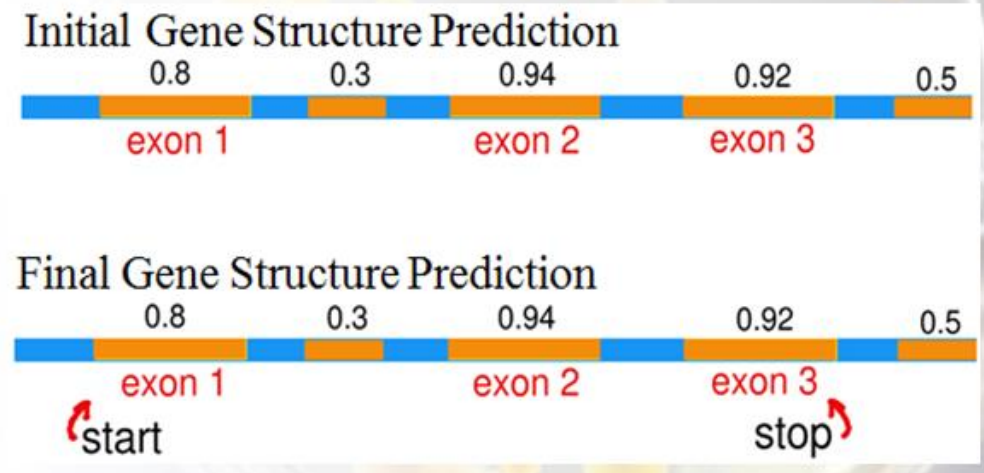
Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Fgenes



Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Fgenes



Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Fgenes

Advantages	Disadvantages
• Alternative gene structures are reported. • Also attempts to identify putative promoter and poly-A sites.	• User cannot train models. • Only human model-based version is available for unrestricted public use.

Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Genscan

- Models for different states (GHMMs)
- State 1 and 2: Exons and Introns
  - Length
  - Composition
- State 3: Splice junctions

Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes - Genscan

- Weight matrix based array to identify consensus sequences
- Weight matrix to identify promoters, poly-A signals and other features.

Ahmad

Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes – Genscan Output

- Gene structure
- Promoter site
- Translation initiation exon
- Internal exons
- Terminal exon (translation termination)

Ahmad

Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes – Genscan

Advantages

- Most accurate of available tools.
- Excellent at identifying internal and terminal exons
- Provides some assistance in identifying putative promoters

Disadvantages

- User cannot train models nor weak parameters.
- Identification of initial exons is weaker than other kinds of exons.
- Promoter identification can be mis-leading.

Genes Predictions in Eukaryotes

Genes Predictions Eukaryotes – Genscan

- Eukaryotic gene structures can be quite complex.
- The best approaches to gene finding in eukaryotes combine different methods.

Lesson 98 : 98_GO functional analysis

- **GO Functional Analysis**

Introduction

After we get a list of genes, it is important to identify interesting Biological patterns

These functions are represented and classified as GO (Gene Ontology) terms

- **GO Functional Analysis**

GO Terms

GO (Gene Ontology) terms are implemented as Directed Acyclic Graphs (DAG)

Three main types

Biological Process (BP) Cellular Components (CC)

Molecule Function (MF)

- **GO Functional Analysis**
- **GO Functional Analysis**

DAVID Tool

The **D**atabase for **A**nnotation, **V**isualization and **I**ntegrated **D**iscovery (**DAVID**) (Huang et al., 2009).

http://david.abcc.ncifcrf.gov/

- **GO Functional Analysis**

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DAVID Tool

David provides a comprehensive set of functional annotation tools to understand biological meaning behind large list of genes.

http://david.abcc.ncifcrf.gov/

- **BP-FAT 954 genes, 98 terms**
- **CC-FAT 954 genes, 15 terms**
- **MF-FAT 954 genes, 62 terms**
- **Associated Pathways**
- **Functional Clustering**
- **Functional Clustering**
- **GO Functional Analysis**

Conclusions

After we get a list of genes, it is important to identify interesting Biological patterns

These functions are represented and classified as GO (Gene Ontology) terms

- **References**

- Trevino, V., Falciani, F., and Barrera-Saldana, H.A. 2007. DNA microarrays: a powerful genomic tool for biomedical and clinical research. Mol Med 13: 527-541.
- Kato, H., Saito, K., and Kimura, T. 2005. A perspective on DNA microarray technology in food and nutritional science. Curr Opin Clin Nutr Metab Care 8: 516-522.

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Lesson 99 : 99_Gene Networks

- **Gene Network**

Introduction

A network is a topology that connects its components based upon some relationship with each other.

- Networks are implemented as graphs.

.

- **Gene Network**

Graphs

Graphs are abstract representation of a set of data objects which are somehow linked with each other.

- Directed graphs have an orientation or order as opposed to undirected graphs
- **Gene Network**

Graph Types

- **Gene Network**

Gene Regulatory Networks

- Genetic Regulatory Networks (GRNs) are the networks of genes based upon their regulatory interactions with each other
- Generally implemented as Directed Graphs
- Two Genes are connected if the expression of one gene modulates the effect of other gene by either activation or inhibition
- **Gene Network**

Modelling Gene Networks

- Correlation (coexpression) Networks
- Bayesian Networks
- Differential Equations
- Boolian Networks
- Gaussian Models
- **Gene Network**

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Correlation Networks (G4 vs G5)

- **Gene Network**

Correlation Networks (G4 vs G5)

- **Gene Network**

Correlation Networks (G1 vs G3)

- **Gene Network**

Correlation Networks (G1 vs G3)

- **Gene Network**

Correlation Networks

- **Gene Network**

Correlation Network as heat Map

- **Gene Network**

Correlation Network as heat Map

- **Gene Network**

Conclusions

Gene networks are abstract representation of relationships of genes based on gene to gene correlations

Genes are nodes while correlations are presented as edges

- **References**

- Trevino, V., Falciani, F., and Barrera-Saldana, H.A. 2007. DNA microarrays: a powerful genomic tool for biomedical and clinical research. Mol Med 13: 527-541.
- Kato, H., Saito, K., and Kimura, T. 2005. A perspective on DNA microarray technology in food and nutritional science. Curr Opin Clin Nutr Metab Care 8: 516-522.

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Lesson 100 : 100_ORF Prediction

ORF Prediction

ORF (Open Reading Frame)

Gene finding, specially in prokaryotes starts form searching for an open reading frames (ORF)

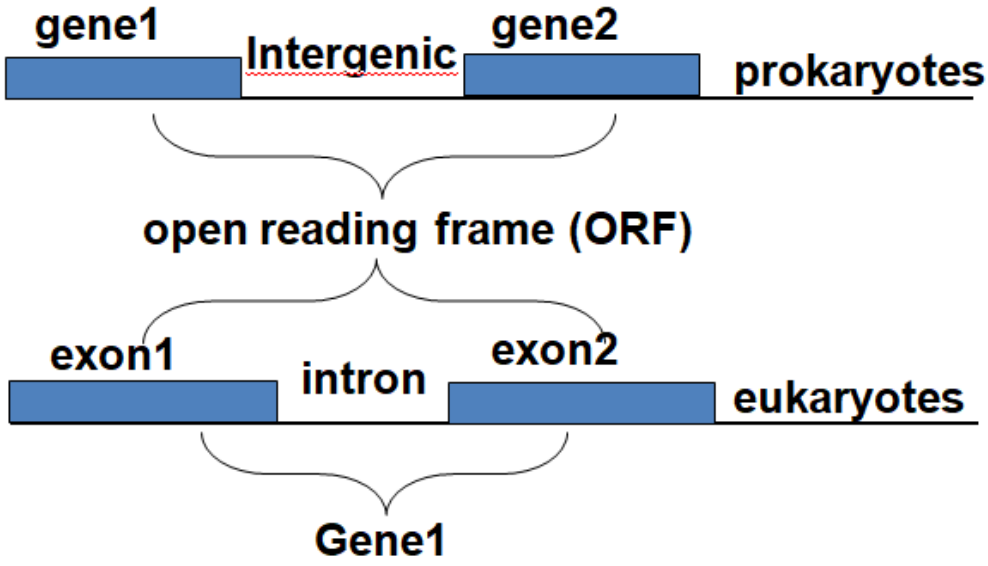
ORF Prediction

ORF (Open Reading Frame)

- An ORF is a sequence of DNA that starts with start codon “ATG” (not always) and ends with any of the three termination codons (TAA, TAG, TGA)

ORF Prediction

Coding regions: exons, genes



Ahmad

ORF Prediction

ORF and gene finding

ORF provide an important evidence in gene finding

- Generally longer ORFs are preferred
- However presence of ORF not necessarily means the region is translated to a functional product

Ahmad

ORF Prediction

ORF and gene finding

ORF provide an important evidence in gene finding

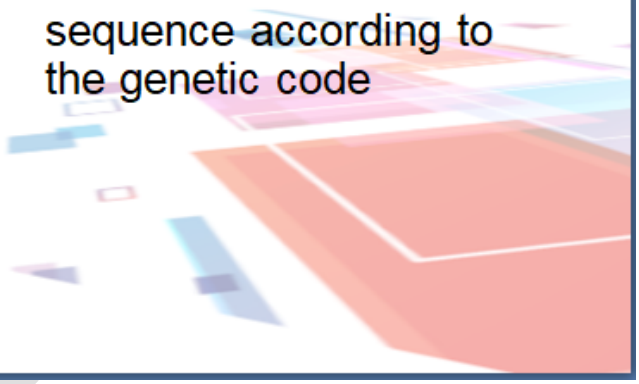
- Generally longer ORFs are preferred
- However presence of ORF not necessarily means the region is translated to a functional product

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ORF Prediction

Reading Frames

Depending on the start point, there are six possible ways of translating any nucleotide sequence into amino acid sequence according to the genetic code



ORF Prediction

Reading Frames

- Three on forward strand and three on complementary strand

ORF Finders

ORF Finding

Long ORF may be a gene

- Expected 64/3 ~ 21 codons before we see a stop codon
- Genes are longer than this
- We might scan for ORF longer than a threshold

ORF Finders

Codon usage and likelihood ratio

An ORF is more 'reliable' if it has 'likely' codons

- We can do sliding window calculations to ORF having 'likely' codon usage

ORF Finders

Codon usage and likelihood ratio

An ORF is more 'reliable' if it has 'likely' codons

- However average vertebrate exon length (130 nucleotides) is too small for reliable peaks

ORF Finders

Codon usage and likelihood ratio

An improvement may be;

- In-Frame hexamer count

i.e frequencies of pairs of consecutive codons

ORF Finders

ORF Finders

Tools are mainly based on pattern finding algorithms

- **NCBI's ORF Finder**
- **ORF Investigator**
- **OrfPredictor**

ORF Finders

Conclusions

ORF provide an important evidence in gene finding

- Generally longer ORFs are preferred
- However presence of ORF not necessarily means the region is translated to a functional product

Lesson 102 : 102_TSS Prediction

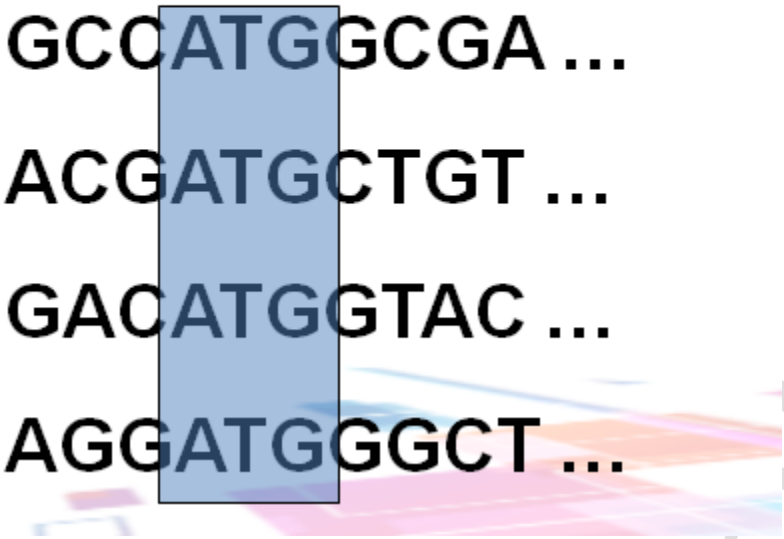
TSS Prediction

Translation Start Site (TSS)

Translation starts with ATG that codes for methionine in a polypeptide

Ahmad

Translation Start Site (TSS)



Assumption

Certain nucleotides prefer to be around TSS than others

- The "biased" nucleotide distribution is information is a basis for translation start prediction

Coding Potential

Hexamer frequencies in coding versus non-coding regions may provide important insights Frequency of X(A,G,C,T) at position i is

$$F_i(X) = \Sigma \log(C_i(X)/N_i(X))$$

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Frequency table (Biased Nucleotide distribution)

ATG

A	17.36,	19.01,	17.36,	48.76,	28.93,	15.70,	21.49,	23.14,	19.83,	21.49,	25.62,	15.70,
C	16.53,	28.93	57.85,	5.79,	39.67,	50.41,	22.31,	38.84,	23.97,	27.27,	31.40,	38.02,
G	46.28,	29.75,	19.01,	42.98,	14.88,	26.45,	42.98,	23.97,	33.88,	32.23,	25.62,	29.75,
T	19.83,	22.31,	5.79,	2.48,	16.53,	7.44,	13.22,	14.05,	22.31,	19.01,	17.36,	16.53}

-4 -3 -2 -1 ATG +3 +4 +5 +6



Example

Which one is more probable to be a Translation Start?

CACC ATA GC

TCGA ATG TT

Solution

We can use frequency table and the scoring function as under;

$$S_i = \sum \log (F_i (X)/0.25)$$

We can call it

Information Content (IC)

Algorithm

- Build a mathematical model, based on collected translation start sequence
- For each candidate translation start sequence, apply the model and get a score
- If the score is larger than zero, predict it is a "translation start"; the higher score, the higher the probability the prediction is true

Conclusions

- TSS prediction can be an important step in gene prediction
- TSS can be predicted while using the frequency of neighboring nucleotides

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Prediction of splice junctions

Splice Junctions

Donor site

- Coding region | GT

Acceptor

- YAG | coding region
- **Canonical form**
  - GT-AG: 99.24%

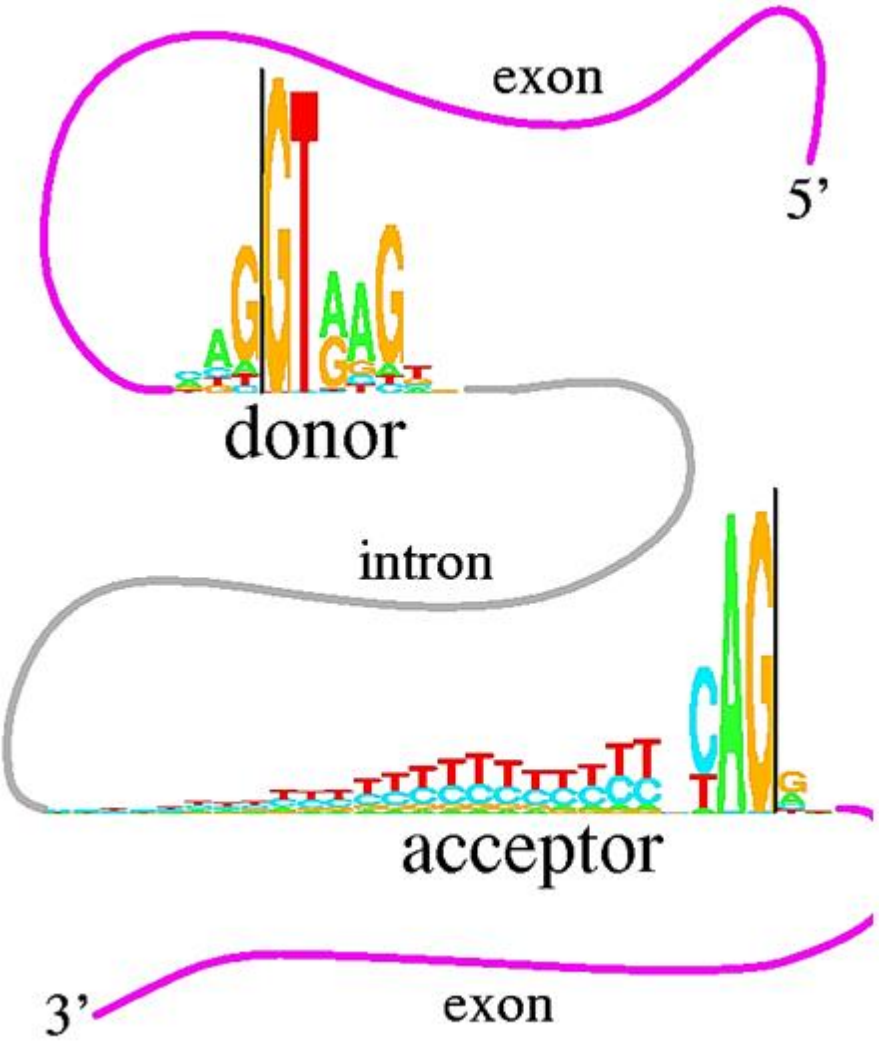
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Splice Junctions

Like TSS, the flanks of splice junctions show "biased" distributions of nucleotides in certain positions

- These biased distributions of nucleotides are the basis for prediction of splice junctions

Ahr Need


exon
5’
donor
intron
acceptor
3’
exon

Sequence LOGOS

A visual representation of a position-specific distribution

Easy for nucleotides, but we need colour to depict up to 20 amino acid proportions.

Sequence LOGOS

Overall height at position is proportional to the information content

Genome Assembly

Outline

- Introduction
- Vocabulary terms
- Overview
- Conclusions

Genome Assembly

The process of reconstruction of original chromosomes based on overlaps between short sequence reads

- It groups reads into **contigs** and Contigs to **Scaffolds**

Vocabulary

- **Contig:** A consensus sequence of DNA that has been assembled from overlapping DNA fragments
- **Scaffolds:** One or more contigs linked together by unknown sequence

Vocabulary

Captured Gap:

A gap within a scaffold

- The order and orientation of the contigs spanning the gap is known

Conclusions

- The process of reconstruction of original chromosomes based on overlaps between short sequence reads
- Reads are assembled into contigs which are then assembled into scaffolds

Lesson 105 : 105_Genome Assembly

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Genome Assembly

The process of reconstruction of original chromosomes based on overlaps between short sequence reads

- It groups reads into **contigs** and Contigs to **Scaffolds**

Vocabulary

- **Contig:** A consensus sequence of DNA that has been assembled from overlapping DNA fragments
- **Scaffolds:** One or more contigs linked together by unknown sequence

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Vocabulary

Captured Gap:

A gap within a scaffold

- The order and orientation of the contigs spanning the gap is known

Saeed

Lesson 106 : 106_Genome Assembly Overview

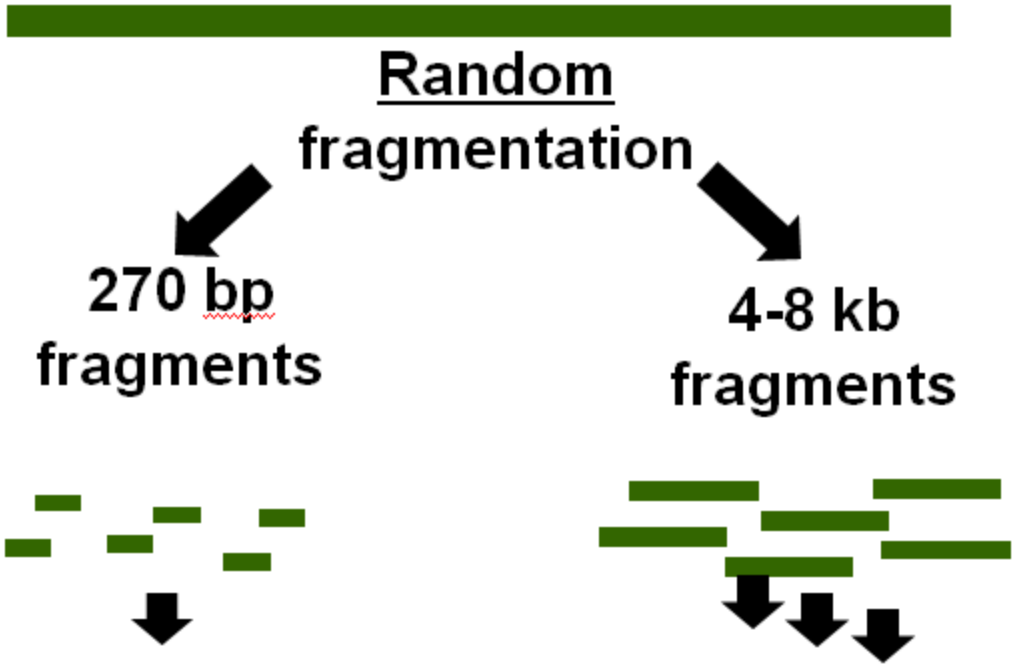
Sample preparation

DNA is collected from the biological sample followed by library preparations



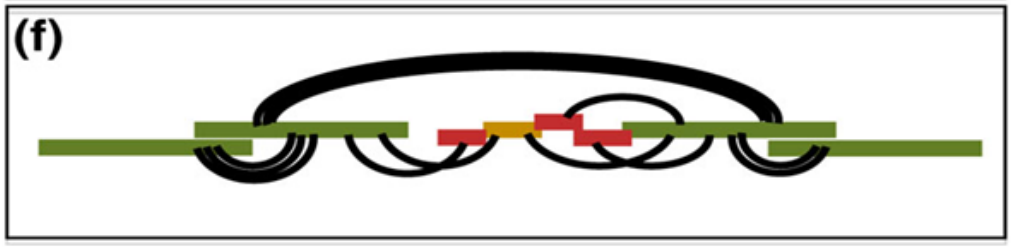
Schatz et al. Genome Biology 2012 13:243

Genomic DNA



Refinement and simplification

Finally, mates, markers and other long-range information are used to order and orient the initial contigs into large scaffolds, as shown as thin black lines connecting the initial contigs



Schatz et al. Genome Biology 2012 13:243

Factors

Following factors may be important in planning an Assembly project;

- **Size of Genome**
- Repeat content
- Related species Assembled
- Strategy for performing Assembly

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Size of the genome

Size of the Genome can be estimated from the ploidy **(N)** of the organism and the DNA content per cell

- Haploid/Monoploid
- Diploid
- Tri/Tetra/-------
- Polyploid

Size of the genome

- This will affect:
  - How many reads will be required to attain sufficient coverage (typically **10x to 100x**)

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Size of the genome

- This will affect:
  - How many reads will be required to attain sufficient coverage (typically **10x to 100x**)

Coverage

The average number of times any given base in the genome is sequenced

It can be derived by dividing the total length of acquired sequences by the genome length

aeed

Coverage

$$C = NL / G$$

N: Total number of reads

L: Length of a Read

G: Genome size

Represented as X

Saeed

Lesson 108 : 108_Effect of repeats

Factors

Following factors may be important in planning an Assembly project;

- Size of Genome
- **Repeat content**
- Related species Assembled
- Strategy for performing Assembly

Normal Sequence



Incorrect Assembly with an Orphan contig (Red)



Salzberg S L , and Yorke J A
Bioinformatics 2005;21:4320-4321

Repeat content

A major proportion of eukaryotic genome is made up of repeats.

Repeats are the most common source of Mis-assemblies

Repeat solution

If sequencing technology produces reads longer than repeat size, impact is much smaller but that's not mostly the choice

- Most common solution is to generate mate pairs with spacing greater than largest known repeat

Ahmad Saeed

Double-barreled Shotgun Assembly

- DNA is randomly sheared into fragments (inserts)
- The ends of each insert are sequenced, resulting in two reads per fragment (double-barreled sequencing)

Lesson 109 : 109_Related Species Assembled

Ahmad Saeed

Factors

Following factors may be important in planning an Assembly project;

- Size of Genome
- Repeat content
- **Related species Assembled**
- Strategies for performing Assembly

Importance

Availability of a related specie's good quality genome having large reliable scaffolds is always a great help

Importance

Related species genomes helps in;

- Guiding the assembly of the target species
- Verifying the completeness of the assembly

Importance

Related species genomes helps in;

- Can themselves be improved in some cases

Caution

- Can cause errors while architectures are different

Ahmad Saeed

Strategies for performing Assembly

Factors

Following factors may be important in planning an Assembly project;

- Size of Genome
- Repeat content
- Related species Assembled
- **Strategies for performing Assembly**

Strategies for performing Assembly

Factors

Following factors may be important in planning an Assembly project;

- Size of Genome
- Repeat content
- Related species Assembled
- **Strategies for performing Assembly**

Strategies for performing Assembly

Conclusions

- Availability of a related specie's good quality genome is always a great help in assembly
- The sequencing approaches and assembly strategies are interdependent

Lesson 110 : 110_Greedy Graph Algo

Greedy Graph Algorithm

Outline

- Introduction
- Steps
- Advantages
- Limitations
- Greedy Graph based Assemblers
- Conclusion

Greedy Graph Algorithm

Introduction

- Greedy Graph algorithms represent the simplest, most intuitive, solution to the assembly problem.

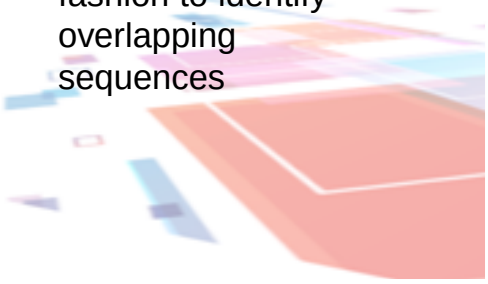


Greedy Graph Algorithm

Steps

Greedy Graph algorithms works as follows:

- Compare all reads or contigs in a pairwise fashion to identify overlapping sequences



Greedy Graph Algorithm

Steps

Greedy Graph algorithms works as follows:

- Merge the sequences that overlap each other the best



Sced

Greedy Graph Algorithm

Steps

- Repeat step 2 until no more sequences can be merged, or the remaining overlaps conflict with existing contigs



Lesson 101 : 101_ORF Finders

ORF Finders

ORF Finding

Long ORF may be a gene

- Expected 64/3 ~ 21 codons before we see a stop codon
- Genes are longer than this
- We might scan for ORF longer than a threshold

Codon usage and likelihood ratio

An ORF is more 'reliable' if it has 'likely' codons

- We can do sliding window calculations to ORF having 'likely' codon usage

Ahmad Saeed

Codon usage and likelihood ratio

An ORF is more 'reliable' if it has 'likely' codons

- However average vertebrate exon length (130 nucleotides) is too small for reliable peaks

Ahmad Saeed

Codon usage and likelihood ratio

An improvement may be;

- In-Frame hexamer count

  i.e frequencies of pairs of consecutive codons

ORF Finders

Tools are mainly based on pattern finding algorithms

- **NCBI's ORF Finder**
- **ORF Investigator**
- **OrfPredictor**

Conclusions

ORF provide an important evidence in gene finding

- Generally longer ORFs are preferred
- However presence of ORF not necessarily means the region is translated to a functional product

Ahmad Saeed

TSS Prediction

Translation Start Site (TSS)

Translation starts with ATG that codes for methionine in a polypeptide

Ahmad

Translation Start Site (TSS)

GCCATGGCGA ...

ACGATGCTGT ...

GACATGGTAC ...

AGGATGGGCT ...

Ahmad Saeed

Assumption

Certain nucleotides prefer to be around TSS than others

- The "biased" nucleotide distribution is information is a basis for translation start prediction

Ahmad Ibrahim

Coding Potential

Hexamer frequencies in coding versus non-coding regions may provide important insights Frequency of X(A,G,C,T) at position i is

$$F_i(X) = \Sigma \log(C_i(X)/N_i(X))$$

Lesson 103 : 103_Splice sites Prediction

Ahmad Ali Khan

Splice Junctions

Donor site

- Coding region | GT

Acceptor

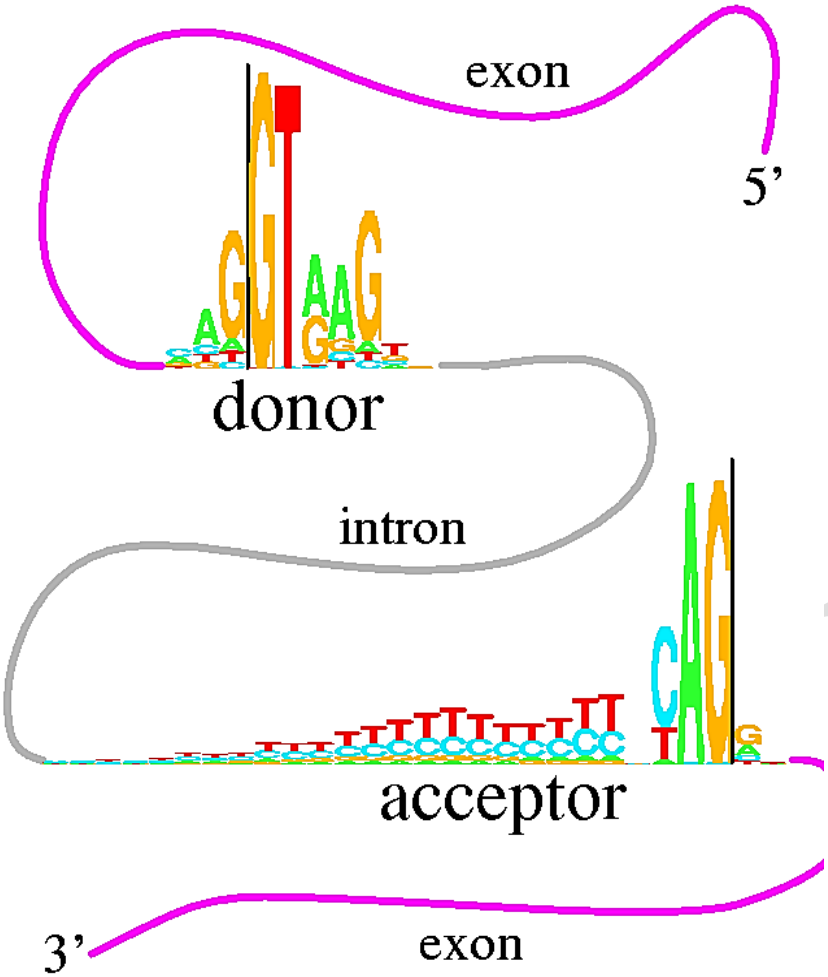
- YAG | coding region
- **Canonical form**
  - GT-AG: 99.24%

Ahmad Naeed

Splice Junctions

Like TSS, the flanks of splice junctions show "biased" distributions of nucleotides in certain positions

- These biased distributions of nucleotides are the basis for prediction of splice junctions



Ahmad Saeed

Sequence LOGOS

A visual representation of a position-specific distribution

Easy for nucleotides, but we need colour to depict up to 20 amino acid proportions.

Ahmad Barclay

Sequence LOGOS

A visual representation of a position-specific distribution

Easy for nucleotides, but we need colour to depict up to 20 amino acid proportions.

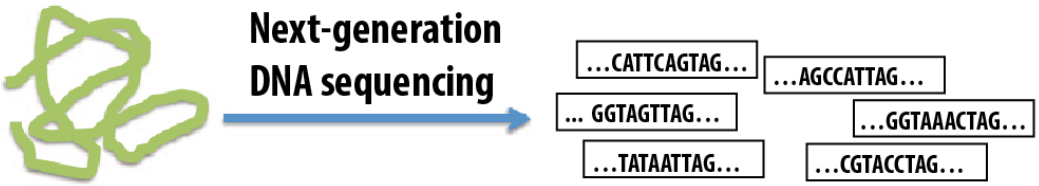
Sequence LOGOS

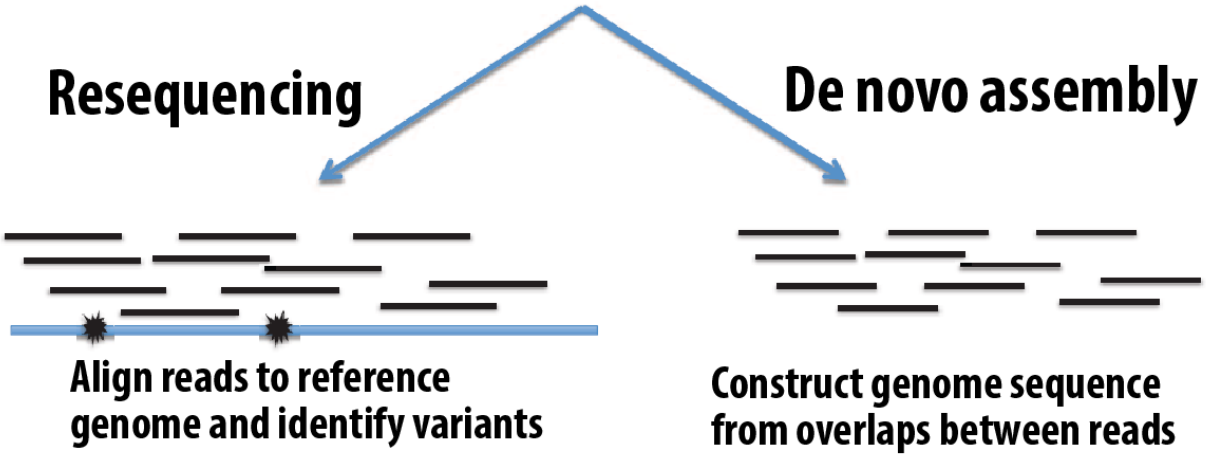
Overall height at position is proportional to the information content

Conclusions

Like TSS, the flanks of splice junctions show "biased" distributions of nucleotides in certain positions

- These biased distributions of nucleotides can be used for prediction of splice junctions





Genome Assembly

The process of reconstruction of original chromosomes based on overlaps between short sequence reads

- It groups reads into contigs and Contigs to **Scaffolds**

Vocabulary

- **Contig:** A consensus sequence of DNA that has been assembled from overlapping DNA fragments
- **Scaffolds:** One or more contigs linked together by unknown sequence

Ahmad Saeed

Vocabulary

Mis-assemblies:

Include regions where a genome is incorrectly re-arranged

- Or the places where large chunks of DNA sequence are simply deleted and the surrounding sequences just crunched together

Ahmad Saeed

Conclusions

- The process of reconstruction of original chromosomes based on overlaps between short sequence reads
- Reads are assembled into contigs which are then assembled into scaffolds

Lesson 105 : 105_Genome Assembly

Ahmad Saeed

Genome Assembly

The process of reconstruction of original chromosomes based on overlaps between short sequence reads

- It groups reads into **contigs** and Contigs to **Scaffolds**

Vocabulary

- **Contig:** A consensus sequence of DNA that has been assembled from overlapping DNA fragments
- **Scaffolds:** One or more contigs linked together by unknown sequence

Saeed

Vocabulary

Captured Gap:

A gap within a scaffold

- The order and orientation of the contigs spanning the gap is known

Vocabulary

Mis-assemblies:

Include regions where a genome is incorrectly re-arranged

- Or the places where large chunks of DNA sequence are simply deleted and the surrounding sequences just crunched together

Saeed

Conclusions

- The process of reconstruction of original chromosomes based on overlaps between short sequence reads
- Reads are assembled into contigs which are then assembled into scaffolds

Ahmad Saeed

Lesson 106 : 106_Genome Assembly Overview

Sample preparation

DNA is collected from the biological sample followed by library preparations



Schatz et al. Genome Biology 2012 13:243

Ahmad Saeed

Sequencing

The output from the sequencer consists of many billions short, unordered DNA fragments (strings of ATGCs) from random positions in genome



Schatz et al. Genome Biology 2012 13:243

Ahmad Saeed

Conclusions

Genome assembly is performed as follows;

- Sample preparation
- Sequencing
- Assembly
- Refinement



Lesson 107 : 107_Genome Assembly Planning

Ahmad Saeed

Factors

Following factors may be important in planning an Assembly project;

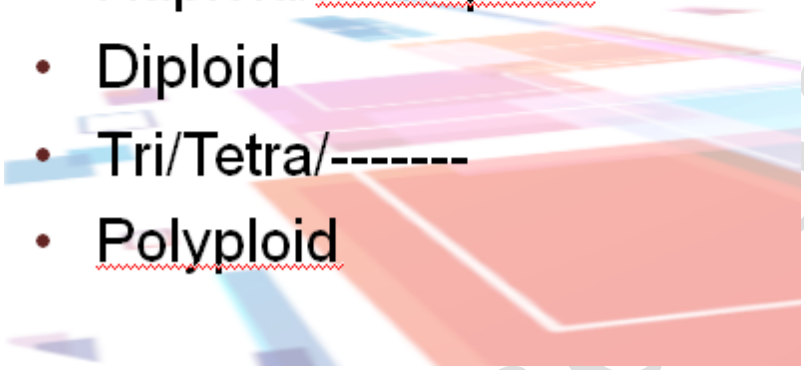
- **Size of Genome**
- Repeat content
- Related species Assembled
- Strategy for performing Assembly

Ahmad Saeed

Size of the genome

Size of the Genome can be estimated from the ploidy **(N)** of the organism and the DNA content per cell

- Haploid/Monoploid
- Diploid
- Tri/Tetra/-------
- Polyploid



Ahmad Saeed

Coverage

The average number of times any given base in the genome is sequenced

It can be derived by dividing the total length of acquired sequences by the genome length

Coverage

$$C = NL / G$$

N: Total number of reads

L: Length of a Read

G: Genome size

Represented as X

Asaeed

Coverage

Example

Illumina produced 1 million reads of 100 bp in length for a species whose genome size is 1 MB. **What is coverage?**

C = NL/G

1000000x100/1000000

= 100 X

Effect of genome size

- This will help decide which sequencing technology to use and what computational resources will be needed

Qasaeed

Conclusions

- Genome size is an important factor in Genome Sequencing.
- This will help decide which sequencing technology and computational resources will be needed

Lesson 108 : 108_Effect of repeats

Ahmia Syed

Factors

Following factors may be important in planning an Assembly project;

- Size of Genome
- **Repeat content**
- Related species Assembled
- Strategy for performing Assembly

Effect of Repeats

Normal Sequence



Incorrect Assembly with an Orphan contig (Red)



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Effect of Repeats

Repeat content

A major proportion of eukaryotic genome is made up of repeats.

Repeats are the most common source of Mis-assemblies

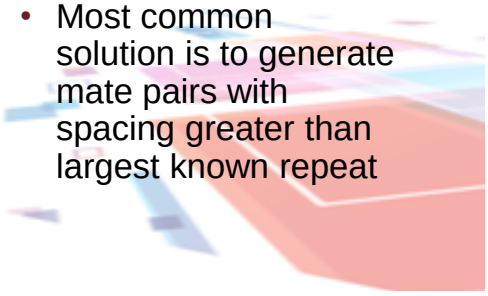


Effect of Repeats

Repeat solution

If sequencing technology produces reads longer than repeat size, impact is much smaller but that's not mostly the choice

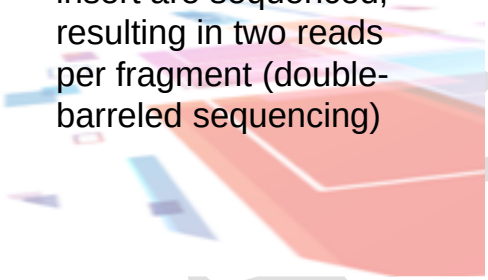
- Most common solution is to generate mate pairs with spacing greater than largest known repeat



Effect of Repeats

Double-barreled Shotgun Assembly

- DNA is randomly sheared into fragments (inserts)
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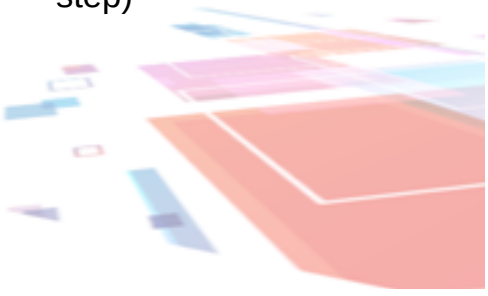


Sliced

Effect of Repeats

Double-barreled Shotgun Assembly

- The original DNA sequence is reconstructed from the reads (assembly step)



Effect of Repeats

Advantage of Double-barreled Shotgun Assembly

- It is unlikely that both reads (mate pairs) are coming from a repeat
- A read in a unique portion of DNA can help find the true repeat where it's mate is residing

Effect of Repeats

Conclusions

- Most of the eukaryotic genomes possess repeats
- Repeats pose special challenges to assembly
- Using mate pair libraries can help fix these issues

Related Species Assembled

Factors

Following factors may be important in planning an Assembly project;

- Size of Genome
- Repeat content
- **Related species Assembled**
- Strategies for performing Assembly

Related Species Assembled

Importance

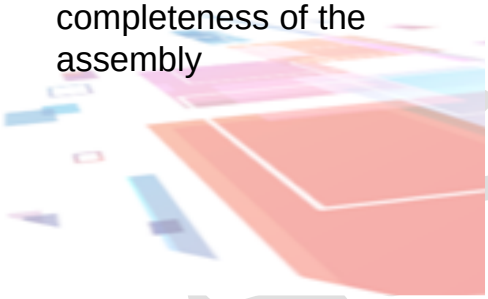
Availability of a related specie's good quality genome having large reliable scaffolds is always a great help

Related Species Assembled

Importance

Related species genomes helps in;

- Guiding the assembly of the target species
- Verifying the completeness of the assembly



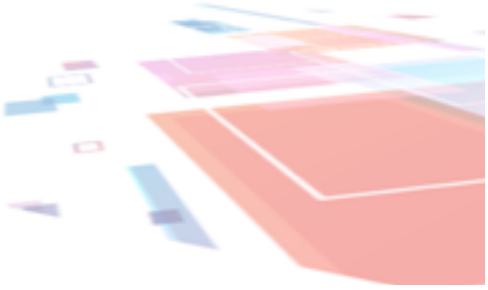
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Related Species Assembled

Importance

Related species genomes helps in;

- Can themselves be improved in some cases



Strategies for performing Assembly

Conclusions

- Availability of a related specie's good quality genome is always a great help in assembly
- The sequencing approaches and assembly strategies are interdependent

Lesson 110 : 110_Greedy Graph Algo

Greedy Graph Algorithm

Introduction

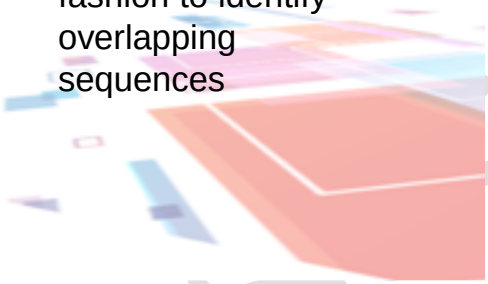
- Greedy Graph algorithms represent the simplest, most intuitive, solution to the assembly problem.

Greedy Graph Algorithm

Steps

Greedy Graph algorithms works as follows:

- Compare all reads or contigs in a pairwise fashion to identify overlapping sequences



Greedy Graph Algorithm

Steps

Greedy Graph algorithms works as follows:

- Merge the sequences that overlap each other the best



Greedy Graph Algorithm

Steps

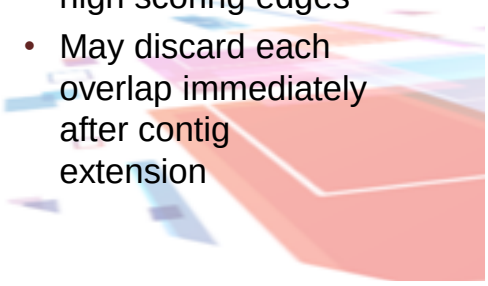
- Repeat step 2 until no more sequences can be merged, or the remaining overlaps conflict with existing contigs



Greedy Graph Algorithm

Advantages:

- Suitable for small size genomes
- They drastically simplify the graph by considering only the high scoring edges
- May discard each overlap immediately after contig extension



ALLPATHS-LG

Introduction

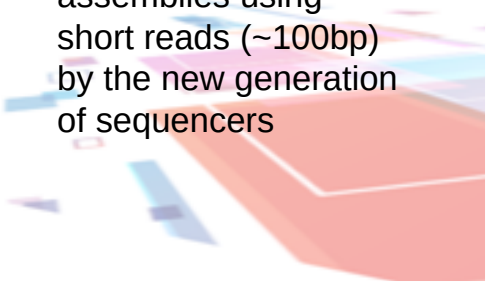
It's a short read genome assembler from the Computational Research and Development group at the Broad Institute



ALLPATHS-LG

Goal

- High quality genome assembly from low cost data
- It can generate high quality genome assemblies using short reads (~100bp) by the new generation of sequencers



ALLPATHS-LG

Difference between ALLPATHS-LG and Traditional Assemblers

- ALLPATHS-LG assemblies are not necessarily linear, but instead are presented in the form of a graph

Sliced

ALLPATHS-LG

Linearized Graph Assemblies

- An assembly consisting of contigs and scaffolds with embedded ambiguity codes

ALLPATHS-LG

Advantages

- Relatively fast runtime
- Can use long reads only for small genomes



ALLPATHS-LG

Conclusion

- It's a short read genome assembler from the Computational Research and Development group at the Broad Institute



Lander Waterman Curve

Introduction

- Lander/Waterman suggested in their 1988 paper that the number of times a base is sequenced follows a Poisson distribution



Lander Waterman Curve

Assumptions

There are two key assumptions they made;

- Reads will be randomly distributed in the genome



Lander Waterman Curve

Assumptions

There are two key assumptions they made;

- The ability to detect an overlap between two truly overlapping reads does not vary from clone to clone

Lander Waterman Curve

Conclusion

- The original paper relates to clone fingerprinting for physical mapping, but the even so the theory is readily applicable to whole genome shotgun projects.

Assembly Pipeline

Strategy

Overall process can be divided into following major phases;

- Quality Screening
- Overlapping
- Unitiging
- Scaffolding
- Repeat resolution

Sced

Assembly Pipeline

Quality screening

Screening is done while satisfying the following matrices;

- Genome properties (GC)
- Library quality
- Sequencing Run Quality
- Read Quality

Assembly Pipeline

Strategy

Overall process can be divided into following major phases;

- Quality Screening
- **Overlapping**
- Unitiging
- Scaffolding
- Repeat resolution

Assembly Pipeline

Strategy

Overall process can be divided into following major phases;

- Quality Screening
- Overlapping
- **Unitiging**
- Scaffolding
- Repeat resolution

Quality of Assembled Genome

Factors

The qualities of genome assemblies are evaluated by;

- Percentage of sequences assembled
- Accuracy of contigs and scaffolds
- Repeat resolution
- Presence of expected genes

Quality of Assembled Genome

N50

Most common measure of assembly quality;

- N50 = length of the shortest contig in a set making up 50% of the assembly length

Quality of Assembled Genome

Assembly tips

- Try different values of key parameters like k-mer size for DBG assemblers, and evaluate the output (some assemblers can do this automatically)

Quality of Assembled Genome

Conclusions

The qualities of genome assemblies are evaluated by;

- Percentage of sequences assembled
- Accuracy of contigs and scaffolds
- Repeat resolution
- Presence of expected genes

Graphs

Introduction

- Graph is a set of node plus set of edges between the nodes
- Nodes and edges may also be called vertices and arc, respectively

Sliced

Graphs

Types of Graphs

- Directed Graphs
- Undirected Graphs

Graphs

Directed Graphs

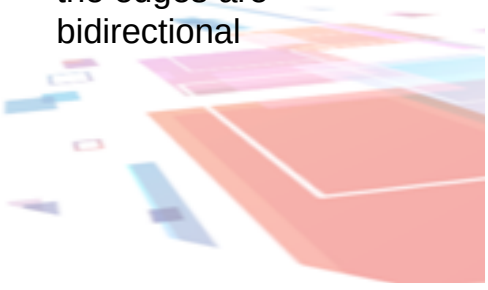
- If the edges may only be traversed in one direction, the graph is known as directed graph



Graphs

Undirected Graphs

- Undirected graph is set of objects (called vertices or nodes) that are connected together, where all the edges are bidirectional



Graphs

Conclusions

- Graph is a set of node plus set of edges between the nodes
- NGS assemblers have been organized into three categories, based on graphs

Lesson 117 : 117_OLC

Overlap Layout Consensus

Introduction

- The OLC strategy is arguably the most successful assembly strategy in practical setting
- 3 steps
  - Overlap
  - Layout
  - Consensus

Overlap Layout Consensus

Overlap stage

- Reads are compared to each other to construct a list of pairwise overlaps
- This information is then used to construct an overlap graph

Lesson 118 : 118_DBG

De Bruijn Graph

Introduction

- This approach for genome assembly is most widely applied to short reads from the Solexa and SOLiD Platforms

De Bruijn Graph

Steps

- Generate overlapping substrings of length k from the reads
- These substrings are called **Kmers**



Scanned

De Bruijn Graph

Steps

- Generate De Bruijn graph
  - Nodes represent all possible fixed length strings or **Kmers**



De Bruijn Graph

Steps

- Edges represent overlap of k-1 nucleotides
- Two nodes are linked with an edge if they share a k-1 mer



Sped

De Bruijn Graph

Steps

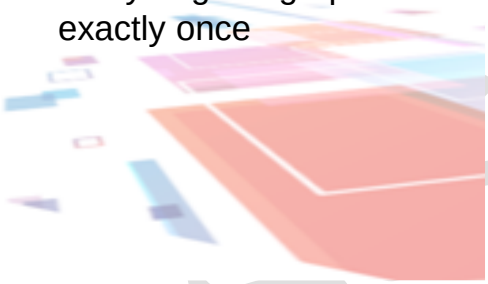
- Connect one Kmer to another if the two k-mers completely overlap except for one nucleotide at each end



De Bruijn Graph

Steps

- Third, look for a Euler cycle, representing a candidate genome because it visits every edge of graph exactly once

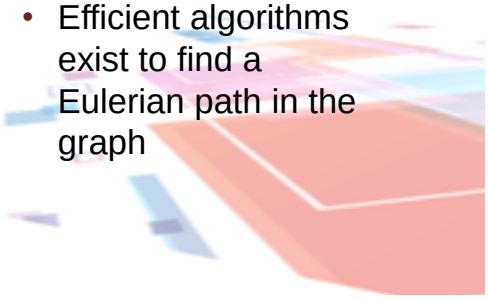


Sped

De Bruijn Graph

Advantages

- Works for short reads
- No pairwise overlap computation between reads which speeds up the process
- Efficient algorithms exist to find a Eulerian path in the graph



De Bruijn Graph

Conclusion

- This approach for genome assembly is most widely applied to short reads from the Solexa and SOLiD Platforms.



Lesson 119 : 119_DBG_Example

Ahmad Saeed

De Bruijn Graph Example

Deconstruction

- The first step of the De Bruijn assembler is to deconstruct the sequencing reads into its constitutive “kmers”.



Sped

De Bruijn Graph Example

Deconstruction

- The first step of the De Bruijn assembler is to deconstruct the sequencing reads into its constitutive “kmers”.



De Bruijn Graph Example

“It was the best of times, it was the worst of times, it was the age of wisdom, it was the age of foolishness, it was the epoch of belief, it was the epoch of incredulity,.... “

Dickens, Charles. A Tale of Two Cities. 1859. London: Chapman Hall
Example Adopted from J. Leipzig 2010

De Bruijn Graph Example

Itwasthebestoftimesitwastheworstoftimesitwas theageofwisdomitwastheageoffoolishness…

Generate random ‘reads’  **How do we assemble?**

fincreduli geoffoolis Itwasthebe Itwasthebe geofwisdom itwastheep epochofinc timesitwas stheepocho nessitwast wastheageo theepochof stheepocho hofincredu estoftimes eoffoolish lishnessit hofbeliefi pochofincr itwasthewo twastheage toftimesit domitwasth ochofbelie eepochofbe eepochofbe astheworst chofincred theageofwi iefitwasth ssitwasthe astheepoch efitwasthe wisdomitwa ageoffooli twasthewor

…etc. to 10’s of millions of reads

De Bruijn Graph Example

Conclusions

- Using an apropriate size Kmer and storing information stored in millions of reads is an important aspact of de Bruijn graph Algorithm



Ahmad Saeed

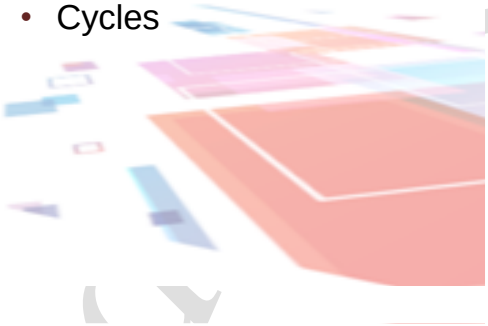
Ahmad Saeed

Lesson 120 : 120_Assembly Issues

Assembly Issues

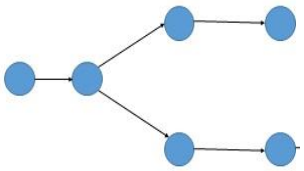
Problems in Graph Construction

- Spurs
- Bubbles
- Converging and diverging paths
- Cycles



Assembly Issues

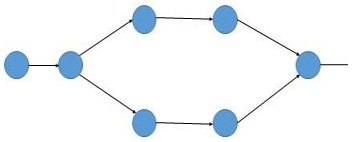
Spurs



- Short dead-end branches (divergences) of the main path
- Possible causes include sequencing errors toward one end of a read, and low coverage

Assembly Issues

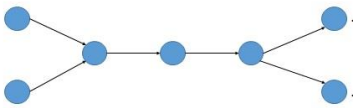
Bubbles



- Divergence of a path into two branches that afterwards join together again into one path
- Possible causes include sequencing errors toward the middle of a read, and by polymorphisms in the target

Assembly Issues

Convergence

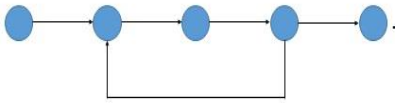


Converging and Diverging Paths

- Inverse definition than for the bubbles, two paths converge into one, that later diverges again into two separate paths
- Possible causes are repeats in the target genome

Assembly Issues

Cycles

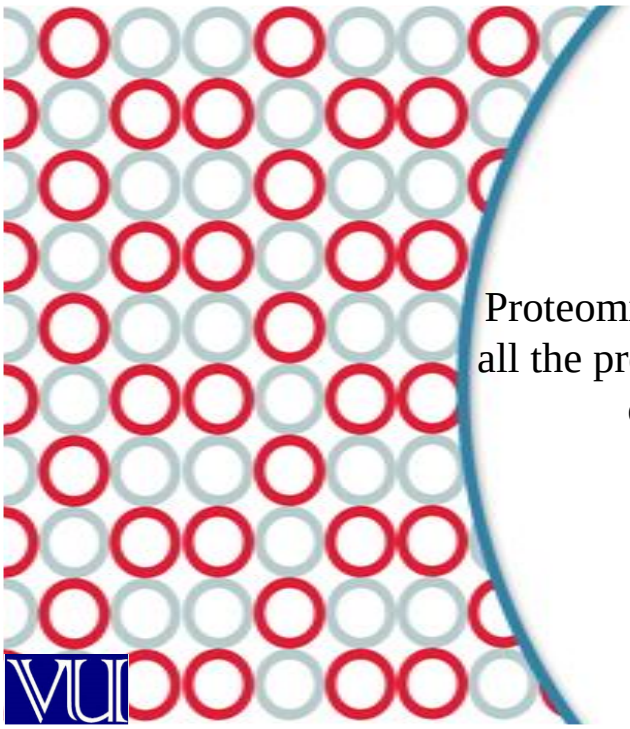


- Paths that converge on themselves.
- Possible causes are repeats in the target genome.

Assembly Issues

Quality Control

- Filter the graph of erroneous occurrences (i.e., bubbles, spurs, cycles) convergences or divergences



Proteomics is the study of all the proteins found in an organism

Lesson-1

Introduction to Proteomic

- The terms “proteomics” and “proteome” were coined by Marc Wilkins and colleagues in 1994 and mirror the terms “genomics” and “genome.



GENOMICS

- Genome: The entire sequence of an organism's hereditary information, including both coding and non-coding regions, encoded in DNA is known as "genome".
- Studying genome of an organism by employing sequencing and genome mapping is known as "genomics".

Lesson 122 : 122 Comparative Proteomics

Comparative Proteomics

Introduction

- For a new Genome, predicted genes are translated
- The collection of protein sequences encoded by the genome makes up the proteome of that individual

Comparative Proteomics

Cluster Analysis

- To sort out the relationships of all related proteins
- Clustering classify the proteins based on some objective criteria e.g
  - E value cut off
  - Distance in alignment

Lesson 123 : 123 Between Proteome Comparison

Between-Proteome comparisons

Introduction

- Each proteome is used as a query in a database similarity search against another proteome or a set of proteomes
- If proteome is not available, EST database may be searched

Between-Proteome comparisons

Significance

- Helps finding orthologs, Gene families and domains
- Proteins with highly significant alignment score are likely to be orthologs
- Mostly the proteins related to core biological functions

Between-Proteome comparisons

Finding true orthologs

- Method 1
  - Reciprocal Hits
  - E < 0.01
  - 60% coverage
- Keep matched pairs with a very conservative P value 10^{-10} to 10^{-100}

Between-Proteome comparisons

Clusters of Orthologous Group (COG)

- Orthologs are assumed to be derived from common ancestor
- Each orthologs might also have paralogs
- Orthologs are clustered to form **COG**

Between-Proteome comparisons

Conclusions

- Proteome comparison helps finding orthologs, gene families and protein domains
- Domain comparison reveals their biological roles

Proteomics, Types and Techniques

Study of Proteomics

- Proteomics is the study of all the proteins found in an organism
- Different Types

Proteomics, Types and Techniques

Proteomics: Proteome Complexity

GENOME	PROTEOME
▪ 4 nucleotides	▪ 20 amino acids
▪ Double helix	▪ Each protein has unique 3D shape
▪ Same in all cells	▪ Differs with cell type

Proteomics, Types and Techniques

Structural Proteomics

- Structural proteomics deals with mapping out the 3-D structure and nature of protein and/or proteins complexes

Proteomics, Types and Techniques

Functional Proteomics

- Posttranslational modifications
- Protein-protein, protein-ligand interactions
- Sequence-structure-function relationships

Proteomics, Types and Techniques

Study of Proteomics

- Protein profiling
- Predicting protein structure
- Protein networks

Proteomics, Types and Techniques

Protein Profiling

- Determination of the proteins that make up a given proteome

Proteomics, Types and Techniques

Protein Profiling

- Proteomes vary by cell type
- Proteomes vary by stage of cell development
- Some proteins abundant, others very rare

Proteomics, Types and Techniques

Protein Profiling Techniques

- Two-dimensional gel electrophoresis
- Chemical protein sequencing
- Protein sequencing by mass spectrometry

Proteomics, Types and Techniques

Proteins Structure Determination Techniques

- X-ray crystallography reliable but slow, not all protein crystallize
- Computer structure-prediction programs not reliable for all proteins

Proteomics, Types and Techniques

Proteomics - Conclusion

- Proteomics is the study of all the proteins found in an organism
- Different Types

Lesson 125 : 125_Characteristics of human proteome

Human Proteome - Characteristics

Proteomics

- Proteomics is the study of proteins that are generated from the genetic code of an organism.

Human Proteome - Characteristics

Proteomics

- Proteomics differs from genomics in that chromosomes for a genome are consistent throughout a multicellular organism, protein output varies from cell to cell.

Human Proteome - Characteristics

Human Proteome is Larger than Human Genome

- The proteome is larger than the genome due to alternative splicing.
- Humans ~ 250,000 proteins. Another estimate 1,000,000 individual proteins

Human Proteome - Characteristics

Human Proteome

- Less than 2 % of human diseases result from a single gene defect
- 98 % complex diseases like cancer are reflected in a modified protein network

Lesson 126 : 126_why proteomics

What do mean by proteomics?

- Proteomics is the large-scale study of proteins, usually by biochemical methods. The word proteomics has been associated
- traditionally with displaying a large number of proteins from a given
- cell line or organism on two -dimensional polyacrylamide gels

Scope of Proteomics

The identification, characterization and quantification of all proteins involved in a particular pathway, organelle, cell, tissue, organ or organism that can be studied in concert to provide accurate and comprehensive data about that system.

Or - A complete description of proteins expressed in any given cell at any given time

Why should we study Proteomics?

- directly contributes to drug development
- Verification of a gene product by proteomic methods
- Modifications of the proteins
- Protein expression level d

Advantages of study of proteomics

 Shows that genetic alterations are not the reaso for all types of diseases

 Helps in determining the proper treatment of diseases

 With the help of three dimensional analysis of proteins we have found that HIV protease is the enzyme which is responsible for AIDS.

 One of the most important use of proteomics in diagnosis is the identification of biomarkers. The study of drugs in proteomics is called pharmacoproteomics.

WHY PROTEOMICS?

Many types of information cannot be obtained from the study of genes alone. For example, proteins, not genes, are responsible for the phenotypes of cells. It is impossible to elucidate mechanisms of disease, aging, and effects of the environment solely by studying the genome.

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Lesson 127 : 127 Importance of Proteins

Biological Importance of proteins

- Cellular Function
- Structural importance
- Enzymes
- Harmones
- Transport protein
- Messenger protein
- Antibodies
- Plasma protein
- Protein in diet

Pharmaceutical Importance of proteins

- Proteins as pharmaceuticals
- Proteins applications
- Whey proteins health effects
- Iron chelate Protein
- Zinc chelate Protein
- Tumor markers

Cellular Function

- Structural support
- Biochemical reactions of cell
- Role of proteins in plasma membrane

Lesson 128 : 128 Chemical composition of proteins

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Chemical composition of proteins

- Proteins are polymers of amino acids.
- They range in size from small to very large.
- All the proteins are made up of Twenty different types of amino acids. So these amino acids are called standard amino acids.

Chemical composition of proteins

- In a protein molecule, each amino acid residue is joined to its neighbour by a specific type of covalent bond which is called Peptide Bond.

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Lesson 129 : 129 Primary structure of proteins

Primary structure of proteins

- Primary structure or covalent structure of protein refers to the amino acid sequence of its polypeptide chain.
- Each type of protein has a unique amino acid sequence.

Primary structure of proteins

Peptide Bond Is Rigid and Planar

- Linus Pauling and Robert Corey carefully analyzed the peptide bond.
- Their findings laid the foundation for our present understanding of protein structure.

Primary structure of proteins

- They demonstrated that the peptide C - N bond is somewhat shorter than the C - N bond in a simple amine.

Primary structure of proteins

- The six atoms of the peptide group are co-planar i.e., lie in a single plane, with the oxygen atom of the carbonyl group and the hydrogen atom of the amide nitrogen trans to each other.

Secondary structure of proteins

- Secondary structure of proteins refers to the local conformation of some part of a polypeptide.

Secondary structure of proteins

- A few types of secondary structures are particularly stable and occur widely in proteins.

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Lesson 131 : 131 alpha Helix

α- Helix

- The simplest arrangement which a polypeptide chain could assume with its rigid peptide bonds is a helical structure, which Pauling and Corey called the **α-helix**.

α- Helix

- The helical twist of the *α*-helix found in all proteins is right-handed.
- The repeating unit is a single turn of the helix, which extends about 5.4 Å (includes 3.6 amino acid residues) along the long axis.

α- Helix

- The amino acid residues in an helix have conformations with psi = –45 to –50 and phi = – 60.
- An helix makes optimal use of internal hydrogen bonds.

Lesson 132 : 132 ß- Pleated Sheets

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β- Pleated Sheets

- Pauling and Corey predicted a second type of secondary structure which they called **β-sheets.**
- This is a more extended conformation of polypeptide chains.

β- Pleated Sheets

- The backbone of the polypeptide chain is extended into a zigzag structure.
- The zigzag polypeptide chains are arranged side by side to form a structure resembling a series of pleats.

Tertiary Structure of Proteins

- The overall three-dimensional arrangement of all atoms in a protein is referred to as the protein's **tertiary structure.**

Tertiary Structure of Proteins

- It includes longer-range aspects of amino acid sequence.
- Amino acids that are far apart in the polypeptide chain may interact within the completely folded structure of a protein.

Tertiary Structure of Proteins

- Interacting segments of polypeptide chains are held in their characteristic tertiary positions by different kinds of weak interactions (and sometimes by covalent bonds) between the segments.

Tertiary Structure of Proteins

- Large polypeptide chains usually fold into two or more globular clusters known as **domains,** which often give these proteins a bi- or multilobal appearance.

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Lesson 134 : 134 Quaternary Structure of Proteins

Quaternary Structure of Proteins

- Some proteins contain two or more separate polypeptide chains or subunits, which may be identical or different.
- The spatial arrangement of these subunits is known as a protein's quaternary structure.

Quaternary Structure of Proteins

- A multi-subunit protein is also referred to as a multimer.
- A multimer with just a few subunits is called as oligomer and a single subunit or a group of subunits, is called a protomer.

Quaternary Structure of Proteins

- Identical subunits of multimeric proteins are generally arranged in a symmetric patterns.
- Oligomers can have either rotational symmetry or helical symmetry.

Quaternary Structure of Proteins

- There are several forms of rotational symmetry. The simplest is cyclic symmetry, involving rotation about a single axis.

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Lesson 135 : 135_ Life and death of a protein

Life and Death of a Protein

Proteins are synthesized by the translation of mRNAs into polypeptides on ribosomes.

In most cases, the initial polypeptide-translation product undergoes some type of modification before it assumes its functional role in a living system.

These changes are broadly termed “posttranslational modifications” and encompass a wide variety of reversible and irreversible chemical reactions.

Approximately 200 different types of posttranslational modifications have been reported. Some of these are summarized in Fig. 1, which depicts the life cycle of a prototypical protein.

Modifications during Protein Cycle

Modifications those occur early in the life of the protein

- Carboxylation of glutamate residues
- Removal of the N-terminal methionine
- Glycosylation
- Addition of Prosthetic groups
- Formation of multisubunit complexes
- Prenylation of cysteine residues assists anchoring of proteins in or on membranes.

These more or less “permanent” modifications and transport ultimately result in the delivery of functional proteins to specific locations in cells.

Cont…

- The activities of many proteins are then controlled by posttranslational modifications.
- The most prominent and best-understood of these is phosphorylation of serine, threonine, or tyrosine residues.
- Phosphorylation may activate or inactivate enzymes, alter proteinprotein interactions and associations, change protein structures, and target proteins for degradation.
- Protein phosphorylation regulates protein function in diverse contexts and appears to be a key switch for rapid on-off control of signaling cascades, cell-cycle control, and other key cellular functions.

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Lesson 136 : 136 Proteins as Modular Structures

Proteins as Modular Structures

Segments of amino acid sequences can be considered as functional building blocks or modules.

The modular units in proteins that confer specific properties and functions are referred to as “motifs” or “domains”.

Motifs and domains are recognizable sequences that confer similar properties or functions when they occur in a variety of proteins.

In some cases, amino acid sequences within motifs and domains are highly conserved and do not vary from protein to protein.

In other cases, some key amino acids occur in a reproducible relationship to each other in a sequence, even though various substitutions in other amino acids occur

Proteins as Modular Structures

Longer amino acid sequences often form domains, which confer specific properties or functions on a protein.

Some domain structures refer simply to sequences that confer a bulk physical property to a segment of the polypeptide, such as transmembrane domains, which simply form helices that span a lipid bilayer membrane.

Other domain structures provide hydrogen bonding or other contacts for key enzyme substrates or prosthetic groups.

In many cases, domains are made up of combinations of units of secondary structure, such as helix-loop-helix domains.

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Lesson 137 : 137_Proteins and Families

Type of Proteins/Families

Protein Structure

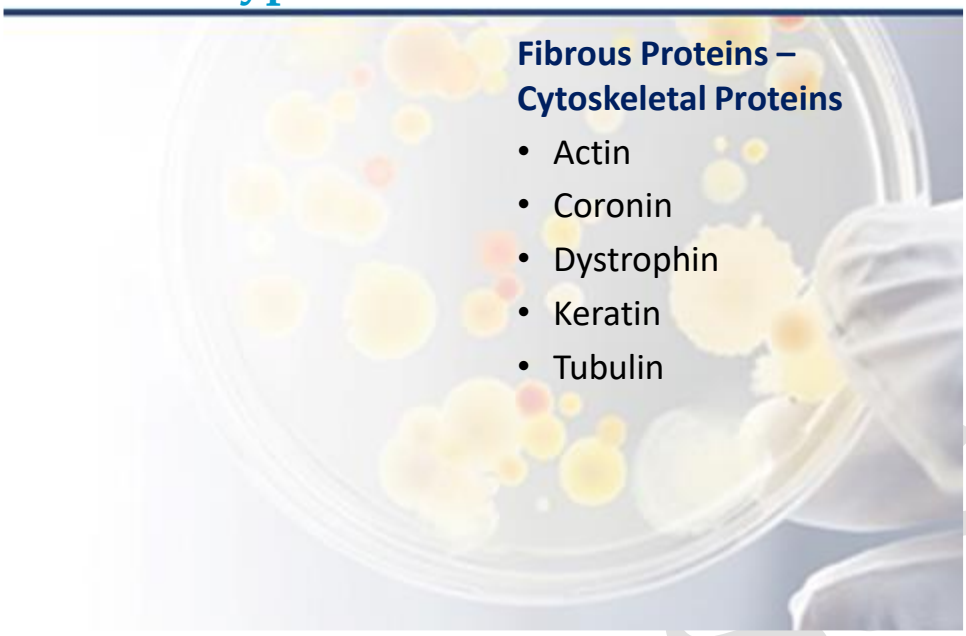
- Amino acids
  - 20 amino acids
  - Hydrophobic / hydrophylic
  - Charged / neutral
- Functions
  - Enzymes
  - Structure protein
  - Channel
  - Other functions

Type of Proteins/Families

Proteins

- Different method of categorizing proteins
- Three major categories
- Fibrous proteins
- Globular proteins
- Complexes with multiple components including proteins

Type of Proteins/Families



Fibrous Proteins – Cytoskeletal Proteins

- Actin
- Coronin
- Dystrophin
- Keratin
- Tubulin

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Lesson 138 : 138 over view of analytical proteomics

Analytical Protein and Peptide Separations

Overview

- At the stage of proteomic analysis, we must do two things

1. First, we must convert proteins to peptides. This is generally done with proteolytic enzymes.
2. Second, we must separate very complex mixtures of proteins or peptides into somewhat less complex mixtures

- This gives the MS instruments a better opportunity to obtain useful data on the

Ahmad Ali

Lesson 139 : 139 Extracting Proteins from Biological Samples

Introduction

- Biological sample can be a piece of tissue, a plate of cultured cells, a flask of bacteria, a leaf, and so on.
- The sample then is usually pulverized, homogenized, sonicated, or otherwise disrupted to yield a soup that contains cells, subcellular components, and other biological debris in an aqueous buffer or suspension.
- For proteomic analysis, the objective here is to recover as much of the protein as possible from this soup with as little contamination by other biomaterials (e.g., lipids, cellulose, nucleic acid, etc.) as possible.

- Protein is generally extracted with least contamination with the aid of

1. Detergents
2. Reductants
3. Denaturing agents
4. Enzymes

1. Detergents

- Detergents e.g SDS, 3- ([3-cholamidopropyl] dimethylammonio)-1propane sulfonate (CHAPS), cholate, Tween can be used for protein extraction.
- These help to solubilize membrane proteins and aid their separation from lipids

2. Reductants

- Reductants e.g. e.g., dithiothreitol [DTT], mercaptoethanol, thiourea are used.
- These reduce disulfide bonds or prevent protein oxidation

3. Denaturing agents

- Examples of denaturing agents are urea and acids etc.
- These agents disrupt protein protein interactions, secondary and tertiary structures by altering solution ionic strength and pH

4. Enzymes

- Enzymes like DNAse and RNAse are used.
- Enzymes digest contaminating nucleic acids, carbohydrates, and lipids present in the soup.

- The three principal separation approaches used with intact proteins are;

1. 1D-SDS-PAGE
2. 2D-SDS-PAGE
3. preparative isoelectric focusing (IEF).

- The protein mixture may be separated into a relatively small number of fractions as in 1D-SDS-PAGE and preparative IEF.
- Or it can be separated into many fractions as in the many spots in 2D-SDS-PAGE.
- The fractions then are taken for proteolytic digestion followed either by further separation of the peptide fragments or direct MS analysis of the peptides.

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Lesson 141 : 141. One-Dimensional SDS-PAGE

Principle

- It is a single most widely used analytical separation in all of protein chemistry.
- protein sample is dissolved in a loading buffer that usually contains a thiol reductant (mercaptoethanol or DTT) and SDS.
- The separation method is based on the binding of SDS to the protein, which imparts negative charge to the protein in proportion to molecular weight.
- When the gel is subjected to high voltage, the protein-SDS complexes migrate through the cross-linked polyacrylamide gel.
- Rate of migration is based on the ability of proteins to penetrate the pore matrix of the gel.
- The proteins thus are resolved into bands in order of molecular weight.

Extent of cross-linking in 1D-SDS PAGE

- In 1D SDS-PAGE the extent of cross linking in gels varies from 5-15%.
- Lower degrees of cross-linking allow easier passage of larger proteins through the gel.
- A sample containing low molecular-weight proteins is better resolved on a more highly cross-linked gel.

Lesson 142 : 142 Two-Dimensional SDS-PAGE

Principle

- It is a single best method for resolving highly complex protein mixtures.
- It involves two different types of separations.

1. In the first, the proteins are resolved on the basis of isoelectric point by IEF.
2. In the second, focused proteins then are further resolved by electrophoresis on a polyacrylamide gel

In past 2D SDS-PAGE was difficult to use because of

- The relative technical difficulty of performing the IEF step.
- Getting the delicate tube gel containing the focused proteins set up to efficiently transfer the proteins in the SDS-PAGE slab gel was a technical challenge.

New designed 2D SDS-PAGE System

- It uses immobilized pH gradient (IPG) strips and relatively foolproof hardware to facilitate the transfer of proteins from the IPG strip into the SDS-PAGE slab gel.
- In this, use of narrow pH ranges facilitates the separation of proteins with highly similar isoelectric points.
- The strip is hydrated with a buffer and the protein is slowly loaded into the strip under voltage.
- Then the voltage is increased to achieve focusing.
- After the focusing step, the strip is treated with a buffer that contains a thiol reductant and SDS and then is joined to the SDS-PAGE slab gel. In this respect, the IPG strip containing the focused proteins acts as a “stacking” gel in 1D-SDS-PAGE.
- The proteins then are resolved on the SDS-PAGE slab gel in the same manner as for 1D-SDS-PAGE.
- Proteins separated by 2D gels are visualized by conventional staining techniques, including silver, Coomassie, and amido black stains.

Problems 1

- The first is the difficulty of performing completely reproducible 2D-SDS-PAGE analyses. This problem becomes important when one wishes to use 2D-SDS-PAGE to compare two samples by comparing the images of the stained gels. Differences in protein migration in either dimension could be mistaken for differences in levels of certain proteins between the two samples.

Problem 2

- A second problem with 2D-SDS-PAGE is the relative incompatibility of some proteins with the first-dimension IEF step. Many large, hydrophobic proteins simply do not behave well in this type of analysis. Marginal solubility leads to protein precipitation and aggregation, which leads to “smearing” of proteins within the IPG strip, rather than clean focusing into discrete bands. When these proteins are subsequently run in the second (SDS-PAGE) dimension, these proteins appear as streaks across a molecular-weight region

Problem 3

- A third problem with 2D-SDS-PAGE is the relatively small dynamic range of protein staining as a detection technique. Spot densities reflect about a 100-fold range of protein concentrations, at best. This means that staining of 2D-gels allows the visualization of abundant proteins, whereas less abundant proteins frequently cannot be detected.

Lesson 144 : 144 Isoelectric Focusing

Ahmad S

Introduction

- The *isoelectric point* is the pH at which the net charge of the protein molecule is neutral.
- Different proteins have different isoelectric points.
- Isoelectric point is found by drawing the sample through a stable pH gradient.
- The range of the gradient determines the resolution of the separation.

- This technique is analogous to the first step in 2D-SDS-PAGE.
- In IEF generation of a pH gradient is achieved with soluble ampholytes, which are polycarboxylic acid compounds that generate a stable pH gradient when voltage is applied across the focusing cell.
- The protein sample then is added, voltage again is applied, and the proteins then are separated by isoelectric point.

- In commercially available apparatus, such as the BioRad Rotofor™ cell, the focusing cell is divided by permeable membranes into a series of chambers.
- After the focusing step, the chambers are quickly and simultaneously emptied by a vacuum sipper that draws the contents of each section of the cell into a separate tube.
- With this type of apparatus, the entire protein mixture is separated into 12–20 fractions

Advantages of solution phase IEF

- It has relatively large sample capacity (milligrams to grams of total protein per run).
- It has relative ease of working with samples in solutions as opposed to gels.
- The ampholytes can be removed from the fractionated samples by dialysis or gel filtration prior to further processing of the proteins.
- Recovery of proteins from solution-phase IEF typically exceeds 85–90%.
- Detergents and chaotropic agents can be used to maintain solubility of hydrophobic proteins.

Introduction

- High-performance liquid chromatography (HPLC; formerly referred to as high-pressure liquid chromatography), is a technique in analytical chemisry used to separate, identify, and quantify each component in a mixture.
- HPLC is a chromatographic technique that can separate a mixture of compounds. It is used in biochemistry and analytical chemistry to identify, quantify and purify the individual components of a mixture.
- Although HPLC of intact proteins has not become a widely used technique for analytical proteomics, it is nevertheless highly applicable as an initial step to fractionate protein mixtures.
- HPLC would appear to be about as useful as preparative IEF for resolving protein mixtures into fractions.

Principle

- HPLC relies on pumps to pass a pressurized liquid solvent containing the sample mixture through a column filled with a solid adsorbent material.
- Each component in the sample interacts slightly differently with the adsorbent material, causing different flow rates for the different components and leading to the separation of the components as they flow out the column

Advantages of HPLC

- The advantage of HPLC is the diversity of separation modes available. Indeed, tandem HPLC separations combine two different types of chromatography. For example, strong cation exchange, followed by RP (Reverse phase), would apply two completely different separation modes.

Lesson 146 : 146 Protein Separations After Digestion

Ahmad S

Introduction

- In this approach, the proteins in the sample are first digested into a mixture of peptides, then the peptides are separated prior to analysis.
- The extreme application of this approach would be to digest a complete cell or tissue extract to peptides and then perform MS analysis on the mixture.
- Indeed, this sort of analysis has been done with considerable success.

Use of HPLC for peptide separation

- The use of microcapillary HPLC with special control adaptations and automated MS instrument control allowed the acquisition of MS data on hundreds or thousands of peptides in a single run.
- The primary rationale for this approach is that it permits one to convert a very heterogeneous mixture of proteins to a more homogeneous mixture of peptides, which can be more easily analyzed.
- If one does elect this approach, the number of available methods to separate peptide mixtures is far more limited.

Use of other methods for peptide separation

- One-dimensional- and 2D-SDS-PAGE are out, as they are not practically useful in resolving peptides from digests, which typically display a much more limited range of pI and molecular weight.
- Although preparative IEF can be performed on peptide mixtures, but it may be of limited utility for resolving peptide mixtures.
- However, little has been done to evaluate preparative IEF as a tool for peptide separations and it cannot be ruled out.

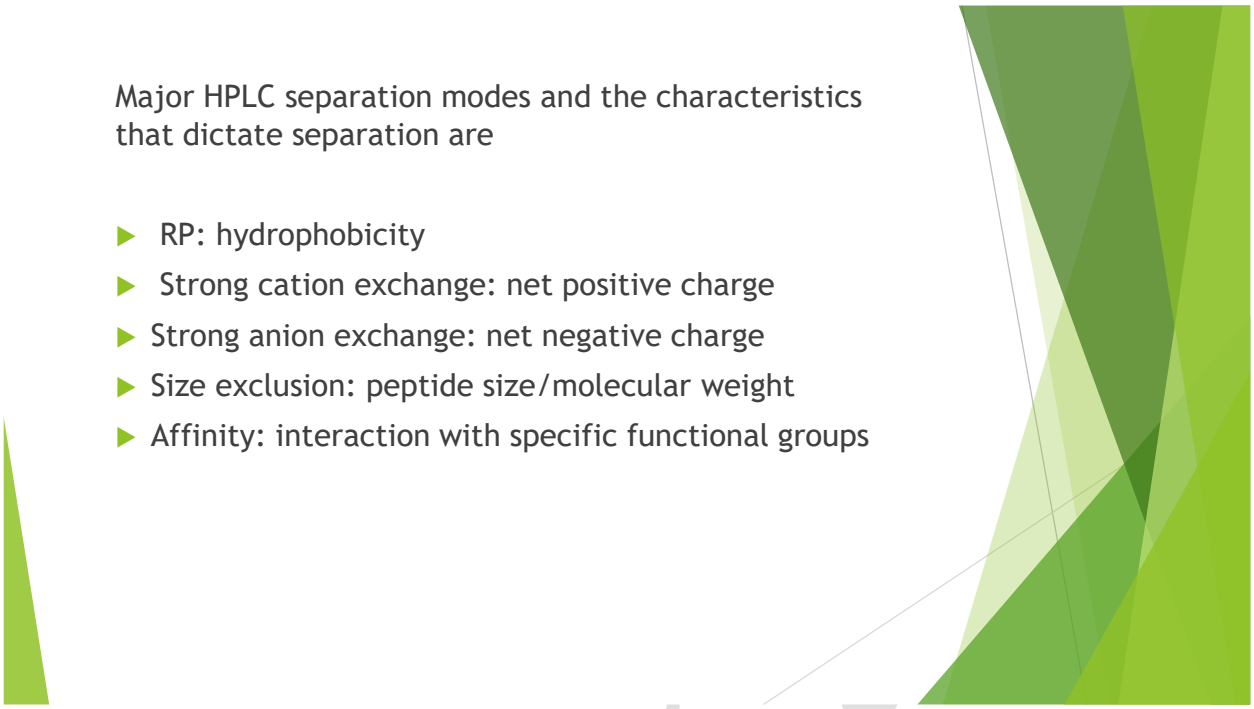
Lesson 147 : 147 Tandem LC Approaches for Peptide Analysis

Introduction

- Certainly the most widely used approach to analysis of peptide mixtures is HPLC.
- The diversity of stationary phases and separation modes gives HPLC considerable resolving power.
- The use of combined separation modes in series is referred to as “tandem HPLC.”
- The idea behind tandem LC is that the combination of dissimilar separation modes allows a greater resolution of peptides in a mixture.

Major HPLC separation modes and the characteristics that dictate separation are

- RP: hydrophobicity
- Strong cation exchange: net positive charge
- Strong anion exchange: net negative charge
- Size exclusion: peptide size/molecular weight
- Affinity: interaction with specific functional groups



Lesson 148 : 148 Capillary Electrophoresis (CE)

Ahmad S

- Capillary electrophoresis (CE) operates on the same general principal as IEF.
- Proteins placed in an electric field will migrate to a point in a pH gradient where they display an overall neutral change.
- The performance of the analysis in a microcapillary tube provides greatly enhanced resolution over the preparative IEF techniques discussed earlier.
- CE offers the greatest resolution of all peptide analytical techniques and can be coupled directly to MS instruments.
- CE thus has great potential as a technique for analytical proteomics.

- The utility of CE is limited at the present time by the lack of commercially available, robust, and reliable CE-MS instrumentation for analytical proteomics.
- Development of instrumentation for this purpose is continuing and CE-MS may become a very useful tool in proteomics analysis in the near future.

Working principle

- Capillary tube is placed between two buffer reservoir, and an electric field is applied, separation depends on electrophoretic mobility & electro-osmosis .
- Defined volume of analysate is introduced in to the capillary by replacing one buffer reservoir with sample vial.
- Electrophoretic separation is measured by detector.

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